Category Theory for Mathematical Physics

Braided, Fusion and Module Categories, in Pictures

Contents

Reading this course 4

I The Language 7

1 Categories: a theory of systems and processes 8
- 1.1 Systems and processes . . . 8
- 1.2 Categories and examples . . . 9
- 1.3 Functors . . . 10
- 1.4 Natural transformations; $\mathrm{Rep}(G) \cong \mathrm{Fun}(BG, \mathrm{Vect}_{\mathrm{fd}})$ . . . 11
- 1.5 Equivalence; V^{**} vs V^{*} . . . 13

2 Universal properties, adjunctions, semisimple categories 17
- 2.1 Universal properties . . . 17
- 2.2 Representables; Yoneda (statement and use) . . . 19
- 2.3 Adjunctions; $\otimes$–Hom . . . 20
- 2.4 Linear and semisimple categories; Schur . . . 21
- 2.5 $\mathrm{Rep}(S_3)$ and a Jordan block . . . 23

3 Monoidal categories and string diagrams 26
- 3.1 The definition . . . 26
- 3.2 Coherence (formal diagrams) . . . 28
- 3.3 String diagrams; interchange; Joyal–Street . . . 28
- 3.4 Eckmann–Hilton . . . 30
- 3.5 Vec_G^{ω}: the $\mathbb{Z}_2$ pentagon . . . 30
- 3.6 Monoidal functors . . . 31

4 Duality: cups, caps and quantum dimension 34
- 4.1 Duals and the snake . . . 34
- 4.2 Bending wires; teleportation as one picture . . . 37
- 4.3 Pivotal, trace, dimension, spherical . . . 37
- 4.4 The Penrose sign and the FS indicator . . . 40
- 4.5 Dagger . . . 41

II Anyons 43

5 Fusion categories: particles and how they fuse 44
5.1 The definition . . . 44
5.2 Fusion rings . . . 46
5.3 FPdim; the canonical spherical structure . . . 47
5.4 Vertices, bubbles, completeness . . . 48
5.5 Fusion spaces . . . 50

6 F-symbols and the pentagon 52
6.1 F-moves . . . 52
6.2 The pentagon equation . . . 54
6.3 Gauge and Ocneanu . . . 55
6.4 Fibonacci solved . . . 55
6.5 Ising: κH verified; κ gauge invariant . . . 57

7 Braiding: hexagons, Yang–Baxter and topological spin 59
7.1 B_n versus S_n . . . 59
7.2 Braided categories and hexagons . . . 60
7.3 Yang–Baxter is Reidemeister III . . . 61
7.4 R-symbols; Fibonacci solved . . . 62
7.5 Topological spin . . . 64
7.6 Pointed braidings; Ising . . . 65

III Quantum Groups and Knots 68

8 Hopf algebras and quantum groups 69
8.1 Bialgebras give monoidal representation categories . . . 69
8.2 Antipodes give duals . . . 70
8.3 Quasitriangularity gives a braiding . . . 71
8.4 Reconstruction, and what it forbids . . . 72
8.5 $U_q(\mathfrak{sl}_2)$ and the XXZ dictionary . . . 72

9 Ribbon categories, Temperley–Lieb and $\mathrm{SU}(2)_k$ 76
9.1 Twists and ribbon categories . . . 76
9.2 Tangles and the Reshetikhin–Turaev functor . . . 78
9.3 Temperley–Lieb and the Kauffman bracket . . . 78
9.4 p_2 and coloured strands . . . 80
9.5 Roots of unity, the negligible quotient, the pivotal flip . . . 81

IV Topological Order 84

10 Modular tensor categories and anyon models 85
10.1 S and T . . . 85
10.2 Premodular, modularity, the killing loop . . . 87
10.3 Verlinde . . . 88
10.4 $\mathrm{SL}(2,\mathbb{Z})$ and the central charge . . . 89
10.5 Fibonacci live; tables of the others . . . 90

V Boundaries and Defects 93

11 Algebras in tensor categories and module categories 94
11.1 Algebras and modules in C . . . 95
11.2 Module categories; the explicit module associator . . . 96
11.3 Internal Hom; Ostrik . . . 98
11.4 Vec_G: (H, ψ), worked for $\mathbb{Z}_2$; $\mathrm{Rep}(G)$ . . . 99

12 Morita equivalence and the Drinfeld centre 101
12.1 Module functors; the dual category; Morita equivalence . . . 101
12.2 The Drinfeld centre . . . 103
12.3 $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$ in full; $D(G)$ and $\mathcal{Z}(\mathrm{Vec}_G)$ . . . 104
12.4 Müger and ENO (statements) . . . 105

13 Boundaries, defects and non-invertible symmetries 108
13.1 The toric-code boundaries . . . 108
13.2 Module categories and boundaries; Lagrangian algebras . . . 109
13.3 TFIM recap; KW at criticality; the continuum line and the lattice caveat . 111
13.4 Full gauging, Rep(G), anomaly (statement) . . . 114

VI Synthesis 116

14 Topological quantum field theory: a synthesis 117
14.1 Atiyah's axioms; 1D . . . 117
14.2 Frobenius algebras; 2D TQFT is commutative Frobenius . . . 118
14.3 Dijkgraaf–Witten . . . 121
14.4 Turaev–Viro and string nets . . . 122
14.5 RT and TV = RT $\circ$ $\mathcal{Z}$; the outlook . . . 124

Diagram reference 126

Bibliography 129

Reading this course

Physics describes systems and processes. Processes can follow one another, and systems can be put together. Category theory records these operations and the relations between them. This course develops the structures needed to describe antiparticles, fusion and exchange, boundaries, and topological quantum field theories. Its examples are finite enough to calculate.

The notes are intended for independent reading. Lecture numbers divide the material into topics. A lecture need not be read in one sitting. The worked examples and exercises are part of the learning path. There is no page or reading-time target.

Starting knowledge

The course assumes linear algebra over $\mathbb{C}$, including dual spaces and tensor products. It also assumes group representations, intertwiners, Schur's lemma, characters and the addition of angular momenta. The relevant background is in *Linear Algebra for Physics*, Lectures 11–12, *Symmetries*, Lectures 3–5, 8 and 12–14, and *Quantum Mechanics I*, Lectures 10–11 and 13.

The optional diagnostic, Sheet 0, turns these prerequisites into tasks. Its questions identify the earlier lectures that supply each tool and the places where it is used here. Difficulty with dual bases points back to linear algebra. Difficulty with characters or spin multiplicities points back to representation theory. Difficulty with partial traces points back to quantum tensor products. The diagnostic has hints, answers and worked solutions. Its purpose is to identify what to revisit.

No category theory, Hopf algebras or knot theory is assumed. The topology used in the examples is introduced through paths, surfaces and gluing. The XXZ comparison in Lecture 8 is optional context for readers of *Integrability*. The chain and lattice models used later are described where their properties are needed.

The route through the course

The first block develops the language. Lectures 1 and 2 describe familiar linear algebra and representations by their maps. A linear category alone forgets how representations combine. Lecture 3 adds a tensor product, and Lecture 4 adds duals and traces. The first practical task is to turn a composite of maps into a diagram and read its source and target back.

The second block describes fusion and exchange. Lecture 5 constructs fusion spaces from allowed intermediate charges. Lecture 6 changes the fusion basis, and Lecture 7 computes exchanges. Fibonacci is the main calculation: the same particles lead to a fusion rule, an F-matrix, R-symbols and a braid operation on a two-dimensional fusion space.

The third block constructs examples. Lecture 8 explains how an algebra can supply tensor products, duals and braidings on its representations. Lecture 9 develops ribbons, link evaluation and the finite categories $\mathrm{SU}(2)_k$. These constructions explain where several of the earlier models come from. The ribbon structure in the first section of Lecture 9 is

also used by Lecture 10. A route focused on anyon data must retain that section even when treating the quantum-group and Temperley–Lieb constructions as further study.

The fourth block, Lecture 10, asks which charges can be detected by braiding. It constructs the S-matrix, recovers the fusion rules and computes central charges. The fifth block, Lectures 11 to 13, studies actions on categories. The two small actions of $\mathrm{Vec}_{\mathbb{Z}_2}$ eventually describe the two boundaries of the toric code. The centre describes the bulk, while a dual category describes boundary excitations. These categories have different roles.

The last block, Lecture 14, relates gluing spaces and spacetimes to linear maps. Duality reappears in one dimension, a commutative Frobenius algebra describes two-dimensional theories, and fusion and modular data supply three-dimensional constructions. The distinction between a state space and a partition function remains essential.

Examples to keep together

The examples have different purposes. The representations of S_3 show which information a categorical structure remembers. The categories Vec_G^{ω} show how an associator changes without changing fusion rules. Fibonacci supplies explicit many-particle calculations. Ising and $\mathrm{SU}(2)_2$ show why equal fusion rules need not give the same associators or spins. The centre of $\mathrm{Vec}_{\mathbb{Z}_2}$ connects a small algebraic calculation to the toric code and its boundaries.

Practice and feedback

After a worked example, reproduce its main construction with the notes closed. Check the types of maps and the allowed labels before doing arithmetic. For a picture, write its source and target and name the rule that permits the next move. A correct final number does not identify a mistaken intermediate step.

Each lecture has a sheet of longer exercises. Every exercise has a hint, an answer and a complete worked method. The three PDF editions contain questions only, hints and answers, or worked solutions. On the website the stages open separately. Attempt the question first. Use a hint to restart, compare the answer, then read the worked method to locate a gap. The starred exercise is an extension. Ordinary exercises do not depend on solving it.

The website also has six short self-checks per lecture. These have explained answers. The companion *Self-checks with Explained Answers* prints the same questions, with all feedback after the questions. The exercise sheets and self-check companion are available in both English and Lithuanian. Each language page provides its matching practice edition. The complete notes include the diagram reference at the end.

The notes, diagnostic and feedback are available at `https://matfiz.github.io/kategorijos/en/guide/`. The Lithuanian reading guide is at `https://matfiz.github.io/kategorijos/guide/`.

The calculation companion provides the finite checks cited in the notes, their code and their scope. These calculations test examples and conventions; they do not replace proofs.

What a proof label means

An unqualified theorem or proposition is proved with the tools stated in its proof. The label *proved given coherence* records the use of the coherence theorem behind strict pictures. A *sketch* gives the central argument without every detail. A *cited* result is an input whose proof is in the stated source. A *physical principle* identifies a physical assumption rather than a mathematical theorem. An example explains how an input is used; it does not change that input's status.

Compound labels distinguish the general input from the part established here. The label *proved by verification* identifies a finite identity check; the proof states which data and identities are covered.

Block I

The Language

Lecture 1

Categories: a theory of systems and processes

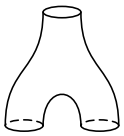
Objectives. *Physics composes processes: one after another, and, from Lecture 3 on, side by side. This lecture fixes the vocabulary that every later lecture uses: categories, functors, natural transformations and equivalences. Its main result says that a representation of a group is a functor, and an intertwiner a natural transformation. Naturality then makes "canonical" precise: $V \to V^{**}$ is natural, and no linear isomorphism $V \to V^{*}$ is.*

By the end of this lecture you will be able to:

- Check the axioms of a category on an example, and read a group as a category with one object.
- Prove that a representation of G is a functor $BG \to \mathrm{Vect}_{\mathrm{fd}}$ and an intertwiner a natural transformation, and slide a box through a natural transformation in a picture.
- Prove that $V \to V^{**}$ is natural and that no family of linear isomorphisms $V \to V^{*}$ is, and say why the Riesz map escapes.
- Prove $\mathrm{Vect}_{\mathrm{fd}} \simeq \mathrm{Mat}_{\mathbb{C}}$, and explain why the two categories are equivalent but not isomorphic.

1.1 Systems and processes

Three theories from this collection share a shape. In quantum mechanics the systems are Hilbert spaces and the processes linear maps. In representation theory the systems are representations of a group G and the processes are intertwiners (*Symmetries*, Lecture 3). In topology the systems are closed curves, finitely many circles, and a process is a surface whose boundary is the incoming circles and the outgoing ones, a *cobordism*. The pair of pants takes two circles to one:



We draw time upward, as throughout the course: a process starts at the bottom and ends at the top.

In each theory a process has an input and an output. Two processes compose when the output of the first is the input of the second. Composing is associative. Every system has a process that does nothing: the identity map, the identity intertwiner, the cylinder. Baez and Stay set the dictionary out in a table (their Table 2). A theory with this shape is a *category*.

What the vocabulary buys is a way to compare theories. A *functor* translates one category into another, and a *natural transformation* compares two translations. The main result

(Theorem 1.9) says that a representation of G is exactly a functor from a category with one object, and an intertwiner exactly a natural transformation. Two further results show what naturality is for: it makes the word "canonical" a theorem (Theorem 1.11), and it lets two categories be "the same" without being equal (Theorem 1.16).

1.2 Categories and examples

Definition 1.1 (Category)**.** A *category* C consists of

- a collection of *objects* $X, Y, Z, \ldots$;
- for each pair of objects a set $\mathrm{Hom}(X, Y)$ of *morphisms* $f\colon X \to Y$;
- for $f \in \mathrm{Hom}(X, Y)$ and $g \in \mathrm{Hom}(Y, Z)$ a *composite* $g \circ f \in \mathrm{Hom}(X, Z)$;
- for each object an *identity* $\mathrm{id}_X \in \mathrm{Hom}(X, X)$,

such that for all $f\colon X \to Y$, $g\colon Y \to Z$, $h\colon Z \to W$

$$h \circ (g \circ f) = (h \circ g) \circ f,$$
$$f \circ \mathrm{id}_X = f = \mathrm{id}_Y \circ f.$$

In $g \circ f$ the morphism f acts first. In a picture a morphism $f\colon X \to Y$ is a box on a strand, X below and Y above, and $g \circ f$ puts g above f. A chain of boxes is drawn without brackets:



$$= h \circ (g \circ f) = (h \circ g) \circ f. \tag{1.1}$$

The picture is sound: by associativity, induction on the length shows that all bracketings of a chain give one morphism, and by the unit laws a box id_X is a plain strand (Selinger, Thm. 2.1, drawn there left to right). That is all a picture of a single strand needs. Pictures in two dimensions need more (Lecture 3).

Example 1.2 (Categories)**.** Each of the following is a category.

- Set: sets and functions.
- Vect: complex vector spaces and linear maps; $\mathrm{Vect}_{\mathrm{fd}}$, the finite-dimensional ones.
- $\mathrm{Hilb}_{\mathrm{fd}}$: finite-dimensional complex inner-product spaces and *all* linear maps. Its morphisms are those of $\mathrm{Vect}_{\mathrm{fd}}$.
- BG, for a group G: one object $*$, with

$$\mathrm{Hom}(*, *) = G,$$
$$g \circ h = gh,$$
$$\mathrm{id}_* = 1.$$

  The axioms are the group axioms.

- $\mathrm{Rep}(G)$: the finite-dimensional complex representations (V, ρ_V) of G, and the intertwiners $T : V \to W$,
$$T\rho_V(g) = \rho_W(g)T \quad \text{for all } g \in G$$
(*Symmetries* Def. 3.11, without asking T to be invertible). Composites of intertwiners are intertwiners.
- The *braid groupoid* Br (Mac Lane's braid category): the objects are $n = 0, 1, 2, \ldots$; a morphism $n \to n$ is a braid of n strands up to isotopy with its ends fixed, and there is none from m to $n \neq m$. Braids compose by stacking, the second on top:



- Cob_2: the objects are finite disjoint unions of oriented circles, a morphism is an oriented surface between them up to diffeomorphism fixing the boundary, composition glues outgoing circles to incoming ones, and the identity is the cylinder (Kock, 1.3.14–1.3.20, for the details). Until Lecture 14 it is used only as pictures.

Definition 1.3 (Isomorphism, groupoid)**.** A morphism $f : X \to Y$ is an *isomorphism* if some $g : Y \to X$ has
$$g \circ f = \mathrm{id}_X,$$
$$f \circ g = \mathrm{id}_Y .$$
Then g is unique, written f^{-1}, and $X \cong Y$. A *groupoid* is a category in which every morphism is an isomorphism.

Uniqueness: if h is another such morphism, then
$$h = h \circ (f \circ g) = (h \circ f) \circ g = g.$$

The isomorphisms of Vect are the invertible linear maps. BG is a groupoid, and so is Br: the inverse of a braid is its mirror image in a horizontal line. The finite-dimensional Hilbert spaces with only the unitary maps form a groupoid; with all linear maps they form $\mathrm{Hilb}_{\mathrm{fd}}$, which is not one.

1.3 Functors

Definition 1.4 (Functor)**.** A *functor* $F : C \to \mathcal{D}$ assigns to each object X of C an object $F(X)$ of $\mathcal{D}$, and to each $f : X \to Y$ a morphism $F(f) : F(X) \to F(Y)$, such that
$$F(g \circ f) = F(g) \circ F(f),$$
$$F(\mathrm{id}_X) = \mathrm{id}_{F(X)} .$$
The *opposite category* C^{op} has the objects of C and a morphism $X \to Y$ for each morphism $Y \to X$ of C, composed in the reverse order. A functor $C^{\mathrm{op}} \to \mathcal{D}$ is a *contravariant* functor from C to $\mathcal{D}$: it reverses composition.

Example 1.5 (Functors)**.** Each of the following is a functor.

- The *forgetful* functor $\mathrm{Rep}(G) \to \mathrm{Vect}_{\mathrm{fd}}$ sends (V, ρ_V) to V and an intertwiner to itself.

- The *dual*: $V^* = \mathrm{Hom}(V, \mathbb{C})$, and $f : V \to W$ goes to $f^* : W^* \to V^*$,
$$f^*(\varphi) = \varphi \circ f$$
(*Linear Algebra*'s dual map T', Def. 11.20). For $g : W \to Z$,
$$\begin{aligned}(g \circ f)^*(\varphi) &= \varphi \circ g \circ f \\ &= f^*\big(g^*(\varphi)\big),\end{aligned}$$
so
$$(g \circ f)^* = f^* \circ g^*,$$
and the dual is a functor $\mathrm{Vect}^{\mathrm{op}} \to \mathrm{Vect}$. Applied twice it is covariant: $V \mapsto V^{**}$, $f \mapsto f^{**}$.
- *Forgetting crossings*: a braid on n strands sends the strand that starts at i to a position $\pi(i)$, and $\pi \in S_n$. This defines a functor from Br to the groupoid with objects n and morphisms $\mathrm{Hom}(n, n) = S_n$. The braid drawn above gives $1 \mapsto 3$, $2 \mapsto 1$, $3 \mapsto 2$. *Symmetries* (Lecture 1) draws permutations as such strands, read upward as here.

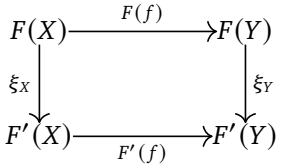
Lemma 1.6. *A functor sends isomorphisms to isomorphisms, with $F(f^{-1}) = F(f)^{-1}$.*

Proof.
$$\begin{aligned}F(f^{-1}) \circ F(f) &= F(f^{-1} \circ f) \\ &= F(\mathrm{id}_X) \\ &= \mathrm{id}_{F(X)},\end{aligned}$$
and likewise in the other order. ∎

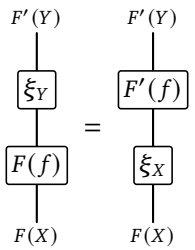
1.4 Natural transformations; $\mathrm{Rep}(G) \cong \mathrm{Fun}(BG, \mathrm{Vect}_{\mathrm{fd}})$

Definition 1.7 (Natural transformation)**.** Let $F, F' : \mathcal{C} \to \mathcal{D}$ be functors. A *natural transformation* $\xi : F \Rightarrow F'$ is a morphism $\xi_X : F(X) \to F'(X)$ for each object X, its *component*, such that for every $f : X \to Y$
$$F'(f) \circ \xi_X = \xi_Y \circ F(f).$$
It is a *natural isomorphism* if every component is an isomorphism.

The condition says that the square



commutes: the two paths from $F(X)$ to $F'(Y)$ agree. Drawn upward, it says that the box $F(f)$ slides through ξ and comes out as $F'(f)$:



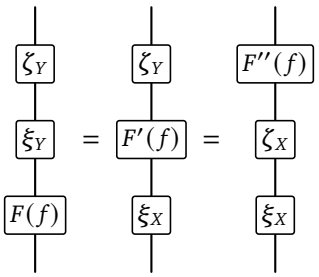
(1.2)

The square names the objects at its corners; the picture puts them on strands and lets the boxes move. From Lecture 3 on we use the picture, because it extends to two dimensions.

Lemma 1.8 (Composing natural transformations). *If $\xi\colon F \Rightarrow F'$ and $\zeta\colon F' \Rightarrow F''$ are natural, so is $\zeta \circ \xi\colon F \Rightarrow F''$, with components*

$$(\zeta \circ \xi)_X = \zeta_X \circ \xi_X.$$

Proof. Slide $F(f)$ through ξ by (1.2), then $F'(f)$ through ζ by (1.2) for ζ:



(1.3)

The outer pictures are the two sides of naturality for $\zeta \circ \xi$. ∎

If the objects of C form a set, as for BG, the functors $C \to \mathcal{D}$ form a category $\mathrm{Fun}(C, \mathcal{D})$: its morphisms are the natural transformations, composed by Lemma 1.8, with identities $(\mathrm{id}_F)_X = \mathrm{id}_{F(X)}$. Associativity and the unit laws hold component by component.

Two categories are *isomorphic*, $C \cong \mathcal{D}$, if there are functors $C \to \mathcal{D} \to C$ whose two composites are the identity functors.

Theorem 1.9 (Representations are functors). *A functor $BG \to \mathrm{Vect}_{\mathrm{fd}}$ is a representation of G, and a natural transformation between two such functors is an intertwiner. Hence*

$$\mathrm{Rep}(G) \cong \mathrm{Fun}(BG, \mathrm{Vect}_{\mathrm{fd}}),$$

an isomorphism of categories that changes no data.

Proof. A functor $F\colon BG \to \mathrm{Vect}_{\mathrm{fd}}$ is a space $V = F(*)$ and a linear map $F(g)\colon V \to V$ for each $g \in G$. Since $g \circ h = gh$ in BG, the axioms read

$$F(gh) = F(g)F(h),$$
$$F(1) = \mathrm{id}_V\,.$$

By Lemma 1.6 each $F(g)$ is invertible, so $\rho = F$ is a homomorphism $G \to \mathrm{GL}(V)$; conversely, every representation satisfies the two axioms.

A natural transformation $\xi\colon F \Rightarrow F'$ has the single component $T = \xi_*\colon V \to V'$, and naturality for the morphism g reads

$$\rho'(g)\,T = T\rho(g).$$

The map T is an intertwiner, and every intertwiner is natural. Natural transformations compose by composing components, as intertwiners do, and the identities agree. ∎

Example 1.10 (S_3 on $\mathbb{C}^3$). Let S_3 permute the coordinates, $\rho(\pi)e_i = e_{\pi(i)}$, and let $\mathbb{C}$ carry the trivial representation, $\pi \mapsto 1$. The sum $s(x) = x_1 + x_2 + x_3$ is natural from ρ to the trivial

representation, since permuting the coordinates does not change their sum. The coordinate $p_1(x) = x_1$ is linear but not natural: for the transposition $(1\,2)$,

$$p_1\big(\rho((1\,2))e_1\big) = p_1(e_2) = 0,$$
$$p_1(e_1) = 1 \neq 0.$$

So the forgetful functor $\mathrm{Rep}(S_3) \to \mathrm{Vect}_{\mathrm{fd}}$ misses morphisms. In the other direction, $u(1) = e_1 + e_2 + e_3$ is natural from the trivial representation to ρ. By Lemma 1.8 so is the composite $u \circ s$, whose matrix has every entry 1: indeed it commutes with every permutation matrix.

1.5 Equivalence; V^{**} vs V^{*}

Linear Algebra (Thm. 11.17) calls $V \to V^{**}$ canonical because it uses no basis; naturality makes this a theorem. To compare V with V^{*}, we must first make the dual covariant. On the groupoid of finite-dimensional spaces and linear isomorphisms, set

$$D(V) = V^{*},$$
$$D(f) = (f^{*})^{-1}.$$

This is a functor, since

$$\big((g \circ f)^{*}\big)^{-1} = (g^{*})^{-1} \circ (f^{*})^{-1}.$$

Theorem 1.11 (The double dual is natural; the dual is not)**.**

(a) The maps $a_V : V \to V^{**}$,

$$a_V(v)(\varphi) = \varphi(v),$$

form a natural transformation $\mathrm{id}_{\mathrm{Vect}} \Rightarrow (-)^{**}$, and a natural isomorphism on $\mathrm{Vect}_{\mathrm{fd}}$.

(b) No family of linear isomorphisms $\theta_V : V \to V^{*}$, one for each finite-dimensional V, is a natural transformation from the identity to D. On $\mathrm{Hilb}_{\mathrm{fd}}$ none is natural even for the unitary maps alone.

Proof. (a) For $f : V \to W$ the double dual map is

$$f^{**}(x) = x \circ f^{*}.$$

For $v \in V$ and $\psi \in W^{*}$,

$$\begin{aligned} f^{**}\big(a_V(v)\big)(\psi) &= a_V(v)(\psi \circ f) \\ &= \psi(f(v)) \\ &= a_W(f(v))(\psi), \end{aligned}$$

so

$$f^{**} \circ a_V = a_W \circ f.$$

For finite-dimensional V, a_V is an isomorphism (*Linear Algebra* Thm. 11.17).

(b) Naturality for $f : V \to W$ reads

$$\theta_W \circ f = (f^{*})^{-1} \circ \theta_V,$$

that is,

$$f^{*} \circ \theta_W \circ f = \theta_V.$$

Take $V \neq 0$ and $f = i\,\mathrm{id}_V$, which is unitary for every inner product. Then $f^{*} = i\,\mathrm{id}_{V^{*}}$, and since θ_V is linear,

$$f^{*} \circ \theta_V \circ f = i \cdot i\,\theta_V = -\theta_V.$$

So naturality demands $-\theta_V = \theta_V$, that is $\theta_V = 0$, which is not an isomorphism. ∎

Example 1.12 (A basis is a choice)**.** A basis (e_i) of V does give an isomorphism $\theta\colon V \to V^*$, $\theta(e_i) = e^i$, onto the dual basis. Take instead the basis $e'_i = 2e_i$. Its dual basis is $e'^{\,i} = e^i/2$, and its isomorphism θ' sends $e_i = e'_i/2$ to $e'^{\,i}/2 = e^i/4$:

$$\theta' = \tfrac{1}{4}\,\theta.$$

The isomorphism changes with the basis, and (b) says more: no way of choosing a basis for every space, however clever, gives a natural family.

Example 1.13 (The Riesz map is not a counterexample)**.** On $\mathrm{Hilb}_{\mathrm{fd}}$ the Riesz map

$$R_V(u) = \langle u, \cdot \rangle$$

is a bijection $V \to V^*$, and for every unitary f

$$f^*\big(R_W(fu)\big) = \langle fu, f\,\cdot\rangle = R_V(u).$$

It escapes (b) because the inner product is antilinear in its first slot (*Linear Algebra* Prop. 11.13), so that

$$R_V(iu) = -i\,R_V(u).$$

In the test of (b) the two factors i cancel instead of multiplying to -1:

$$\langle iu, iw\rangle = (-i)\,i\,\langle u, w\rangle = \langle u, w\rangle\,.$$

Hence R_V is not a morphism of Vect. Lecture 4 builds duals from ev and coev, never from an inner product.

Theorem 1.11 compares objects inside one category. The last question compares two categories. $\mathrm{Vect}_{\mathrm{fd}}$ has a whole class of objects, all the spaces; matrices need only the numbers $0, 1, 2, \ldots$ Linear algebra treats the two as interchangeable, and the next definition says in what sense.

Definition 1.14 (Equivalence)**.** An *equivalence* $C \simeq \mathcal{D}$ is a pair of functors $F\colon C \to \mathcal{D}$ and $G\colon \mathcal{D} \to C$ with natural isomorphisms

$$G \circ F \cong \mathrm{id}_C,$$
$$F \circ G \cong \mathrm{id}_{\mathcal{D}}\,.$$

Definition 1.15 (Full, faithful, essentially surjective)**.** A functor $F\colon C \to \mathcal{D}$ is *faithful* or *full* if each map

$$\mathrm{Hom}(X, Y) \to \mathrm{Hom}(F(X), F(Y))$$

is injective or surjective, respectively. It is *essentially surjective* if every object of $\mathcal{D}$ is isomorphic to some $F(X)$.

The forgetful functor of Example 1.10 is faithful but not full: it keeps every intertwiner, and misses p_1. Let $\mathrm{Mat}_{\mathbb{C}}$ be the category with objects $n = 0, 1, 2, \ldots$, whose morphisms $m \to n$ are the complex $n \times m$ matrices,

$$\mathrm{Hom}(m, n) = \mathbb{C}^{n \times m},$$

composed by the matrix product. It is *skeletal*: isomorphic objects are equal, because an invertible matrix is square. A skeletal category equivalent to C is a *skeleton* of C.

Theorem 1.16 (Vector spaces are matrices, up to equivalence). *The functor* $M\colon \mathrm{Mat}_{\mathbb{C}} \to \mathrm{Vect}_{\mathrm{fd}}$,

$$M(n) = \mathbb{C}^n,$$
$$M(A) = (x \mapsto Ax),$$

is an equivalence. So $\mathrm{Vect}_{\mathrm{fd}} \simeq \mathrm{Mat}_{\mathbb{C}}$*, and* $\mathrm{Mat}_{\mathbb{C}}$ *is a skeleton of* $\mathrm{Vect}_{\mathrm{fd}}$.

Proof. Choose for every V a basis, as an isomorphism $b_V\colon \mathbb{C}^{\dim V} \to V$, with $b_{\mathbb{C}^n} = \mathrm{id}$. Set

$$G(V) = \dim V,$$
$$G(f) = b_W^{-1} \circ f \circ b_V,$$

the matrix of $f\colon V \to W$. For $g\colon W \to X$ the b's cancel,

$$\begin{aligned} G(g) \circ G(f) &= b_X^{-1} \circ g \circ b_W \\ &\quad \circ b_W^{-1} \circ f \circ b_V \\ &= G(g \circ f), \end{aligned}$$

so G is a functor, and $G \circ M = \mathrm{id}$ because $b_{\mathbb{C}^n} = \mathrm{id}$. The components b_V form a natural isomorphism $M \circ G \Rightarrow \mathrm{id}$: by the definition of $G(f)$,

$$f \circ b_V = b_W \circ MG(f).$$

■

Proposition 1.17 (· *sketched*). *A functor that is full, faithful and essentially surjective is an equivalence.*

The proof is the one above, with choices in place of bases. For each object Y of $\mathcal{D}$ choose an object $G(Y)$ and an isomorphism $b_Y\colon F(G(Y)) \to Y$. For $g\colon Y \to Y'$, fullness and faithfulness give exactly one morphism $G(g)$ with

$$F(G(g)) = b_{Y'}^{-1} \circ g \circ b_Y,$$

and b is then a natural isomorphism $F \circ G \Rightarrow \mathrm{id}$. For the other one, full faithfulness lifts each $b_{F(X)}^{-1}$ to an isomorphism $X \to G(F(X))$, natural in X (Leinster, Prop. 1.3.18 and Ex. 1.3.32). Like the bases, the choices are the price of an equivalence; an isomorphism of categories needs none.

Example 1.18 (Equivalent, not isomorphic). $\mathrm{Vect}_{\mathrm{fd}}$ and $\mathrm{Mat}_{\mathbb{C}}$ are not isomorphic. An isomorphism of categories is injective on objects and keeps isomorphic objects isomorphic (Lemma 1.6). The line $\{(z, 0)\} \subset \mathbb{C}^2$ and $\mathbb{C}$ are different, isomorphic objects of $\mathrm{Vect}_{\mathrm{fd}}$; in the skeletal $\mathrm{Mat}_{\mathbb{C}}$ no two different objects are isomorphic. So categories are compared up to equivalence: Lecture 2 finds $\mathrm{Rep}(S_3)$ equivalent to three copies of $\mathrm{Vect}_{\mathrm{fd}}$, and Lecture 3 replaces every monoidal category by an equivalent strict one.

Summary.

- *Handed on:* categories, functors, natural transformations and equivalences; pictures with time running upward, in which a chain of boxes needs no brackets, (1.1).
- *The main result:* a representation is a functor $BG \to \mathrm{Vect}_{\mathrm{fd}}$ and an intertwiner a natural transformation, so $\mathrm{Rep}(G) \cong \mathrm{Fun}(BG, \mathrm{Vect}_{\mathrm{fd}})$ (Theorem 1.9); in pictures, naturality is sliding, (1.2).

- *Canonical, made precise:* $V \to V^{**}$ is natural, and no linear isomorphism $V \to V^{*}$ is, even for unitaries (Theorem 1.11). The Riesz map escapes only by being antilinear (Example 1.13).
- *The guard:* $\mathrm{Vect}_{\mathrm{fd}} \simeq \mathrm{Mat}_{\mathbb{C}}$, but the two are not isomorphic (Theorem 1.16 and Example 1.18).
- *Next:* Lecture 2 characterises $\oplus$ and $\otimes$ by universal properties, and shows why $\mathrm{Rep}(G)$ looks like copies of $\mathrm{Vect}_{\mathrm{fd}}$.

References

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Lecture 2

Universal properties, adjunctions, semisimple categories

Objectives. *Every textbook builds $V \oplus W$ and $V \otimes W$ in its own way, with its own bases, and every textbook gets the same answer. This lecture says why: both are characterised by a universal property, which determines them up to a unique isomorphism compatible with the universal maps. The same idea gives the adjunction between $\otimes$ and* Hom*, which Lecture 4 turns into bending wires. It then explains the shape of* $\mathrm{Rep}(G)$*: a semisimple category is equivalent to n copies of* $\mathrm{Vect}_{\mathrm{fd}}$*, one for each simple object.*

By the end of this lecture you will be able to:

- State the universal properties of products, coproducts and the tensor product, and prove that a representing object is unique up to a unique isomorphism compatible with the universal maps.
- Prove $\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, \mathrm{Hom}(V, W))$, naturally, and read it as an adjunction with its unit and counit.
- Prove that a semisimple category with simple objects $X_1, \dots, X_n$ is equivalent to $\mathrm{Vect}_{\mathrm{fd}}^n$, and compute its Hom spaces from multiplicities.
- Decompose the permutation representation of S_3, and show that a Jordan block in $\mathrm{Rep}(\mathbb{Z})$ is not a direct sum of simple objects.

2.1 Universal properties

A universal property describes an object by the morphisms into it, or out of it, rather than by its elements. The simplest kind asks for exactly one.

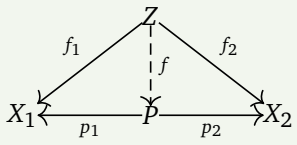
Definition 2.1 (Terminal and initial objects)**.** An object T is *terminal* if every object X has exactly one morphism $X \to T$. It is *initial* if every X has exactly one morphism $T \to X$.

In Set a one-point set is terminal and the empty set is initial. In Vect the zero space is both.

Lemma 2.2. *Any two terminal objects are isomorphic by a unique isomorphism, and so are any two initial objects.*

Proof. Let T and T' be terminal. There is exactly one morphism $a \colon T \to T'$ and one $b \colon T' \to T$. Then $b \circ a$ and id_T are both morphisms $T \to T$, and there is only one, so $b \circ a = \mathrm{id}_T$; likewise $a \circ b = \mathrm{id}_{T'}$. The initial case is the same with the arrows reversed. ∎

Definition 2.3 (Product, coproduct)**.** A *product* of X_1 and X_2 is an object P with morphisms $p_k\colon P \to X_k$ such that for every object Z and morphisms $f_k\colon Z \to X_k$ there is exactly one $f\colon Z \to P$ with $p_k \circ f = f_k$ for $k = 1, 2$:



A *coproduct* is the same with every arrow reversed: morphisms $i_k\colon X_k \to C$ such that every pair $g_k\colon X_k \to Z$ factors as $g \circ i_k$ through exactly one $g\colon C \to Z$.

The proof of Lemma 2.2 applies word for word: a product is unique up to a unique isomorphism compatible with the projections. So "the" product is well defined, whatever construction a textbook uses. In Set the product is $X_1 \times X_2$ and the coproduct the disjoint union, and the two differ: for sets of 2 and 3 elements they have 6 and 5 elements. In Vect they coincide.

Definition 2.4 (Biproduct)**.** In a category whose Hom sets are vector spaces, composed bilinearly, a *biproduct* of X_1 and X_2 is an object $X_1 \oplus X_2$ with morphisms $i_k\colon X_k \to X_1 \oplus X_2$ and $p_k\colon X_1 \oplus X_2 \to X_k$ such that

$$p_k \circ i_l = \delta_{kl}\ \mathrm{id}_{X_k},$$
$$i_1 \circ p_1 + i_2 \circ p_2 = \mathrm{id}\,.$$

The direct sum of vector spaces, with its inclusions and projections, is a biproduct, and a biproduct is both a product and a coproduct: given $f_k\colon Z \to X_k$, the map $f = i_1 \circ f_1 + i_2 \circ f_2$ has $p_k \circ f = f_k$, and any f with this property equals $(i_1 p_1 + i_2 p_2) \circ f$, which is the same map. The coproduct is the mirror argument. This holds for finitely many summands only: the sequence $(1, 1, 1, \dots)$ lies in the product of countably many copies of $\mathbb{C}$ but not in their direct sum, whose elements have finitely many non-zero entries.

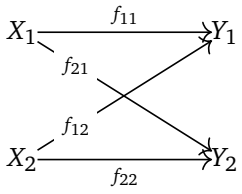
A biproduct turns morphisms into matrices. A morphism $f\colon X_1 \oplus X_2 \to Y_1 \oplus Y_2$ is determined by its four components $f_{jk} = p_j \circ f \circ i_k\colon X_k \to Y_j$, since

$$f = \sum_{j,k} i_j \circ f_{jk} \circ p_k, \tag{2.1}$$

and composites multiply like matrices,

$$(g \circ f)_{lk} = \sum_j g_{lj} \circ f_{jk}.$$

Drawn as arrows between the summands, f is the picture



and composing two such pictures sums over the paths through the middle column. We use this calculus of matrices of morphisms in § 2.4.

2.2 Representables; Yoneda (statement and use)

The tensor product is not a product: *Linear Algebra* (Prop. 12.5) characterises it by its maps out. For every bilinear map $B\colon U\times V\to W$ there is exactly one linear map $\widetilde{B}\colon U\otimes V\to W$ with

$$\widetilde{B}(u\otimes v)=B(u,v).$$

In other words, linear maps out of $U\otimes V$ are the same as bilinear maps, naturally in W. Universal properties of this shape have a name.

Definition 2.5 (Representable functor)**.** A functor $\Phi\colon C\to\mathrm{Set}$ is *represented* by an object X if there is a natural isomorphism

$$\mathrm{Hom}(X,-)\cong\Phi.$$

Here $\mathrm{Hom}(X,-)$ sends Y to $\mathrm{Hom}(X,Y)$ and $g\colon Y\to Y'$ to $f\mapsto g\circ f$.

So $U\otimes V$ represents $W\mapsto\{\text{bilinear maps } U\times V\to W\}$, and the coproduct $X_1\oplus X_2$ represents $Z\mapsto\mathrm{Hom}(X_1,Z)\times\mathrm{Hom}(X_2,Z)$. (The product represents a functor of the other variance, $Z\mapsto\mathrm{Hom}(Z,X_1)\times\mathrm{Hom}(Z,X_2)$, by $\mathrm{Hom}(-,P)$; everything below holds for it with the arrows reversed.)

Theorem 2.6 (Yoneda · *cited*)**.** *For every functor $\Phi\colon C\to\mathrm{Set}$ and object X, the natural transformations $\mathrm{Hom}(X,-)\Rightarrow\Phi$ correspond bijectively to the elements of $\Phi(X)$, by $\xi\mapsto\xi_X(\mathrm{id}_X)$ (Leinster, Thm. 4.2.1 and Cor. 4.3.3).*

We need only one consequence, and we prove it directly.

Proposition 2.7 (A representing object is unique)**.** *Let $\xi\colon\mathrm{Hom}(X,-)\Rightarrow\mathrm{Hom}(X',-)$ be a natural isomorphism. Then $\varphi=\xi_X(\mathrm{id}_X)\colon X'\to X$ is an isomorphism, and it is the only morphism with*

$$\xi_Y(f)=f\circ\varphi$$

for every Y and every $f\colon X\to Y$.

Proof. Naturality for $f\colon X\to Y$, applied to id_X, gives

$$\begin{aligned}\xi_Y(f)&=\xi_Y(f\circ\mathrm{id}_X)\\&=f\circ\xi_X(\mathrm{id}_X)=f\circ\varphi.\end{aligned}$$

Any φ' with this property has $\varphi'=\xi_X(\mathrm{id}_X)=\varphi$. Likewise $\psi=\xi_{X'}^{-1}(\mathrm{id}_{X'})\colon X\to X'$ has $\xi_Y^{-1}(g)=g\circ\psi$. Hence

$$\begin{aligned}\mathrm{id}_X&=\xi_X^{-1}(\varphi)=\varphi\circ\psi,\\\mathrm{id}_{X'}&=\xi_{X'}(\psi)=\psi\circ\varphi.\end{aligned}$$

∎

If X and X' both represent Φ, composing one natural isomorphism with the inverse of the other gives such an ξ. So two constructions of $U\otimes V$, or of $X_1\oplus X_2$, are isomorphic by exactly one isomorphism that respects the universal property. That is the precise sense in which every textbook's tensor product is "the same".

2.3 Adjunctions; ⊗–Hom

A universal property relates morphisms of one kind to morphisms of another. When this happens for all objects at once, the two functors involved are adjoint.

Definition 2.8 (Adjunction)**.** Functors $F\colon C \to \mathcal{D}$ and $G\colon \mathcal{D} \to C$ are *adjoint*, $F \dashv G$, if there are bijections

$$\begin{aligned}&\operatorname{Hom}_{\mathcal{D}}(F(A), B)\\ &\quad\cong \operatorname{Hom}_{C}(A, G(B)),\end{aligned}$$

natural in A and in B. The image of $\mathrm{id}_{F(A)}$ is the *unit* $A \to G(F(A))$; the preimage of $\mathrm{id}_{G(B)}$ is the *counit* $F(G(B)) \to B$.

Fix a finite-dimensional space V, and write $\operatorname{Hom}(V, W)$ for the space of linear maps, itself an object of $\mathrm{Vect}_{\mathrm{fd}}$.

Theorem 2.9 (⊗ and Hom are adjoint)**.** *The maps*

$$\begin{aligned}\Theta\colon\ &\operatorname{Hom}(U \otimes V, W)\\ &\to \operatorname{Hom}(U, \operatorname{Hom}(V, W)),\end{aligned}$$

$$\Theta(f)(u)(v) = f(u \otimes v),$$

are bijections, natural in U and in W. So $- \otimes V \dashv \operatorname{Hom}(V, -)$ on $\mathrm{Vect}_{\mathrm{fd}}$.

Proof. $\Theta(f)(u)$ is linear in v, and $\Theta(f)$ is linear in u, because f is linear and $\otimes$ is bilinear. Conversely, for $g\colon U \to \operatorname{Hom}(V, W)$ the map $(u, v) \mapsto g(u)(v)$ is bilinear, so by the universal property it is $f(u \otimes v)$ for exactly one linear f, and $\Theta(f) = g$. Since the $u \otimes v$ span $U \otimes V$, f is determined by $\Theta(f)$, so Θ is a bijection.

Naturality in U: for $a\colon U' \to U$, put $f' = f \circ (a \otimes \mathrm{id}_V)$; then

$$\begin{aligned}\Theta(f')(u')(v) &= f(a(u') \otimes v)\\ &= \big(\Theta(f) \circ a\big)(u')(v).\end{aligned}$$

Naturality in W: for $b\colon W \to W'$, $\Theta(b \circ f)(u) = b \circ \Theta(f)(u)$, both sending v to $b(f(u \otimes v))$. ∎

For a concrete instance, take $U = V = \mathbb{C}^2$ and $W = \mathbb{C}$. In the ordered tensor basis $e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2$, let the linear functional f have matrix

$$f = \begin{pmatrix} 1 & 2 & 3 & -1 \end{pmatrix}.$$

Currying changes the organisation of the inputs, without changing the value:

$$\Theta(f)(u) = \begin{pmatrix} u_1 + 3u_2 & 2u_1 - u_2 \end{pmatrix}.$$

Thus $\Theta(f)(u)$ is a linear functional on V, rather than a number. It becomes a number only after evaluation at v. For $u = (2, -1)$ and $v = (1, 3)$, both evaluations give

$$\Theta(f)(u)(v) = \begin{pmatrix} -1 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = 14 = f(u \otimes v).$$

The same pattern in Set is currying: a function $X \times B \to C$ of two variables is a function of the first with values in functions of the second, so $- \times B \dashv (-)^B$; for finite sets both sides have $|C|^{|X||B|}$ elements.

The unit and the counit of Theorem 2.9 are maps every physicist has used: the unit $U \to \mathrm{Hom}(V, U \otimes V)$ and the counit $\mathrm{Hom}(V,W) \otimes V \to W$,

$$u \mapsto (v \mapsto u \otimes v),$$
$$g \otimes v \mapsto g(v).$$

They satisfy the two *triangle identities* that every adjunction satisfies (Leinster Lemma 2.2.2): inserting a vector and evaluating gives back what one started with,

$$u \otimes v \mapsto (v' \mapsto u \otimes v') \otimes v \mapsto u \otimes v,$$
$$g \mapsto (v \mapsto g \otimes v) \mapsto (v \mapsto g(v)) = g.$$

The counit is evaluation. For $W = \mathbb{C}$ it is the pairing $V^* \otimes V \to \mathbb{C}$, $\varphi \otimes v \mapsto \varphi(v)$, which Lecture 4 calls ev_V and draws as a cap; there the adjunction becomes the bending of a wire.

2.4 Linear and semisimple categories; Schur

From now on the categories have the structure of $\mathrm{Vect}_{\mathrm{fd}}$ and $\mathrm{Rep}(G)$: morphisms can be added and scaled.

Definition 2.10 (Linear category)**.** A $\mathbb{C}$-*linear* category has a complex vector space structure on each $\mathrm{Hom}(X,Y)$, with composition bilinear. It is *additive* if it has a zero object and a biproduct of any two objects.

$\mathrm{Vect}_{\mathrm{fd}}$, $\mathrm{Rep}(G)$ and $\mathrm{Mat}_{\mathbb{C}}$ are $\mathbb{C}$-linear and additive. So is $\mathrm{Vect}_{\mathrm{fd}}^n$, whose objects are n-tuples $(V_1, \dots, V_n)$ of finite-dimensional spaces and whose morphisms are n-tuples of linear maps, composed and added slot by slot.

Definition 2.11 (Semisimple category)**.** A $\mathbb{C}$-linear additive category with finite-dimensional Hom spaces is *semisimple*, with *simple objects* $X_1, \dots, X_n$, all nonzero, if

(S1) $\mathrm{Hom}(X_i, X_j) = 0$ for $i \neq j$, and $\mathrm{Hom}(X_i, X_i) = \mathbb{C}\,\mathrm{id}$;

(S2) every object is isomorphic to a finite direct sum of copies of the X_i.

For $\mathrm{Rep}(G)$ with G finite, (S1) is Schur's lemma and (S2) is Maschke's theorem (*Symmetries* Thms. 4.12–4.13 and 4.8), and there are as many irreducible representations as conjugacy classes (*Symmetries* Thm. 5.12). $\mathrm{Rep}(\mathrm{SU}(2))$, with continuous representations, satisfies both conditions for its irreducible representations, one for each spin $j = 0, \frac{1}{2}, 1, \dots$ (*Symmetries* Cor. 12.5, Thm. 14.3 and Schur, Thms. 4.12–4.13), but there are infinitely many of them: it is semisimple only in the wider sense that allows an infinite list. We take (S1) and (S2) as the definition, since they are all that is used.

Lemma 2.12 (Multiplicities)**.** *If $X \cong \bigoplus_i X_i^{\oplus m_i}$, then $m_i = \dim \mathrm{Hom}(X_i, X)$. So the multiplicities do not depend on the decomposition.*

Proof. By (2.1), a morphism into a biproduct is a column of morphisms into its summands, so

$$\mathrm{Hom}\Big(X_i, \bigoplus_j X_j^{\oplus m_j}\Big)$$
$$\cong \bigoplus_j \mathrm{Hom}(X_i, X_j)^{\oplus m_j}.$$

By (S1) only $j = i$ contributes, and it gives $\mathbb{C}^{m_i}$. ∎

Theorem 2.13 (Schur, categorically)**.** *Let $\mathcal{C}$ be semisimple with simple objects $X_1, \dots, X_n$. The functor*

$$\begin{aligned} H &\colon \mathcal{C} \to \mathrm{Vect}_{\mathrm{fd}}^{n}, \\ H(X) &= \big(H_i(X)\big)_{i=1}^{n}, \\ H_i(X) &= \mathrm{Hom}(X_i, X), \end{aligned}$$

with $H(f)$ acting by $g \mapsto f \circ g$ in each slot, is an equivalence. In particular

$$\begin{aligned} &\mathrm{Hom}(X, Y) \\ &\quad\cong \bigoplus_i \mathrm{Hom}\big(H_i(X), H_i(Y)\big) \\ &\quad\cong \bigoplus_i H_i(X)^* \otimes H_i(Y), \end{aligned}$$

and $\dim \mathrm{Hom}(X,Y) = \sum_i m_i(X)\, m_i(Y)$, with m_i the multiplicities.

Proof. H is fully faithful. Choose decompositions $X \cong \bigoplus_i X_i^{\oplus m_i}$ and $Y \cong \bigoplus_i X_i^{\oplus k_i}$ (S2). By (2.1) a morphism $f\colon X \to Y$ is a matrix of morphisms between copies of the X_i; by (S1) the only non-zero blocks join copies of the same X_i, and each entry is a scalar. So f is a family of complex $k_i \times m_i$ matrices A_i, one for each i. In the bases of $\mathrm{Hom}(X_i, X) \cong \mathbb{C}^{m_i}$ and $\mathrm{Hom}(X_i, Y) \cong \mathbb{C}^{k_i}$ given by the inclusions of the copies of X_i, $H(f)$ acts in slot i by the same matrix A_i. Hence $f \mapsto H(f)$ is a bijection.

H is essentially surjective. Given $(V_1, \dots, V_n)$, the object $\bigoplus_i X_i^{\oplus \dim V_i}$ has H-image isomorphic to it, by Lemma 2.12.

An inverse. Choose a basis of every finite-dimensional space. Let $G(V) = \bigoplus_i X_i^{\oplus \dim V_i}$, and let G send a tuple of linear maps to the block matrix of their matrices. This is a functor because matrices multiply.

In $H(G(V))$, slot i has the basis of inclusions of the copies of X_i, and sending it to the chosen basis of V_i is a natural isomorphism $H \circ G \Rightarrow \mathrm{id}$, since $G(g)$ is the matrix of g.

In the other direction, the chosen basis vectors b_{ij} of $H_i(X)$ are morphisms $X_i \to X$; together they form a morphism $\zeta_X\colon G(H(X)) \to X$ whose restriction to the j-th copy of X_i is b_{ij}.

For invertibility, observe that $H(\zeta_X)$ sends the inclusion basis to the basis (b_{ij}), so it is invertible; fullness gives an h with $H(h) = H(\zeta_X)^{-1}$, and faithfulness gives $h \circ \zeta_X = \mathrm{id}$ and $\zeta_X \circ h = \mathrm{id}$.

For naturality, $f \circ \zeta_X$ and $\zeta_Y \circ G(H(f))$ agree on each copy of X_i, where both give $f \circ b_{ij}$, so ζ is a natural isomorphism $G \circ H \Rightarrow \mathrm{id}$.

The Hom formula is full faithfulness, followed by $\mathrm{Hom}(A, B) \cong B \otimes A^* \cong A^* \otimes B$ (*Linear Algebra* Prop. 12.18). Its dimension is $\sum_i k_i m_i$. ∎

Example 2.14 (Endomorphisms with multiplicity)**.** In $\mathrm{Rep}(S_3)$ let $X = \mathbf{1} \oplus \mathbf{1} \oplus \mathrm{std}$, with $\mathbf{1}$ trivial and std the standard representation on the sum-zero plane (*Symmetries* §4.4). The theorem gives

$$\mathrm{End}(X) \cong M_2(\mathbb{C}) \times \mathbb{C},$$

of dimension $2^2 + 1^2 = 5$: any linear map between the two trivial summands is an intertwiner, and on std only scalars are. Multiplicity two makes $\mathrm{End}(X)$ non-commutative.

For the same object X, an intertwiner can mix the two copies of $\mathbf{1}$. It cannot send either copy into std, since $\mathrm{Hom}(\mathbf{1}, \mathrm{std}) = 0$. For example, choose endomorphisms f and g whose multiplicity-space maps are

$$H(f) = \left(\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, 3\right), \qquad H(g) = \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, -1\right),$$
$$H(g \circ f) = \left(\begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}, -3\right).$$

The scalar 3 means $3\,\mathrm{id}_{\mathrm{std}}$, which acts on a two-dimensional space. Thus the underlying map f has a matrix of size 4×4, whereas $H(f)$ records one block of size 2×2 and one scalar. The block sizes count copies of a simple object, not its vector-space dimension. Composition multiplies the blocks separately, in the order $H(g)H(f)$.

The theorem answers the second half of the question: up to equivalence, a semisimple category is a list of vector spaces, one for each simple object, and a morphism is a list of matrices. Only the number of simple objects survives. The abelian group $\mathbb{Z}_3$ also has three irreducible representations, the characters $k \mapsto \omega^{jk}$ with $\omega = e^{2\pi i/3}$, so

$$\mathrm{Rep}(S_3) \simeq \mathrm{Rep}(\mathbb{Z}_3) \simeq \mathrm{Vect}_{\mathrm{fd}}^3$$

as linear categories, although S_3 is not abelian and has a two-dimensional irreducible representation. The linear category cannot tell the two groups apart. What can is the tensor product of representations, which Lecture 3 adds to the structure, and the dimensions of the underlying spaces, which an equivalence is free to forget.

2.5 $\mathrm{Rep}(S_3)$ and a Jordan block

Example 2.15 ($\mathrm{Rep}(S_3)$)**.** S_3 has three conjugacy classes, so $\mathrm{Rep}(S_3)$ has three simple objects: the trivial representation $\mathbf{1}$, the sign representation sgn, and the two-dimensional standard representation std on the plane $x_1 + x_2 + x_3 = 0$ (*Symmetries* §4.4, with the basis $e_1 - e_2$, $e_2 - e_3$). For the permutation representation $\mathbb{C}^3$ of Lecture 1,

$$\dim \mathrm{Hom}(\mathbf{1}, \mathbb{C}^3) = 1,$$
$$\dim \mathrm{Hom}(\mathrm{sgn}, \mathbb{C}^3) = 0,$$
$$\dim \mathrm{Hom}(\mathrm{std}, \mathbb{C}^3) = 1,$$

so $\mathbb{C}^3 \cong \mathbf{1} \oplus \mathrm{std}$, and by Theorem 2.13 $\dim \mathrm{End}(\mathbb{C}^3) = 1^2 + 0^2 + 1^2 = 2$. So $\mathrm{End}(\mathbb{C}^3)$ is spanned by the identity I and Lecture 1's composite $u \circ s$, the matrix of ones. With the projections $P_{\mathbf{1}} = (u \circ s)/3$ onto the line $\mathbb{C}(1,1,1)$ and $P_{\mathrm{std}} = I - P_{\mathbf{1}}$ onto the plane,

$$aI + b\,(u \circ s) = (a + 3b)\,P_{\mathbf{1}} + a\,P_{\mathrm{std}},$$

so H sends it to the pair of scalars $(a + 3b, a)$, one for each simple summand. The regular representation $\mathbb{C}[S_3]$ is $\mathbf{1} \oplus \mathrm{sgn} \oplus \mathrm{std}^{\oplus 2}$, so

$$\dim \mathrm{Hom}(\mathbb{C}^3, \mathbb{C}[S_3]) = 1 \cdot 1 + 0 \cdot 1 + 1 \cdot 2 = 3.$$

The dimensions of the simple objects, 1, 1 and 2, are not seen by H, but they recover the dimension of every representation from its multiplicities:

$$\dim \mathbb{C}^3 = 1 \cdot 1 + 0 \cdot 1 + 1 \cdot 2 = 3,$$
$$\dim \mathbb{C}[S_3] = 1 \cdot 1 + 1 \cdot 1 + 2 \cdot 2 = 6.$$

In Lecture 4 these numbers return as quantum dimensions, computed inside the category by closing a loop.

Semisimplicity is a hypothesis, and it can fail. Maschke's theorem needs G finite.

Example 2.16 (A Jordan block)**.** Let $\mathbb{Z}$ act on $\mathbb{C}^2$ by

$$\rho(k) = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix},$$

a representation since $\rho(k+l) = \rho(k)\rho(l)$ (*Symmetries* Ex. 4.9). The line $\mathbb{C}e_1$ is invariant. An invariant complement would be a second invariant line, an eigenline of $\rho(1)$; but $\rho(1)$ has the single eigenvalue 1 with the single eigenline $\mathbb{C}e_1$. So $\mathbb{C}^2$ is not a direct sum of two nonzero representations: it is *indecomposable*. Nor does it satisfy (S1) on its own, since $\mathrm{End}(\mathbb{C}^2)$ contains $\rho(1) - I \neq 0$ besides the scalars. So $\mathbb{C}^2$ is not a direct sum of objects satisfying (S1), $\mathrm{Rep}(\mathbb{Z})$ does not satisfy (S2), and Theorem 2.13 does not apply.

The fusion categories of Lecture 5 are semisimple by definition, which excludes such examples; every count of simple objects and every multiplicity from there on rests on (S1) and (S2).

Summary.

- *Universal properties:* terminal objects, products, coproducts and the tensor product are determined up to a unique isomorphism compatible with the universal maps (Lemma 2.2 and Proposition 2.7); in Vect the direct sum is a biproduct, and morphisms between biproducts are matrices of morphisms, (2.1).
- *Adjunction:* $\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, \mathrm{Hom}(V, W))$ naturally, so $- \otimes V \dashv \mathrm{Hom}(V, -)$, with evaluation as the counit (Theorem 2.9).
- *The main result:* a semisimple category with simple objects $X_1, \dots, X_n$ is equivalent to $\mathrm{Vect}_{\mathrm{fd}}^n$, and $\dim \mathrm{Hom}(X, Y) = \sum_i m_i(X) m_i(Y)$ (Theorem 2.13); $\mathrm{Rep}(S_3)$ has three simples, and $\mathbb{C}^3 \cong \mathbf{1} \oplus \mathrm{std}$.
- *The guard:* a Jordan block in $\mathrm{Rep}(\mathbb{Z})$ is indecomposable but not simple, so semisimplicity can fail (Example 2.16).
- *Next:* Lecture 3 adds the tensor product to the category, and asks what makes putting systems side by side associative.

References

Leinster, *Basic Category Theory*: adjunctions, Def. 2.1.1, p. 41, unit and counit, §2.2, p. 51; initial and terminal objects, Def. 2.1.7 and Lemma 2.1.8, pp. 48–49; representables, Def. 4.1.3, p. 84; Yoneda, Thm. 4.2.1, p. 94, and Cor. 4.3.3, p. 100; products, Def. 5.1.1, p. 108; the direct sum as product and sum, Exs. 5.1.5 and 5.2.5, pp. 110 and 127–128. Mac Lane, *Categories for the Working Mathematician*: the tensor–hom adjunction, §IV.1, p. 80;

biproducts, §VIII.2, p. 194. Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: additive and linear categories, Defs. 1.2.1–1.2.2, pp. 1–2; simple and semisimple objects and Schur's lemma, Def. 1.5.1 and Lemma 1.5.2, p. 5; $\mathrm{End}(X) = k$, Prop. 1.8.4, p. 9.

Lecture 3

Monoidal categories and string diagrams

Objectives. *Lecture 2 showed that a semisimple category alone cannot tell* $\mathrm{Rep}(S_3)$ *from* $\mathrm{Rep}(\mathbb{Z}_3)$*: what can is the tensor product. This lecture adds it. Putting systems side by side is a functor* $\otimes$ *with an associator and unitors, subject to two axioms, the pentagon and the triangle. Coherence says that these axioms are all one needs, and string diagrams turn it into a calculus of pictures. Associators need not be trivial: on* $\mathbb{Z}_2$ *there are two monoidal categories with the same fusion rules and different associators.*

By the end of this lecture you will be able to:

- State the definition of a monoidal category, check it on Vect, $\mathrm{Rep}(G)$ and the matrices, and draw the pentagon and the triangle.
- Read and draw string diagrams, prove the interchange law, and use sliding and the commutativity of scalars in pictures.
- Show that the pentagon of Vec_G^ω is the 3-cocycle condition, and solve it for $\mathbb{Z}_2$.
- Define a monoidal functor, and show that $\omega(s,s,s) = \pm 1$ cannot be changed by one that is the identity on objects.

3.1 The definition

For objects of Vect, $(U \otimes V) \otimes W$ and $U \otimes (V \otimes W)$ are different spaces, identified by $(u \otimes v) \otimes w \mapsto u \otimes (v \otimes w)$. A monoidal category keeps track of such identifications.

Definition 3.1 (Monoidal category)**.** A *monoidal category* is a category C with

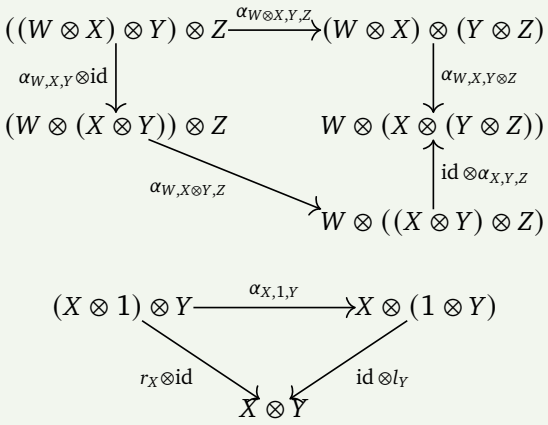
- a *tensor product*: an object $X \otimes Y$ for all objects and a morphism $f \otimes g : X \otimes Y \to X' \otimes Y'$ for all $f : X \to X'$ and $g : Y \to Y'$, with $\mathrm{id}_X \otimes \mathrm{id}_Y = \mathrm{id}_{X \otimes Y}$ and

$$\begin{aligned}(g \otimes k) \circ (f \otimes h) \\ = (g \circ f) \otimes (k \circ h);\end{aligned}$$

- a *unit object* 1;
- natural isomorphisms, the *associator* and the *unitors*,

$$\begin{aligned}\alpha_{X,Y,Z} &: (X \otimes Y) \otimes Z \\ &\to X \otimes (Y \otimes Z), \\ l_X &: 1 \otimes X \to X, \\ r_X &: X \otimes 1 \to X,\end{aligned}$$

such that the *pentagon* and the *triangle* commute for all objects:



It is *strict* if α, l and r are identities.

The first condition says that $\otimes$ is a functor of two variables; we return to it in § 3.3.

Read the pentagon from the upper-left object to the middle-right object. The upper route first groups Y with Z, then groups their pair with X. The lower route first groups X with Y, then regroups that pair with Z, splitting off W, then groups Y with Z inside the last three factors. Each arrow changes brackets only. The order W, X, Y, Z stays fixed. The axiom says that these two composites, with two and three arrows, agree.

Example 3.2 (Monoidal categories)**.**

- $(\mathrm{Vect}, \otimes, \mathbb{C})$, the state spaces of composite quantum systems (*QM-I* Post. 10.1), with α as above and $l(z \otimes v) = zv = r(v \otimes z)$. Both axioms hold on pure tensors, which span: in the pentagon both paths send

$$\begin{aligned} &((w \otimes x) \otimes y) \otimes z \\ &\mapsto w \otimes (x \otimes (y \otimes z)). \end{aligned}$$

  (*Linear Algebra* Rem. 12.6 writes the inverse of our α.)
- $(\mathrm{Set}, \times, \{*\})$, with $\alpha((x, y), z) = (x, (y, z))$.
- $\mathrm{Rep}(G)$: g acts on $V \otimes W$ by $\rho_V(g) \otimes \rho_W(g)$, a representation by the first condition; the unit is the trivial representation $\mathbb{C}$, and α, l, r of Vect are intertwiners. For SU(2) two spin-$\frac{1}{2}$ representations give the singlet plus the triplet, $\Pi_{1/2} \otimes \Pi_{1/2} \cong \Pi_0 \oplus \Pi_1$ (*Symmetries* Thm. 14.12).
- $\mathrm{Mat}_{\mathbb{C}}$ (Lecture 1) with $m \otimes n = mn$ and the Kronecker product of matrices: it is *strict*, because

$$(A \otimes B) \otimes C = A \otimes (B \otimes C),$$
$$1 \otimes A = A = A \otimes 1$$

  hold as equalities of matrices. The quantum circuits of *Quantum Computing* compute in it. They are drawn left to right with the top wire as the first factor; turned a quarter turn anticlockwise they become the pictures of this course, with time upward and the first factor on the left.

3.2 Coherence (formal diagrams)

In $W \otimes X \otimes Y \otimes Z$ there are five bracketings and many composites of associators between them. The pentagon equates two of them; Mac Lane's theorem says this is enough.

Theorem 3.3 (Coherence · *cited*).

(a) Every monoidal category is monoidally equivalent (§ 3.6) to a strict one.

(b) Take a word $x_1 \otimes \cdots \otimes x_n$ in formal variables. Any two formal *morphisms between two of its bracketings, built from α, l, r, their inverses and identities by $\circ$ and $\otimes$, are equal, and stay equal when objects of C are put in for the variables.*

(Mac Lane, §VII.2 and §XI.3; EGNO Thms. 2.8.5 and 2.9.2.)

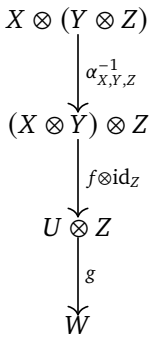
What must be formal is the morphism, not just its source and target. Mac Lane points out that two formally different bracketings may become the same object, and then a morphism between them need not be formal. The category Vec_G^ω on $\mathbb{Z}_2$ has simple objects 1 and s, with $s \otimes s = 1$ (§ 3.5). For $\omega(s,s,s) = -1$,

$$(s \otimes s) \otimes s = s = s \otimes (s \otimes s),$$

yet $\alpha_{s,s,s} = -\mathrm{id}_s$: the diagram formed by $\alpha_{s,s,s}$ and id_s does not commute. There is no contradiction: id_s compares the two bracketings only because they happen to be the same object, and it is not a formal morphism from $(x \otimes x) \otimes x$ to $x \otimes (x \otimes x)$. Some textbooks state coherence as "every diagram commutes"; that is false.

What (a) and (b) give us: in a formal computation the brackets and the associators can be dropped, and we write $X \otimes Y \otimes Z$.

Dropping brackets is a convention for writing composites, not an assertion that the chosen category has identity associators. For example, let $f\colon X \otimes Y \to U$ and $g\colon U \otimes Z \to W$. The picture that applies f to the first two strands and then applies g has the following interpretation if its source is bracketed as $X \otimes (Y \otimes Z)$:



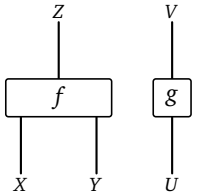
In a strict category the first arrow is an identity. In a skeletal category such as Vec_G^ω, the first arrow can be a nontrivial scalar even when its source and target are the same object. One must insert that scalar when evaluating the picture in that model. Strictification supplies an equivalent category; it does not replace $\alpha_{s,s,s} = -\mathrm{id}_s$ by an identity inside the original category.

3.3 String diagrams; interchange; Joyal–Street

In a strict monoidal category the pictures of Lecture 1 acquire a second direction. A morphism $f\colon X_1 \otimes \cdots \otimes X_m \to Y_1 \otimes \cdots \otimes Y_n$ is a box with m strands entering from below and n leaving

above, the first factor on the left. Composition stacks boxes upward, $\otimes$ places pictures side by side, id_X is a strand, and the unit 1 is the empty picture: it has no strand. For $f\colon X \otimes Y \to Z$ and $g\colon U \to V$,

$$f \otimes g \;=\; \tag{3.1}$$



a morphism $X \otimes Y \otimes U \to Z \otimes V$.

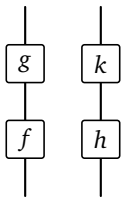
Theorem 3.4 (Interchange). *For composable f, g and composable h, k,*

$$\begin{aligned}(g \otimes k) \circ (f \otimes h)\\ = (g \circ f) \otimes (k \circ h).\end{aligned}$$

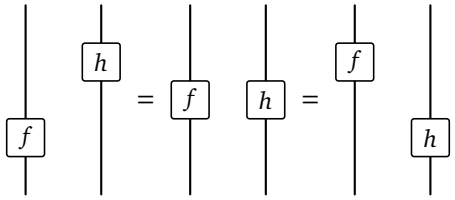
Proof. This is the first condition of Definition 3.1: $\otimes$ is a functor on pairs of morphisms, which compose slot by slot. ∎

The theorem is what makes a picture unambiguous. The diagram

$$= (g \otimes k) \circ (f \otimes h) \tag{3.2}$$



can be read row by row or column by column, and the two readings agree. As a special case, boxes on different strands slide past each other:



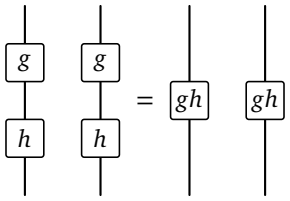
(3.3)

that is,

$$\begin{aligned}(\mathrm{id} \otimes h) \circ (f \otimes \mathrm{id}) &= f \otimes h\\ &= (f \otimes \mathrm{id}) \circ (\mathrm{id} \otimes h).\end{aligned}$$

The heights of boxes on different strands carry no meaning.

A first proof by pictures: the action of G on $V \otimes W$ in Example 3.2 is a representation. Writing g for $\rho_V(g)$ and for $\rho_W(g)$, interchange and $\rho(g)\rho(h) = \rho(gh)$ on each strand give



(3.4)

that is,

$$\begin{aligned}(\rho_V(g) \otimes \rho_W(g))(\rho_V(h) \otimes \rho_W(h))\\ = \rho_V(gh) \otimes \rho_W(gh).\end{aligned}$$

Theorem 3.5 (Joyal–Street · *cited*)**.** *In a strict monoidal category, two string diagrams that can be deformed into one another in the plane, keeping the ends of the strands fixed, never letting strands cross, and never letting a strand turn downward, denote the same morphism (Heunen–Vicary Thm. 1.8; Selinger Thm. 3.1).*

By coherence (Theorem 3.3) the theorem applies to any monoidal category: a picture determines the morphism once its source and target are bracketed, and by (b) every way of inserting associators to match them gives the same result. *Quantum Computing*'s soundness theorem for circuits (L2, Thm. 2.2) is the case of $\mathrm{Mat}_{\mathbb{C}}$, read a quarter turn round.

3.4 Eckmann–Hilton

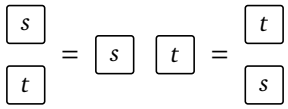
The endomorphisms of the unit, $\mathrm{End}(1)$, are the *scalars* of the category. In Vect, $\mathrm{End}(\mathbb{C}) = \mathbb{C}$. In a picture a scalar is a box with no strands.

Theorem 3.6 (Scalars commute · *proved given coherence*)**.** *In a monoidal category, $s \circ t = t \circ s$ for all $s, t \in \mathrm{End}(1)$.*

Proof. By Theorem 3.3(a) it suffices to take C strict, so that $1 \otimes 1 = 1$ and $s \otimes \mathrm{id}_1 = s = \mathrm{id}_1 \otimes s$. Then, by interchange,

$$\begin{aligned} s \circ t &= (s \otimes \mathrm{id}_1) \circ (\mathrm{id}_1 \otimes t) \\ &= s \otimes t \\ &= (\mathrm{id}_1 \otimes t) \circ (s \otimes \mathrm{id}_1) = t \circ s. \end{aligned}$$

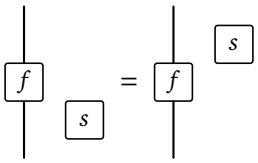
In pictures, two boxes with no strands slide past each other:



(3.5)

■

A box with no strands can slide up and down beside a strand, on the same side of it,



(3.6)

Moving it to the other side of the strand is not a planar deformation, and in a general monoidal category $s \otimes \mathrm{id}_X$ and $\mathrm{id}_X \otimes s$ can differ (Heunen–Vicary Ex. 1.9). In Vect, and in the fusion categories of Lecture 5, whose tensor product is bilinear, a scalar is a number, $s \otimes f = sf = f \otimes s$, and it floats anywhere.

3.5 Vec_G^ω: the $\mathbb{Z}_2$ pentagon

The associator is data. The simplest place to see that it can be non-trivial is a category whose simple objects form a group.

Definition 3.7 (Vec_G^ω)**.** Let G be a finite group and $\omega : G^3 \to \mathbb{C}^\times$ a function. Vec_G^ω is the skeletal semisimple category with one simple object δ_g for each $g \in G$, tensor product

$\delta_g \otimes \delta_h = \delta_{gh}$, unit δ_1, and associator

$$\alpha_{g,h,k} = \omega(g,h,k)\ \mathrm{id}_{\delta_{ghk}}$$

on simple objects, extended to direct sums by linearity, with the tensor product of linear maps on morphisms; the unitors are identities. We write g for δ_g. $\mathrm{Vec}_G = \mathrm{Vec}_G^1$ is equivalent to the category of finite-dimensional G-graded vector spaces. Here ω will be a cocycle, not a symplectic form, a quantum state, or the root of unity of Lecture 2.

Naturality is automatic, since the only morphisms between simple objects are scalars. The two axioms constrain ω.

Theorem 3.8 (The pentagon is a cocycle condition). *Vec_G^ω is a monoidal category if and only if*

$$\omega(g,1,h) = 1$$

for all g,h (the triangle; ω is normalised*) and, for all $g,h,k,m \in G$,*

$$\begin{aligned}\omega(gh,k,m)\,\omega(g,h,km)\\ = \omega(g,h,k)\,\omega(g,hk,m)\\ \cdot\,\omega(h,k,m)\end{aligned}$$

(the pentagon; ω is a 3-cocycle*).*

Proof. By linearity it suffices to check the axioms on simple objects, where every arrow is a scalar times an identity and $\otimes\,\mathrm{id}$ does not change the scalar. In the pentagon for g,h,k,m the two-arrow path is $\omega(gh,k,m)\,\omega(g,h,km)$, and the three-arrow path is $\omega(g,h,k)\,\omega(g,hk,m)\,\omega(h,k,m)$. In the triangle for g,h, the path through $\alpha_{g,1,h}$ is $\omega(g,1,h)$, and the other is 1. ∎

Example 3.9 ($\mathbb{Z}_2$). Let $G = \mathbb{Z}_2 = \{1,s\}$. Substituting $h = 1$ in the cocycle condition and using $\omega(g,1,k) = 1$ gives

$$\omega(g,k,m) = \omega(g,k,m)\,\omega(1,k,m).$$

Since the values are nonzero, $\omega(1,k,m) = 1$. Substituting $k = 1$ likewise gives $\omega(g,h,1) = 1$. Only $\omega(s,s,s)$ is free. Of the sixteen pentagons only (s,s,s,s) is not automatic, and since $s^2 = 1$ it reads

$$1 = \omega(s,s,s)^2.$$

So $\omega(s,s,s) = \pm 1$: there are two monoidal categories Vec_G^ω on $\mathbb{Z}_2$, with the same fusion rule $s \otimes s = 1$.

3.6 Monoidal functors

Are the two categories of Example 3.9 really different, or the same category described twice? To compare monoidal categories we need functors that respect $\otimes$.

Definition 3.10 (Monoidal functor). A *monoidal functor* $(F, J, J_0) : C \to \mathcal{D}$ is a functor F with natural isomorphisms

$$J_{X,Y} : F(X) \otimes F(Y) \to F(X \otimes Y),$$
$$J_0 : 1 \to F(1),$$

such that

$$\begin{aligned} &F(\alpha_{X,Y,Z}) \circ J_{X\otimes Y,Z} \\ &\quad \circ (J_{X,Y} \otimes \mathrm{id}) \\ &= J_{X,Y\otimes Z} \circ (\mathrm{id} \otimes J_{Y,Z}) \\ &\quad \circ \alpha_{FX,FY,FZ} \end{aligned}$$

and

$$F(l_X) \circ J_{1,X} \circ (J_0 \otimes \mathrm{id}) = l_{F(X)},$$
$$F(r_X) \circ J_{X,1} \circ (\mathrm{id} \otimes J_0) = r_{F(X)}.$$

It is a *monoidal equivalence* if F is an equivalence. Between $\mathbb{C}$-linear categories we take F to be $\mathbb{C}$-linear on every Hom space, as all functors are from here on.

The forgetful functor $\mathrm{Rep}(G) \to \mathrm{Vect}_{\mathrm{fd}}$, with J and J_0 identities, is monoidal.

The tensor product already answers the question left open in Lecture 2. $\mathrm{Rep}(\mathbb{Z}_3)$ has the fusion rules of Vec_G for $G = \mathbb{Z}_3$, since its characters multiply, $\chi_i \otimes \chi_j = \chi_{i+j}$: every simple object is invertible. In $\mathrm{Rep}(S_3)$ the standard representation is not: with the notation of Lecture 2,

$$\mathrm{std} \otimes \mathrm{std} \cong \mathbf{1} \oplus \mathrm{sgn} \oplus \mathrm{std},$$

of dimension $4 = 1 + 1 + 2$. So the two categories, equivalent as linear categories, are not equivalent as monoidal ones: a monoidal equivalence would send simple objects to simple objects and respect their tensor products.

Back to Example 3.9. For a functor $\mathrm{Vec}_G^{\omega} \to \mathrm{Vec}_G^{\omega'}$ that is the identity on objects, with $J_{g,h} = j(g,h)\,\mathrm{id}$ and $J_0 = j_0\,\mathrm{id}$, the first condition reads

$$\begin{aligned} &\omega'(g,h,k)\, j(h,k)\, j(g,hk) \\ &\quad = j(g,h)\, j(gh,k)\, \omega(g,h,k), \end{aligned}$$

and the two unit conditions, with trivial unitors, read $j(1,g) = j(g,1) = j_0^{-1}$.

For a general j the first condition says $\omega' = \omega\,(\delta j)^{-1}$, where the *coboundary* of j is

$$(\delta j)(g,h,k) = \frac{j(h,k)\, j(g,hk)}{j(gh,k)\, j(g,h)}.$$

If ω and ω' are both normalised, the condition at $(g,1,h)$ gives $j(g,1) = j(1,h)$, so such a j satisfies the unit conditions with $j_0 = j(1,1)^{-1}$. Normalised cocycles that differ by a coboundary therefore give equivalent categories. Their classes form $H^3(G,\mathbb{C}^\times)$, the 3-cocycles modulo coboundaries. Every class contains a normalised cocycle (EGNO Rem. 2.6.3); that is why Example 3.9 could normalise from the start.

Proposition 3.11. *A monoidal functor $\mathrm{Vec}_G^{\omega} \to \mathrm{Vec}_G^{\omega'}$ on $\mathbb{Z}_2$ that is the identity on objects exists only if $\omega'(s,s,s) = \omega(s,s,s)$.*

Proof. At $(g,h,k) = (s,s,s)$, with $s^2 = 1$, the first condition reads

$$\begin{aligned} &\omega'(s,s,s)\, j(s,s)\, j(s,1) \\ &\quad = j(s,s)\, j(1,s)\, \omega(s,s,s), \end{aligned}$$

and $j(s,1) = j(1,s)$ by the unit conditions, or, for normalised cocycles, by the condition at $(s,1,s)$. ∎

By EGNO Prop. 2.6.1 · *cited*, two categories Vec_G^{ω} and $\mathrm{Vec}_G^{\omega'}$ are equivalent by a $\mathbb{C}$-linear monoidal functor exactly when ω' is obtained from ω by an automorphism of G and a coboundary; their classes form $H^3(G, \mathbb{C}^\times)$ up to automorphisms of G. The only automorphism of $\mathbb{Z}_2$ is the identity, and Proposition 3.11 shows that no coboundary between normalised cocycles changes $\omega(s,s,s)$. So the two categories of Example 3.9 are not equivalent: the same fusion rules carry two different associators. Lecture 6 computes associators as F-symbols, and for Vec_G^{ω} they are the values of ω; Lecture 7 finds two braidings for each.

Summary.

- *Handed on:* a monoidal category is $(C, \otimes, 1, \alpha, l, r)$ with the pentagon and the triangle (Definition 3.1); by coherence, formal diagrams commute and brackets can be dropped (Theorem 3.3), but non-formal ones need not.
- *Pictures:* boxes and strands with time upward; interchange makes them unambiguous, boxes on different strands slide, and planar deformations preserve the morphism (Theorems 3.4 and 3.5).
- *Scalars commute* (Theorem 3.6).
- *The main example:* in Vec_G^{ω} the pentagon is the 3-cocycle condition; on $\mathbb{Z}_2$, $\omega(s,s,s) = \pm 1$, and no monoidal functor that is the identity on objects relates the two (Theorem 3.8 and Proposition 3.11).
- *Next:* Lecture 4 adds duals: cups, caps and loops.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: monoidal categories, Def. 2.1.1, p. 21, and Def. 2.2.8, p. 25; Vec_G^{ω}, Ex. 2.3.8, p. 28; monoidal functors, Def. 2.4.1, p. 30; End(1) commutative, Prop. 2.2.10, p. 26; strictification, Thm. 2.8.5, p. 36; coherence, Thm. 2.9.2, p. 40; the classification, Prop. 2.6.1 and Rem. 2.6.3, p. 33. Mac Lane, *Categories for the Working Mathematician*: §VII.1–2, pp. 161–169 ("formal" diagrams, p. 165; his associator runs the other way), and §XI.3, p. 257. Heunen and Vicary, *Categories for Quantum Theory*: Def. 1.1, p. 30; interchange, Thm. 1.7, p. 32; the graphical calculus, Thm. 1.8, p. 34; boxes cannot pass strands, Ex. 1.9, p. 35; scalars, Lemma 2.3, p. 62. Selinger, *A survey of graphical languages for monoidal categories*, arXiv:0908.3347, §3, Thm. 3.1, p. 11 (drawn left to right).

Lecture 4

Duality: cups, caps and quantum dimension

Objectives. *A particle has an antiparticle, and a pair can be created from the vacuum and annihilated back into it. In a monoidal category this is a* dual*: an object X^* with a cup $1 \to X \otimes X^*$ and a cap $X^* \otimes X \to 1$ that cancel in a zigzag. This lecture gives the axioms, proves that they determine the dual, bends wires with them (the adjunction of Lecture 2, made literal), and closes loops to get traces and quantum dimensions.*

By the end of this lecture you will be able to:

- State the snake equations, prove that duals are unique up to a unique isomorphism, and compute f^* as a rotated box.
- Bend wires to prove $\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, W \otimes V^*)$, and read teleportation as one snake.
- Define a pivotal structure, the trace and the quantum dimension, and prove cyclicity and multiplicativity by pictures.
- Show that for spin-$\frac{1}{2}$ with the singlet cup the snake is straight only with the cap $-\Omega^\dagger$, which closes the loop to -2, a pivotal choice, and identify the sign as a Frobenius–Schur indicator.

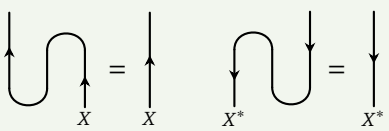
4.1 Duals and the snake

By coherence we work as if C were strict. A strand pointing up is an object X; a strand pointing down is its dual.

Definition 4.1 (Duals)**.** A *left dual* of X is an object X^* with morphisms

$$\mathrm{ev}_X : X^* \otimes X \to 1,$$
$$\mathrm{coev}_X : 1 \to X \otimes X^*,$$

drawn as a cap and a cup, such that the two *snakes* are straight:



(4.1)

that is, $(\mathrm{id}_X \otimes \mathrm{ev}_X) \circ (\mathrm{coev}_X \otimes \mathrm{id}_X) = \mathrm{id}_X$ and $(\mathrm{ev}_X \otimes \mathrm{id}_{X^*}) \circ (\mathrm{id}_{X^*} \otimes \mathrm{coev}_X) = \mathrm{id}_{X^*}$. A *right dual* ${}^*X$ has $\mathrm{ev}'_X : X \otimes {}^*X \to 1$ and $\mathrm{coev}'_X : 1 \to {}^*X \otimes X$ with the mirror-image snakes. C is *rigid* if every object has both.

Heunen–Vicary and Selinger write A^* for the *right* dual; ours is EGNO's.

A snake in coordinates

In $V = \mathbb{C}^2$, choose a basis e_1, e_2 and its dual basis e^1, e^2. The cup creates $e_1 \otimes e^1 + e_2 \otimes e^2$. The cap evaluates a functional on a vector, so $\mathrm{ev}_V(e^i \otimes e_j) = \delta_{ij}$. For $v = v_1 e_1 + v_2 e_2$ and $\lambda = \lambda_1 e^1 + \lambda_2 e^2$, the two composites give

$$(\mathrm{id}_V \otimes \mathrm{ev}_V)(\mathrm{coev}_V \otimes \mathrm{id}_V)(v) = \sum_{i,j=1}^{2} v_j e_i \delta_{ij} = v,$$

$$(\mathrm{ev}_V \otimes \mathrm{id}_{V^*})(\mathrm{id}_{V^*} \otimes \mathrm{coev}_V)(\lambda) = \sum_{i,j=1}^{2} \lambda_j \delta_{ji} e^i = \lambda.$$

The first cap consumes the middle functional and the original input vector. The other leg of the cup is the output vector. No inner product or complex conjugation enters this calculation. The cap and cup express duality, even before a dagger is available.

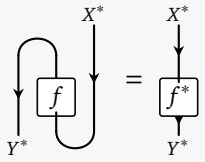
Theorem 4.2 (Duals are unique; the rotated box · *proved given coherence*)**.**

(a) If $(X^*, \mathrm{ev}, \mathrm{coev})$ *and* $(X', \mathrm{ev}', \mathrm{coev}')$ *are left duals of* X*, then*

$$\varphi = (\mathrm{ev} \otimes \mathrm{id}_{X'}) \circ (\mathrm{id}_{X^*} \otimes \mathrm{coev}')$$

is the unique isomorphism $X^* \to X'$ *with* $\mathrm{ev}' \circ (\varphi \otimes \mathrm{id}_X) = \mathrm{ev}$.

(b) For $f : X \to Y$ *the* dual morphism $f^* : Y^* \to X^*$*, the box rotated by a half turn,*



(4.2)

satisfies $(g \circ f)^* = f^* \circ g^*$ *and* $\mathrm{id}^* = \mathrm{id}$*: the dual is a contravariant functor.*

Proof. (a) First expand the compatibility equation. By interchange and the primed snake,

$$\begin{aligned} &\mathrm{ev}' \circ (\varphi \otimes \mathrm{id}_X) \\ &\quad = \mathrm{ev} \circ \big(\mathrm{id}_{X^*} \otimes (\mathrm{id}_X \otimes \mathrm{ev}')\big) \\ &\qquad \circ (\mathrm{id}_{X^*} \otimes \mathrm{coev}' \otimes \mathrm{id}_X) \\ &\quad = \mathrm{ev}\,. \end{aligned}$$

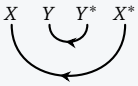
Let $\psi = (\mathrm{ev}' \otimes \mathrm{id}_{X^*}) \circ (\mathrm{id}_{X'} \otimes \mathrm{coev})$. Using this compatibility,

$$\begin{aligned} \psi \circ \varphi &= (\mathrm{ev} \otimes \mathrm{id}_{X^*}) \circ (\mathrm{id}_{X^*} \otimes \mathrm{coev}) \\ &= \mathrm{id}_{X^*}, \end{aligned}$$

by the other snake. Likewise $\varphi \circ \psi = \mathrm{id}_{X'}$. Any φ' satisfying the cap compatibility equation equals $(\mathrm{ev} \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \mathrm{coev}')$ after inserting a snake, which is φ.

(b) Stack the pictures of f^* and g^*; the cap of one and the cup of the other form a snake, which straightens to give the picture of $(g \circ f)^*$. ∎

Proposition 4.3 (The dual of a tensor product · *proved given coherence*)**.** *If X^* and Y^* are left duals of X and Y, then $Y^* \otimes X^*$, in that order, is a left dual of $X \otimes Y$, with the nested cup and cap*

$$\mathrm{coev}_{X\otimes Y} = \tag{4.3}$$


and the corresponding nested cap.

Proof. Each snake for $X \otimes Y$ is a snake for Y inside a snake for X; straighten the inner one, then the outer one. ∎

The order reverses, as for the inverse of a product of matrices, and the cups nest rather than cross.

Example 4.4 (Duals)**.**

- $\mathrm{Vect}_{\mathrm{fd}}$: X^* is the dual space, $\mathrm{ev}(\varphi \otimes v) = \varphi(v)$, and $\mathrm{coev}(1) = \sum_i e_i \otimes e^i$ for any basis (e_i) and its dual basis; it is the identity of X under $\mathrm{Hom}(X, X) \cong X \otimes X^*$, so it does not depend on the basis. The rotated box is the transpose, the dual map of Lecture 1.
- $\mathrm{Rep}(G)$: the dual is the contragredient representation, $g \mapsto (\rho(g)^{-1})^*$ (*Symmetries* Prop. 3.10); ev and coev are those of $\mathrm{Vect}_{\mathrm{fd}}$, and are intertwiners.
- Vec_G^ω: $\delta_g^* = \delta_{g^{-1}}$. With $\mathrm{coev}_g = \mathrm{id}$ and the unitors identities, the first snake equals $\mathrm{ev}_g\ \omega(g, g^{-1}, g)$, from the associator in the middle; so $\mathrm{ev}_g = \omega(g, g^{-1}, g)^{-1}\,\mathrm{id}$. The second snake then holds by the cocycle condition. For the $\mathbb{Z}_2$ category of Lecture 3 with $\omega(s, s, s) = -1$, the naive choice ev = coev = id gives a snake equal to $-\mathrm{id}_s$.

The associator inside a snake

The strict picture of a snake suppresses brackets. To evaluate that picture in the chosen skeletal model Vec_G^ω, put the brackets back before composing the maps. Write $X = \delta_g$ and $X^* = \delta_{g^{-1}}$. With identity unitors the first snake follows this path:

$$\begin{aligned} X &\xrightarrow{\mathrm{coev}_X \otimes \mathrm{id}_X} (X \otimes X^*) \otimes X \\ &\xrightarrow{\alpha_{X,X^*,X}} X \otimes (X^* \otimes X) \\ &\xrightarrow{\mathrm{id}_X \otimes \mathrm{ev}_X} X. \end{aligned}$$

The middle arrow is multiplication by $\omega(g, g^{-1}, g)$. Thus choosing $\mathrm{coev}_g = \mathrm{id}$ forces the cap to contribute its inverse. Dropping brackets in the graphical language does not set this scalar to 1 in a fixed skeletal model. For the other snake, the middle map is $\alpha^{-1}_{X^*,X,X^*}$. Apply the normalized cocycle equation to (g, g^{-1}, g, g^{-1}). The entries with a unit argument are 1, giving

$$\omega(g^{-1}, g, g^{-1})\,\omega(g, g^{-1}, g) = 1,$$

so that snake also has coefficient 1. In the twisted $\mathbb{Z}_2$ example, the middle scalar and the correct cap are both -1. Their product is $+1$; replacing the cap by the identity removes one of these two signs and fails the snake equation.

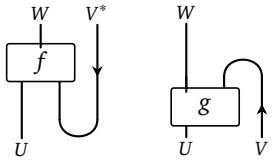
4.2 Bending wires; teleportation as one picture

Theorem 4.5 (Bending wires · *proved given coherence*). *For a left dual V^* of V, the maps*

$$f \mapsto (f \otimes \mathrm{id}_{V^*}) \circ (\mathrm{id}_U \otimes \mathrm{coev}_V),$$
$$g \mapsto (\mathrm{id}_W \otimes \mathrm{ev}_V) \circ (g \otimes \mathrm{id}_V)$$

are inverse bijections $\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, W \otimes V^*)$*, natural in U and W.*

Proof. In pictures the first map bends the V strand of f up to the right, and the second bends it back down:



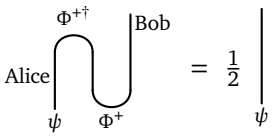
(4.4)

Bending up and then down leaves a snake on the V strand, which straightens by (4.1); the other order straightens the V^* snake. Naturality holds because boxes on U and W slide past the cup and cap. ∎

So $- \otimes V$ is left adjoint to $- \otimes V^*$. In $\mathrm{Vect}_{\mathrm{fd}}$, where $\mathrm{Hom}(V, W) \cong W \otimes V^*$ (*Linear Algebra* Prop. 12.18), this is Lecture 2's adjunction

$$\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, \mathrm{Hom}(V, W)).$$

Teleportation. In $\mathrm{Hilb}_{\mathrm{fd}}$ with an orthonormal basis, coev becomes $\sum_i |ii\rangle = \sqrt{2}\,|\Phi^+\rangle$ for a qubit, after identifying X^* with X by the basis (*Quantum Computing*, L4). Alice holds an unknown $|\psi\rangle$ and half of $|\Phi^+\rangle$; Bob holds the other half. If Alice's Bell measurement returns Φ^+, the picture is the snake: the cup creates the pair, Alice's cap closes her two strands, and Bob's strand carries $|\psi\rangle$, with amplitude $\frac{1}{2}$ from the two normalisations,



(4.5)

that is,

$$(\langle\Phi^+| \otimes \mathrm{id})(|\psi\rangle \otimes |\Phi^+\rangle) = \tfrac{1}{2}\,|\psi\rangle\,.$$

The other three outcomes put a Pauli matrix on Bob's strand, which he undoes.

4.3 Pivotal, trace, dimension, spherical

To close a loop we need a cap on $X \otimes X^*$, the other way round. In a rigid category X^{**} is a left dual of X^*, so we need to compare X with X^{**}.

Definition 4.6 (Pivotal structure, trace, dimension). A *pivotal structure* is a natural isomorphism $a_X : X \to X^{**}$ with $a_{X\otimes Y} = a_X \otimes a_Y$ (under $(X \otimes Y)^{**} \cong X^{**} \otimes Y^{**}$). It gives a second

cap and cup,

$$\widetilde{\mathrm{ev}}_X : X \otimes X^* \to 1,$$
$$\widetilde{\mathrm{ev}}_X = \mathrm{ev}_{X^*} \circ (a_X \otimes \mathrm{id}_{X^*}),$$
$$\widetilde{\mathrm{coev}}_X : 1 \to X^* \otimes X,$$

the second defined likewise from coev_{X^*} and a_X^{-1}; so X^* is also a right dual. For $h : X \to X$ the *left* and *right traces* close the strand to the right and to the left,

$$\mathrm{tr}_L(h) = \widetilde{\mathrm{ev}}_X \circ (h \otimes \mathrm{id}_{X^*}) \circ \mathrm{coev}_X,$$
$$\mathrm{tr}_R(h) = \mathrm{ev}_X \circ (\mathrm{id}_{X^*} \otimes h) \circ \widetilde{\mathrm{coev}}_X,$$

elements of $\mathrm{End}(1)$. The *quantum dimension* is $d_X = \mathrm{tr}_L(\mathrm{id}_X)$, a closed loop. The structure is *spherical* if $\mathrm{tr}_L = \mathrm{tr}_R$.

The four typed bends

The order of the two endpoints identifies the map in Figure 4.1. The untilded cup and cap are the chosen left-dual data. The tilded pair uses the pivotal structure to make the same object X^* a right dual. In full, the second cup is

$$\widetilde{\mathrm{coev}}_X = (\mathrm{id}_{X^*} \otimes a_X^{-1}) \circ \mathrm{coev}_{X^*} .$$

The cap ev_{X^*} initially consumes $X^{**} \otimes X^*$. The map a_X changes its first input from X to X^{**}, giving $\widetilde{\mathrm{ev}}_X$. For the second cup, a_X^{-1} changes the second output back from X^{**} to X. The arrows record X and X^*; time still runs upward even along a strand whose arrow points down.

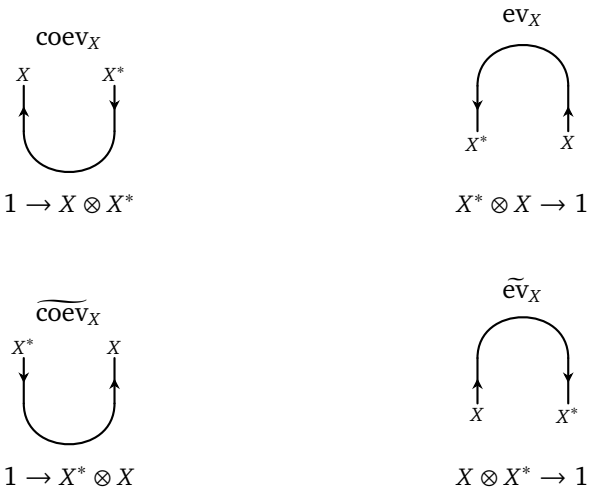
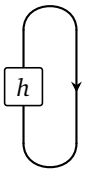
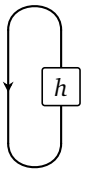


Figure 4.1: The four bends after choosing a_X. Read tensor factors from left to right and maps from bottom to top. The lower pair depends on the pivotal structure.

A left snake pairs coev_X with ev_X and leaves one strand open. A left trace pairs coev_X with $\widetilde{\mathrm{ev}}_X$ and leaves no strand open. These caps have different input orders. Replacing one by the other is not a planar deformation of a typed diagram.

In pictures the two traces close the strand on opposite sides:

$\mathrm{tr}_L(h) =$  $\qquad \mathrm{tr}_R(h) =$  (4.6)

In $\mathrm{Vect}_{\mathrm{fd}}$ the canonical a_V of Lecture 1 is a pivotal structure, both traces are the ordinary trace, and $d_V = \dim V$:

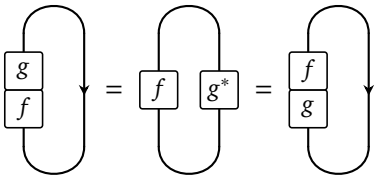
 $= d_X.$ (4.7)

Theorem 4.7 (Traces · *proved given coherence*)**.** *In a rigid $\mathbb{C}$-linear category with bilinear $\otimes$, $\mathrm{End}(1) = \mathbb{C}$ and a pivotal structure, for $f\colon X \to Y$, $g\colon Y \to X$ and $h\colon X \to X$, $h'\colon X' \to X'$,*

$$\mathrm{tr}_L(g \circ f) = \mathrm{tr}_L(f \circ g),$$
$$\mathrm{tr}_L(h \otimes h') = \mathrm{tr}_L(h)\,\mathrm{tr}_L(h'),$$

so $d_{X\otimes Y} = d_X\, d_Y$ and $d_1 = 1$; likewise for tr_R.

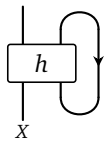
Proof. Cyclicity: slide g around the loop. Rotated by the cup and the cap it becomes g^* on the down-going strand, by (4.2) and the naturality of a, and slides back up on the other side:

 (4.8)

Multiplicativity: $a_{X\otimes X'} = a_X \otimes a_{X'}$ makes the loop on $X \otimes X'$ two nested loops, one around each factor, and the inner one, a number, can be moved out of the picture (Lecture 3). ∎

In $\mathrm{Rep}(G)$ with its canonical pivotal structure, a closed loop gives $\dim V$. For the simple representations of S_3 these values are 1, 1 and 2.

Closing only the right strand of $h\colon X \otimes Y \to X \otimes Y$ is the *partial trace* over Y, the operation of *QM-I* (Def. 11.2) that gives the state of a subsystem:

$\mathrm{tr}_Y(h) =$  (4.9)

Dimensions need not be positive integers. On Vec_G for $G = \mathbb{Z}_2$, the choice $a_s = -\mathrm{id}$ is also a pivotal structure, and it gives $d_s = -1$; still $d_{s\otimes s} = d_s^2 = 1 = d_1$, as the theorem requires. The pivotal structure is a choice, and the quantum dimension depends on it. A character χ of G gives one on Vec_G with $d_g = \chi(g)$ (EGNO Ex. 4.7.16). For $G = \mathbb{Z}_3 = \{1, r, r^2\}$ and $\chi(r) = e^{2\pi i/3}$,

$$d_{r^*} = d_{r^2} = e^{4\pi i/3} \neq e^{2\pi i/3} = d_r,$$

so this pivotal structure is not spherical. A spherical one has $d_X = d_{X^*}$: the right trace of id_X, the loop drawn the other way round, is the left trace of id_{X^*} (EGNO Prop. 4.7.3, Ex. 4.7.9), so $\mathrm{tr}_L = \mathrm{tr}_R$ gives $d_X = d_{X^*}$.

4.4 The Penrose sign and the FS indicator

Let $V = \mathbb{C}^2$ be spin-$\frac{1}{2}$. The SU(2)-invariant form Ω, with $\Omega(e_1, e_2) = 1 = -\Omega(e_2, e_1)$, identifies V with V^*, and as a cup

$$\Omega\colon \mathbb{C} \to V \otimes V, \quad 1 \mapsto e_1 \otimes e_2 - e_2 \otimes e_1,$$

which is $\sqrt{2}$ times the singlet of *Symmetries* (L14). It is an intertwiner, since $(U \otimes U)\Omega = \det(U)\,\Omega = \Omega$ for $U \in \mathrm{SU}(2)$.

Theorem 4.8 (The Penrose sign)**.** *A cap $c\colon V \otimes V \to \mathbb{C}$ satisfies the snake equation with the cup Ω if and only if $c = -\Omega^\dagger$. Then both snakes are straight and the loop is $c \circ \Omega = -2$. With the cap $\Omega^\dagger$ instead, the snake is $-\mathrm{id}$ and the loop is $+2$.*

Proof. Write $c(e_i \otimes e_j) = c_{ij}$. The snake $(\mathrm{id} \otimes c)(\Omega \otimes \mathrm{id})$ sends e_k to $e_1 c_{2k} - e_2 c_{1k}$, so it is the identity exactly when $c_{21} = 1$, $c_{12} = -1$ and $c_{11} = c_{22} = 0$; that is $c = -\Omega^\dagger$, since $\Omega^\dagger(e_1 \otimes e_2) = 1 = -\Omega^\dagger(e_2 \otimes e_1)$. The loop is $-\Omega^\dagger\Omega = -\|\Omega\|^2 = -2$, and the other snake is checked the same way. With $c = \Omega^\dagger$ every sign flips. ∎

One singlet cup and two pivotal closures

The self-duality identification changes the types of the preceding maps. Let $j\colon V \to V^*$ be given by $j(v) = \Omega(\cdot, v)$. Transporting each dual factor through j gives

$$\begin{aligned}(\mathrm{id}_V \otimes j^{-1}) \circ \mathrm{coev}_V &= \Omega,\\ \mathrm{ev}_V \circ (j \otimes \mathrm{id}_V) &= -\Omega^\dagger,\\ \widetilde{\mathrm{ev}}_V \circ (\mathrm{id}_V \otimes j) &= \Omega^\dagger.\end{aligned}$$

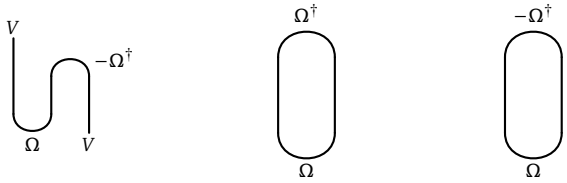
Here the tilded map uses the canonical pivotal structure a. The dual cap transports j on the first input. The closing cap transports j on the second input. The alternating form changes sign when its arguments exchange places. This explains the two signs without changing the cup.

Keep this left-dual cup and cap fixed. The pivotal change takes place in $\mathrm{Rep}(\mathrm{SU}(2))$. The central group element $-I$ acts by $+1$ on integer-spin representations and by -1 on half-integer-spin representations. Its action is a monoidal natural automorphism, so composing it with a gives another pivotal structure a'. On the spin-$\frac{1}{2}$ object V, it gives $a'_V = -a_V$. Thus the tilded closing cap changes sign, while the untilded dual data and their snakes stay fixed. This spin-dependent change is a choice on $\mathrm{Rep}(\mathrm{SU}(2))$; it is not a natural pivotal change on all of $\mathrm{Vect}_{\mathrm{fd}}$. Indeed, naturality with every map $\mathbb{C} \to V$ forces a natural automorphism of the identity functor on $\mathrm{Vect}_{\mathrm{fd}}$ to act by the same scalar on every vector. A monoidal automorphism fixes the unit, so that scalar is 1. The three composites are shown in Figure 4.2.

The canonical closure gives $d_V = 2$. The changed pivotal closure gives $d'_V = -2$. Both choices retain the same straight left-dual snake. The Hilbert-space rule $\widetilde{\mathrm{ev}}_V = \mathrm{coev}_V^\dagger$ concerns the closing cap (Theorem 4.10). It does not replace the dual cap in the snake by the dagger of the singlet cup.

Definition 4.9 (Frobenius–Schur indicator)**.** Let X be a simple object in a pivotal category with $\mathrm{End}(X) = \mathbb{C}$, and $j\colon X \to X^*$ an isomorphism. Then $j^* \circ a_X\colon X \to X^*$ is another, so it defines the *Frobenius–Schur indicator* κ_X by

$$j^* \circ a_X = \kappa_X\, j.$$



id_V 2 −2

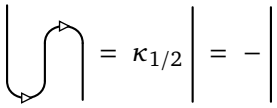
Figure 4.2: The fixed singlet cup with its dual cap gives the open snake. The two closed diagrams use the canonical and changed pivotal closing caps, respectively. All dual factors have been identified with V through j.

Applying the dual again gives, by naturality of a and $a_{X^*} = (a_X^*)^{-1}$ (EGNO Ex. 4.7.9),

$$\begin{aligned}\kappa_X^2 j &= (j^* \circ a_X)^* \circ a_X \\ &= a_X^* \circ j^{**} \circ a_X \\ &= a_X^* \circ a_{X^*} \circ j = j.\end{aligned}$$

Thus $\kappa_X = \pm 1$. Rescaling j leaves this sign unchanged, since the dual is linear.

For spin-$\frac{1}{2}$, $j(v) = \Omega(\cdot, v)$ has the antisymmetric matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, and j^* is its transpose, $-j$: so $\kappa_{1/2} = -1$. With the canonical pivotal structure on representations, an object with a symmetric invariant form has $\kappa = +1$. Bonderson's flags record the self-duality identification j used when bending a strand. Here the flagged cup is Ω and the same-way flagged cap is $\Omega^\dagger$. Reversing either flag multiplies its morphism by κ_X, so opposed flags give a straight snake. Same-way flags give



(4.10)

and the sign returns in Lecture 6 as an F-symbol, κ_a / d_a.

4.5 Dagger

A *dagger category* has an operation $f \mapsto f^\dagger$, reversing morphisms, with $(g \circ f)^\dagger = f^\dagger \circ g^\dagger$, $\mathrm{id}^\dagger = \mathrm{id}$ and $f^{\dagger\dagger} = f$; in $\mathrm{Hilb}_{\mathrm{fd}}$ it is the adjoint.

Theorem 4.10 ($\mathrm{Hilb}_{\mathrm{fd}}$ is dagger pivotal · *sketched*)**.** *With X^* given the inner product for which dual bases of orthonormal bases are orthonormal, the caps of $\mathrm{Hilb}_{\mathrm{fd}}$ are the daggers of its cups: $\widetilde{\mathrm{ev}}_X = \mathrm{coev}_X^\dagger$ and $\mathrm{ev}_X = \widetilde{\mathrm{coev}}_X^\dagger$ (Heunen–Vicary Def. 3.51 and Prop. 3.52).*

In an orthonormal basis,

$$\begin{aligned}&\mathrm{coev}^\dagger(e_j \otimes e^k) \\ &\quad = \Big\langle \sum_i e_i \otimes e^i, e_j \otimes e^k \Big\rangle \\ &\quad = \delta_{jk}.\end{aligned}$$

Spin-$\frac{1}{2}$ does not contradict it: there the cup is Ω, which identifies V^* with V by an antisymmetric form, not by the inner product.

Summary.

- *Handed on:* cups and caps with straight snakes; duals are unique up to a unique isomorphism, and f^* is the rotated box (Theorem 4.2).
- *Bending:* $\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, W \otimes V^*)$ (Theorem 4.5); teleportation is a snake.
- *Loops:* a pivotal structure closes loops; traces are cyclic and multiplicative, and the quantum dimension d_X is a loop (Theorem 4.7).
- *The sign:* the straight Penrose cap is $-\Omega^\dagger$, with loop -2 (Theorem 4.8); the canonical pivotal structure has $\kappa_{1/2} = -1$.
- *Next:* Lecture 5 fuses particles: semisimple rigid categories with finitely many simple objects, and their fusion rules.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: duals, Defs. 2.10.1–2.10.2 and Prop. 2.10.5, pp. 40–41; examples, Exs. 2.10.12–2.10.14, pp. 42–43; bending, Prop. 2.10.8, p. 42; traces, pivotal structures and dimensions, Defs. 4.7.1, 4.7.7, 4.7.11, 4.7.14 and Thm. 4.7.15, pp. 73–75 (Ex. 2.10.14 prints $\omega(g, g^{-1}, g)$; the inverse is right). Heunen and Vicary, *Categories for Quantum Theory*: duals and snakes, Def. 3.1, pp. 89–90 (their A^* is our right dual); teleportation, §3.2, p. 99; pivotal, Def. 3.24, p. 107; dagger pivotal, Def. 3.51 and Prop. 3.52, p. 117. Kitaev, *Anyons in an exactly solved model and beyond*, cond-mat/0506438, App. E, pp. 74–77 (the Frobenius–Schur indicator). Bonderson, PhD thesis (Caltech, 2007), (2.22)–(2.27), pp. 20–21 (flags). Selinger, *A survey of graphical languages*, §4.1, p. 18. Kauffman and Lins, *Temperley–Lieb Recoupling Theory*, §§8.1–8.2, pp. 77–79 (the loop -2 as Penrose's theory).

Block II

Anyons

Lecture 5

Fusion categories: particles and how they fuse

Objectives. *Lectures 1–4 built the language: semisimple categories, tensor products, duals and loops. This lecture puts them together into the setting of anyon models. A fusion category has finitely many particle types. Two particles fuse into a sum of types, counted by multiplicities N_{ab}^{c}. States with a fixed total charge form* fusion spaces. *For repeated Fibonacci particles with total charge τ, their dimensions are Fibonacci numbers. We also count even numbers of Ising particles with vacuum charge.*

By the end of this lecture you will be able to:

- State the definition of a fusion category and give the fusion rules of Vec_G, $\mathrm{Rep}(S_3)$, Fibonacci, Ising and $\mathrm{SU}(2)_k$.
- Prove that the multiplicities form an associative fusion ring with $N_{ab}^{c} = N_{bc^*}^{a^*}$.
- Compute Frobenius–Perron and quantum dimensions and the global dimension.
- Use the trivalent vertex, the bubble and completeness relations, and count fusion spaces with fusion trees.

5.1 The definition

Definition 5.1 (Fusion category)**.** A *fusion category* is a $\mathbb{C}$-linear semisimple category with finitely many simple objects (Definition 2.11), with a monoidal structure whose tensor product is bilinear on morphisms, which is rigid (Definition 4.1), and whose unit 1 is simple. We take it pivotal (Definition 4.6), as all our examples are; then $X^{**} \cong X$.

The simple objects $a, b, c, \ldots$ are the *particle types*, or anyons; 1 is the vacuum, and the dual a^*, which we also write $\bar{a}$, is the antiparticle. The tensor product of two simple objects decomposes,

$$a \otimes b \cong \bigoplus_{c} N_{ab}^{c}\, c,$$

with *fusion multiplicities* $N_{ab}^{c} = \dim \mathrm{Hom}(a \otimes b, c)$ (Lemma 2.12).

Particle types and state spaces

A particle type is a simple object. A state with prescribed particle types belongs to a vector space of morphisms. For fixed external types $a_1, \dots, a_n$ and total charge c, this is the *fusion space*

$$V_c^{a_1 \cdots a_n} = \mathrm{Hom}(c, a_1 \otimes \cdots \otimes a_n).$$

Choose a bracketing of the tensor product. In a unitary model the dagger gives this space an inner product. The number of independent states is its ordinary integer dimension. The quantum dimension of a simple object, defined by the loop in Definition 4.6, is a different quantity.

The familiar angular-momentum example separates the two kinds of space. Write V_j for the spin-j irreducible representation of SU(2). Adding two and then three spin-$\frac{1}{2}$ particles gives (*QM-I* Thm. 13.2)

$$V_{1/2} \otimes V_{1/2} \cong V_0 \oplus V_1,$$
$$V_{1/2}^{\otimes 3} \cong 2V_{1/2} \oplus V_{3/2}.$$

For the second line, the first pair can have spin 0 or spin 1. Each choice allows total spin $\frac{1}{2}$, so $\mathrm{Hom}_{\mathrm{SU}(2)}(V_{1/2}, V_{1/2}^{\otimes 3})$ has dimension 2. This counts the copies of the spin multiplet. Each copy of $V_{1/2}$ itself has dimension 2. Thus the full total-spin-$\frac{1}{2}$ sector has dimension 4, and the whole three-spin Hilbert space has dimension 8. The fusion space keeps the multiplicity, rather than the magnetic states inside each multiplet.

Rigidity means that every particle can be created from the vacuum together with its antiparticle, by coev_a, and annihilated with it, by ev_a. Rep(SU(2)) has every property but finiteness, since its spins $j = 0, \frac{1}{2}, 1, \dots$ do not stop. The category $\mathrm{Vect}_{\mathrm{fd}}$ is the trivial fusion category, whose only particle is the vacuum.

Example 5.2 (Fusion categories).

- Vec_G^{ω} for a finite group G (Lecture 3): simple objects $g \in G$, $g \otimes h = gh$, and $g^* = g^{-1}$. Every simple object is invertible; such a category is *pointed*.
- $\mathrm{Rep}(G)$ for a finite group (Lectures 2 to 4): the fusion multiplicities are the Clebsch–Gordan multiplicities, computed from characters since $\chi_{V \otimes W} = \chi_V \chi_W$ (*Symmetries* Prop. 8.2), and V^* is the dual representation. For S_3, with simple objects $\mathbf{1}$, sgn, std, all self-dual (*Symmetries* Ex. 8.4):

$$\mathrm{sgn} \otimes \mathrm{sgn} \cong \mathbf{1}, \quad \mathrm{sgn} \otimes \mathrm{std} \cong \mathrm{std},$$
$$\mathrm{std} \otimes \mathrm{std} \cong \mathbf{1} \oplus \mathrm{sgn} \oplus \mathrm{std}.$$

- *Fibonacci*: simple objects $1, \tau$, with

$$\tau \otimes \tau \cong 1 \oplus \tau.$$

- *Ising*: simple objects $1, \sigma, \psi$, with

$$\sigma \otimes \sigma \cong 1 \oplus \psi,$$
$$\sigma \otimes \psi \cong \sigma, \quad \psi \otimes \psi \cong 1.$$

- $\mathrm{SU}(2)_k$: simple objects $j = 0, \frac{1}{2}, \dots, \frac{k}{2}$, with the truncated Clebsch–Gordan rule

$$j_1 \otimes j_2 \cong \bigoplus_{j=|j_1-j_2|}^{\min(j_1+j_2,\, k-j_1-j_2)} j,$$

  in integer steps; without the truncation it is *Symmetries*' Thm. 14.12.

In the last three every simple object is self-dual: $N_{aa}^{1} = 1$ for each a. For Fibonacci, Ising and $\mathrm{SU}(2)_k$ we have so far only the rules; that categories with these rules exist is shown in Lecture 6, by solving the pentagon for Fibonacci and verifying a solution for Ising, and for $\mathrm{SU}(2)_k$ in Lecture 9, from a cited construction.

Two channels for three Fibonacci particles

The Fibonacci rule lists the possible total charges of a pair. The decomposition $\tau \otimes \tau \cong 1 \oplus \tau$ does not by itself choose a state or its amplitudes. If the total charge is fixed to 1, the two-particle fusion space is a line. If the total charge is fixed to τ, it is another line. For three particles with total charge τ, both intermediate pair charges are allowed, so

$$\dim V_{1}^{\tau\tau} = \dim V_{\tau}^{\tau\tau} = 1, \qquad \dim V_{\tau}^{\tau\tau\tau} = 2.$$

A basis for this last space is labelled by whether the first pair has charge 1 or τ. The total charge and all three external particle types stay fixed. A state can have two complex amplitudes on these two intermediate channels. The fusion-tree construction in § 5.5 will realize that basis explicitly.

The quantum dimension computed below is $d_\tau = \varphi$, where $\varphi = (1+\sqrt{5})/2$. This noninteger value does not mean that a single τ carries an ordinary Hilbert space of dimension φ. It describes the growth of the integer dimensions of many-particle fusion spaces. The recurrence proved in § 5.5 and the Binet formula in Sheet 5 make that statement precise.

5.2 Fusion rings

Theorem 5.3 (The fusion ring · *proved given coherence*)**.** *Let $K(C)$ be the free abelian group on the simple objects, with*

$$[a][b] = \sum_{c} N_{ab}^{c}\,[c].$$

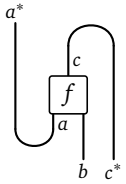
This product is associative with unit $[1]$, and

$$N_{ab}^{1} = \delta_{b,a^*},$$
$$N_{ab}^{c} = N_{bc^*}^{a^*},$$
$$N_{ab}^{c} = N_{b^*a^*}^{c^*}.$$

Proof. Associativity: $(a \otimes b) \otimes c \cong a \otimes (b \otimes c)$, and the multiplicity of e on the two sides is $\sum_x N_{ab}^{x} N_{xc}^{e}$ and $\sum_y N_{bc}^{y} N_{ay}^{e}$. The unit: $1 \otimes a \cong a$. Bending wires (Theorem 4.5) gives $\mathrm{Hom}(a \otimes b, 1) \cong \mathrm{Hom}(a, b^*)$, which by Schur is $\mathbb{C}$ if $a \cong b^*$ and 0 otherwise (b^* is simple, since the rotated box is a bijection $\mathrm{End}(b) \to \mathrm{End}(b^*)$); since $b^{**} \cong b$, $a \cong b^*$ exactly when $b \cong a^*$. For the second identity, bend c down and then a across, using its right dual (also a^*, since C is pivotal):

$$\begin{aligned}\mathrm{Hom}(a \otimes b, c) &\cong \mathrm{Hom}(a \otimes b \otimes c^*, 1)\\ &\cong \mathrm{Hom}(b \otimes c^*, a^*),\end{aligned}$$

and compare dimensions. In pictures, f goes to



whose inverse bends the two strands back. For the third, the rotated box gives $\mathrm{Hom}(a\otimes b, c) \cong \mathrm{Hom}(c^*, (a\otimes b)^*)$, and $(a\otimes b)^* \cong b^*\otimes a^*$ (Proposition 4.3); by semisimplicity the multiplicity of c^* in $b^*\otimes a^*$ is the dimension of either Hom space between them. ∎

For Fibonacci the two bracketings of three τ's give

$$(\tau\otimes\tau)\otimes\tau \cong (1\oplus\tau)\otimes\tau \cong 1\oplus 2\tau,$$
$$\tau\otimes(\tau\otimes\tau) \cong \tau\otimes(1\oplus\tau) \cong 1\oplus 2\tau.$$

The ring records that the two sides are isomorphic, but not which isomorphism the associator is; that is the content of Lecture 6.

The *fusion matrix* N_a is the matrix of left multiplication by $[a]$, $(N_a)_{cb} = N_{ab}^c$. For Fibonacci and Ising, in the bases $1, \tau$ and $1, \sigma, \psi$,

$$N_\tau = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad N_\sigma = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

and associativity reads $N_aN_b = \sum_c N_{ab}^c N_c$. The matrices need not commute: in Vec_G for $G = S_3$, $N_gN_h = N_{gh}$. The rings of our anyon models are commutative, because $a\otimes b \cong b\otimes a$ once there is a braiding (Lecture 7). A ring with a basis of this kind, non-negative integer structure constants and the properties of the theorem is a *fusion ring*.

Different categories can share a ring. For $k = 2$ the truncated rule gives $\frac{1}{2}\otimes\frac{1}{2} \cong 0\oplus 1$, $\frac{1}{2}\otimes 1 \cong \frac{1}{2}$ and $1\otimes 1 \cong 0$: under $0, \frac{1}{2}, 1 \mapsto 1, \sigma, \psi$ this is the Ising ring, though Lecture 6 shows that the two categories differ. For $k = 3$, $1\otimes 1 \cong 0\oplus 1$, since the truncation removes $j = 2$; the objects 0 and 1 fuse like 1 and τ.

The fusion ring does not determine the category. Vec_G and Vec_G^ω for $G = \mathbb{Z}_2$ have the same ring and are not equivalent (Lecture 3). For groups: the dihedral group of order 8 and the quaternion group have the same character table (*Symmetries* Ex. 5.15, which calls the dihedral group D_4), hence the same fusion ring, yet their representation categories are not equivalent as fusion categories (Calaque–Etingof, Cor. 3.12 · *cited*).

5.3 FPdim; the canonical spherical structure

The fusion matrices determine the Frobenius–Perron dimensions.

Theorem 5.4 (Frobenius–Perron dimension · *sketched given coherence*)**.** *For each simple a, the matrix N_a has non-negative entries and a largest real eigenvalue $\mathrm{FPdim}(a) \geq 1$ with a positive eigenvector. The map $a \mapsto \mathrm{FPdim}(a)$ is multiplicative,*

$$\mathrm{FPdim}(a)\,\mathrm{FPdim}(b) = \sum_c N_{ab}^c\,\mathrm{FPdim}(c),$$

and it is the only such function that is positive on every simple object (EGNO Prop. 3.3.6).

Sketch. Let $R = \sum_a [a]$, and let M be the matrix of right multiplication by R. Its entries $\sum_a N_{ba}^c = \sum_a \dim \mathrm{Hom}(a, b^* \otimes c)$ are positive, since $b^* \otimes c \neq 0$ (1 is a summand of $b \otimes b^*$, so c is one of $b \otimes b^* \otimes c$). By the Perron–Frobenius theorem M has a positive eigenvector v, unique up to scale, for its largest eigenvalue. Left and right multiplications commute, by associativity, so each N_a maps this eigenline to itself: $N_a v = \lambda_a v$, and $\lambda_a > 0$ because N_a is non-negative and non-zero. In fact $\lambda_a \geq 1$: every column of N_a has an entry at least 1, since $a \otimes b \neq 0$, so the entries of $N_a v$ add up to at least those of v. A non-negative matrix with a positive eigenvector has that eigenvalue as its largest, so $\lambda_a = \mathrm{FPdim}(a)$; applying $N_a N_b = \sum_c N_{ab}^c N_c$ to v gives multiplicativity. Conversely, a positive multiplicative f makes $(f(c^*))_c$ a positive eigenvector of M, so it is proportional to v; with $f(1) = 1$ this pins f down. ∎

For Fibonacci N_τ has eigenvalues $\varphi = (1+\sqrt{5})/2$ and $-1/\varphi$, so $\mathrm{FPdim}(\tau) = \varphi$; for Ising $\mathrm{FPdim}(\sigma) = \sqrt{2}$ and $\mathrm{FPdim}(\psi) = 1$; in $\mathrm{Rep}(G)$ it is the dimension. Uniqueness can be seen directly for Fibonacci: a multiplicative function has $d_\tau^2 = 1 + d_\tau$, whose only positive root is φ. The other root $-1/\varphi$ returns in Lecture 6 as the Yang–Lee category. For $\mathrm{SU}(2)_k$ (Sheet 5),

$$\mathrm{FPdim}(j) = \frac{\sin\big((2j+1)\pi/(k+2)\big)}{\sin\big(\pi/(k+2)\big)};$$

for $k = 2$ it gives $1, \sqrt{2}, 1$, Ising's values, and for $k = 3$ it gives $\mathrm{FPdim}(\frac{1}{2}) = \mathrm{FPdim}(1) = \varphi$.

The quantum dimension d_a of Lecture 4 is defined by a loop and depends on the pivotal structure. It too is multiplicative: $d_a d_b = d_{a\otimes b}$ (Theorem 4.7), and the trace is additive on the direct sum $a \otimes b \cong \bigoplus_c N_{ab}^c\, c$, so $d_a d_b = \sum_c N_{ab}^c d_c$. A fusion category is *pseudo-unitary* if $\sum_a d_a^2 = \sum_a \mathrm{FPdim}(a)^2$ for some spherical structure (EGNO Def. 9.4.4).

A fusion category is *unitary* if it has an antilinear dagger (§ 4.5) with $(f \otimes g)^\dagger = f^\dagger \otimes g^\dagger$ and unitary associators and unitors. For every simple a and object X, require the formula

$$\langle v, w \rangle\ \mathrm{id}_a = v^\dagger w, \qquad v, w \in \mathrm{Hom}(a, X),$$

to define a positive inner product. Unitary fusion categories are pseudo-unitary (EGNO Rem. 9.4.7 · *cited*).

Theorem 5.5 (The canonical spherical structure · *cited*)**.** *A pseudo-unitary fusion category has a unique spherical structure with $d_a = \mathrm{FPdim}(a)$ for every simple a (EGNO Prop. 9.5.1).*

All the anyon numerics of the course use this canonical structure (so $d_\tau = \varphi$, $d_\sigma = \sqrt{2}$). Other pivotal structures exist: on Vec_G for $G = \mathbb{Z}_2$ the choice $a_s = -\mathrm{id}$ of Lecture 4 gives $d_s = -1$. The *global dimension* is

$$\dim C = \sum_a d_a^2,$$

with the canonical d: $|G|$ for Vec_G and for $\mathrm{Rep}(G)$ (*Symmetries* Thm. 5.11), $2+\varphi$ for Fibonacci, 4 for Ising.

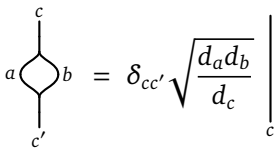
5.4 Vertices, bubbles, completeness

From here on the categories are unitary, with the canonical spherical structure (so every $d_a > 0$), and *multiplicity-free*, every $N_{ab}^c \in \{0, 1\}$, as all the course's anyon models are. Then

$\mathrm{Hom}(c, a \otimes b)$ is a line when $N_{ab}^{c} = 1$, and we choose a basis vector, the *splitting vertex* ψ_c^{ab}, drawn with c below; its dagger, the *fusion vertex* $\psi_c^{ab\dagger}$, is the picture reflected top to bottom:



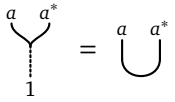
The vertex is fixed up to a scalar, and we fix its size by the *isotopy-invariant normalisation* (Bonderson (2.14)–(2.19); Kitaev eqs. 195–196): a fusion vertex on a splitting vertex is



(5.1)

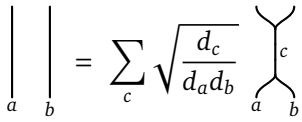
This can be arranged: by Schur the left side is a multiple of id_c, zero unless $c = c'$, and positive since it has the form $\psi^\dagger\psi$; rescaling ψ by a positive number sets it. A phase remains free, the gauge of Lecture 6.

The vacuum channels are fixed by Lecture 4. Choosing $\psi_a^{1a} = \psi_a^{a1} = \mathrm{id}_a$ satisfies (5.1). For $a \otimes a^* \to 1$, the bubble asks $\psi^\dagger\psi = \sqrt{d_a d_{a^*}} = d_a$. Choose the duality maps *standard*: each cap is the dagger of a cup, $\mathrm{coev}_a^\dagger = \widetilde{\mathrm{ev}}_a$, as in $\mathrm{Hilb}_{\mathrm{fd}}$ (Theorem 4.10); this can be done with the canonical dimensions (Kitaev eqs. (190)–(191) · *cited*). Then $\mathrm{coev}_a^\dagger\,\mathrm{coev}_a$ is the loop d_a, and we may take the cup itself:



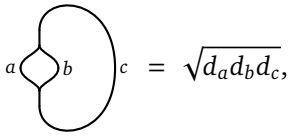
(5.2)

For a self-dual a this identifies a^* with a; when $\kappa_a = -1$, as for spin $\frac{1}{2}$ in $\mathrm{SU}(2)_k$, the cup carries the flag of Lecture 4. Since $a \otimes b$ is the direct sum of its channels, the identity of $a \otimes b$ is the sum of the projections onto them:



(5.3)

the coefficient making each term an idempotent by (5.1). Closing the c strand of (5.1) into a loop gives the *theta graph*,



(5.4)

symmetric in a, b, c. Taking the trace of (5.3), the left side becomes two loops, $d_a d_b$; by cyclicity of the trace each term on the right becomes $\sqrt{d_c/d_a d_b}$ times a theta graph, that is d_c. So

$$d_a d_b = \sum_c N_{ab}^{c}\, d_c,$$

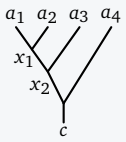
recovering the multiplicativity of § 5.3.

In $\mathrm{Rep}(S_3)$, for std ⊗ std, a splitting vertex is an isometric intertwiner $c \to \mathrm{std} \otimes \mathrm{std}$ times $(4/d_c)^{1/4}$, and both relations can be checked as matrix identities. With std real orthogonal on an orthonormal basis e_1, e_2, the vacuum vertex is $1 \mapsto e_1 \otimes e_1 + e_2 \otimes e_2$, which is $\mathrm{coev}_{\mathrm{std}}$ under the intertwiner $\mathrm{std}^* \to \mathrm{std}$, $e^i \mapsto e_i$, and its bubble is $2 = d_{\mathrm{std}}$. Kitaev's App. E.1 uses unit-norm vertices instead, $\psi^\dagger\psi = \mathrm{id}$, with no square roots in completeness; the two conventions differ by $(d_c/d_a d_b)^{1/4}$ per vertex, and a formula must not mix them.

5.5 Fusion spaces

The fusion space $V_c^{a_1\cdots a_n}$ was defined at the start of the lecture. We now construct its basis by decomposing the tensor product one pair at a time. As in the spin example, the intermediate labels count multiplicities at fixed total charge.

Theorem 5.6 (Fusion trees). *Fix the left-associated bracketing. The* fusion trees *of this shape with root c and leaves $a_1, \dots, a_n$, labelled by intermediate simple objects $x_1, \dots, x_{n-2}$ with every vertex allowed, form a basis of $V_c^{a_1\cdots a_n}$:*



So

$$\dim V_c^{a_1\cdots a_n} = \sum_x N_{a_1a_2}^{x_1} N_{x_1a_3}^{x_2} \cdots N_{x_{n-2}a_n}^{c}.$$

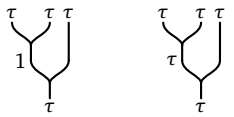
Proof. For $n = 2$ the space is a line or zero. For $n > 2$, decompose $a_1 \otimes \cdots \otimes a_{n-1}$ into simple objects by induction: by semisimplicity,

$$\begin{aligned}\mathrm{Hom}(c, a_1 \otimes \cdots \otimes a_n) &\cong \bigoplus_x \mathrm{Hom}(c, x \otimes a_n) \\ &\quad \otimes \mathrm{Hom}(x, a_1 \otimes \cdots \otimes a_{n-1}),\end{aligned}$$

and a basis is the vertex $\psi_c^{xa_n}$ followed by the trees of the second factor. ∎

Bracketing the leaves differently gives another basis of the same space; the change of basis is the F-move of Lecture 6.

Three τ's with total charge τ have two trees, with $x_1 = 1$ and $x_1 = \tau$:



In general each intermediate label of n τ's is 1 or τ, and a 1 cannot follow a 1. Write f_n for $\dim V_\tau^{\tau\cdots\tau}$ with n leaves; the last vertex is $\tau \otimes \tau \to \tau$ or $1 \otimes \tau \to \tau$, and a 1 after $n-1$ leaves must come from τ after $n-2$, so

$$f_n = f_{n-1} + f_{n-2}, \qquad f_1 = f_2 = 1.$$

The dimensions are $1, 1, 2, 3, 5, 8, \dots$, the Fibonacci numbers, and they grow like φ^n (Sheet 5): a large degenerate space, protected by topology. For Ising, $2m$ copies of σ with total charge 1 have 2^{m-1} states. For $m = 2$ the trees have $x_1 \in \{1, \psi\}$ and then $x_2 = \sigma$: four σ's carry one qubit.

Summary.

- *Handed on:* a fusion category is semisimple, rigid and pivotal, with finitely many simples, a simple unit and bilinear $\otimes$ (Definition 5.1); Fibonacci, Ising, $\mathrm{SU}(2)_k$, Vec_G^ω and $\mathrm{Rep}(G)$ are the examples.
- *The fusion ring:* N_{ab}^c form an associative ring with $N_{ab}^1 = \delta_{b,a^*}$ and $N_{ab}^c = N_{bc^*}^{a^*}$ (Theorem 5.3); it does not determine the category.
- *Dimensions:* FPdim is the Perron–Frobenius eigenvalue, φ for τ and $\sqrt{2}$ for σ; for a pseudo-unitary C with its canonical spherical structure $d = \mathrm{FPdim}$, and $\dim C = \sum_a d_a^2$ (Theorems 5.4 and 5.5).
- *Pictures:* the trivalent vertex with the bubble $\sqrt{d_a d_b / d_c}$ and completeness (5.1), (5.3); admissibly labelled trees of one fixed shape form a basis, with Fibonacci numbers for τ (Theorem 5.6).
- *Next:* Lecture 6 computes the associator on fusion trees, the F-symbols, and solves the pentagon for Fibonacci.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: fusion categories, Def. 4.1.1, p. 65; based and fusion rings, Defs. 3.1.3, 3.1.7 and Prop. 3.1.6, pp. 49–50; the Frobenius–Perron dimension, Def. 3.3.3 and Prop. 3.3.6, p. 53; the Yang–Lee (Fibonacci), Ising and Verlinde rings, Exs. 3.1.9 and 4.10.6, pp. 51 and 79; pseudo-unitary categories, Def. 9.4.4, p. 283, and unitary ones, Rem. 9.4.7, pp. 283–284; the canonical spherical structure, Prop. 9.5.1, p. 284. Kitaev, *Anyons in an exactly solved model and beyond*, cond-mat/0506438, App. E: fusion rules, p. 67, unit-norm vertices, eqs. 174–176, p. 68, the standard cups and caps, eqs. (190)–(191), p. 76, the isotopy-invariant normalisation, eqs. 195–196, p. 77, quantum dimensions as Perron–Frobenius eigenvalues, pp. 79–80. Bonderson, PhD thesis (Caltech, 2007), (2.1)–(2.19), pp. 13–19. Simon, *Topological Quantum*, Ch. 8, pp. 93–102 (Fibonacci counting, p. 98). Calaque and Etingof, *Lectures on tensor categories*, Cor. 3.12, p. 24.

Lecture 6

F-symbols and the pentagon

Objectives. *Lecture 5 gave each anyon model its particles and fusion rules, and a basis of fusion trees for the states of many particles. It said nothing about the associator, which Lecture 3 made part of the definition. In a fusion category the associator is a finite table of numbers, the F-symbols: the change of basis between two bracketings of three particles. Mac Lane's pentagon becomes a system of polynomial equations for them. This lecture writes it down and solves it for Fibonacci, where the fusion rules leave almost no freedom.*

By the end of this lecture you will be able to:

- Define the F-symbols on fusion trees and prove their unitarity for a unitary category with normalised vertices.
- Derive the pentagon equation from Mac Lane's pentagon.
- Compute how a change of vertices acts on the F-symbols, and find gauge invariants among them.
- Solve the pentagon for Fibonacci, and verify the two Ising solutions and why they are inequivalent.

6.1 F-moves

As in § 5.4, C is unitary and multiplicity-free. Its dagger gives positive inner products, and its associators are unitary. The vertices ψ_c^{ab} are normalised by (5.1). Three particles a, b, c with total charge d have two bases of fusion trees (Theorem 5.6, and its mirror image). The *left trees* fuse a and b first, through a channel e:

$$\begin{aligned} L_e &= (\psi_e^{ab} \otimes \mathrm{id}_c) \circ \psi_d^{ec} \\ &\in \mathrm{Hom}\big(d, (a \otimes b) \otimes c\big). \end{aligned}$$

The *right trees* fuse b and c first, through a channel f:

$$\begin{aligned} R_f &= (\mathrm{id}_a \otimes \psi_f^{bc}) \circ \psi_d^{af} \\ &\in \mathrm{Hom}\big(d, a \otimes (b \otimes c)\big). \end{aligned}$$

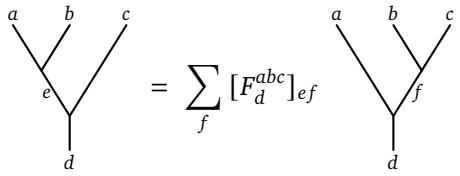
The associator $\alpha_{a,b,c}$ carries the first space to the second, and the R_f are a basis there.

Definition 6.1 (F-symbols)**.** The *F-symbols* $[F_d^{abc}]_{ef}$ are the numbers with

$$\alpha_{a,b,c} \circ L_e = \sum_f [F_d^{abc}]_{ef}\, R_f.$$

The *F-matrix* F_d^{abc} has rows indexed only by allowed left channels e and columns only by allowed right channels f, in the order of the simple objects. For sums over all simple labels, extend the indexed symbols by 0 when a tree does not exist. These zeros are not extra matrix rows or columns.

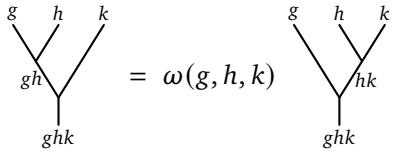
In pictures the associator is invisible (Lecture 3), and the definition is the *F-move*:



(6.1)

The two trees in (6.1) are different branching graphs. The first joins a to b at a vertex; the second joins b to c. A planar isotopy with fixed labelled endpoints preserves which leaves meet at a vertex, so it cannot turn one tree into the other. The equality is a linear relation supplied by the associator and the chosen vertex bases. It may require a sum over several right trees. Suppressing the associator in strict picture notation does not suppress these coefficients. For example, the next diagram can carry a nontrivial scalar even when there is only one admissible channel.

For Vec_G^ω every space is a line, and the associator of Lecture 3 gives a single F-symbol for each triple, $F^{ghk}_{ghk} = \omega(g,h,k)$:



(6.2)

For the twisted $\mathbb{Z}_2$ category of Example 3.9, this is $F_s^{sss} = -1$. An F-move is needed whenever an operation acts on two particles that the current tree does not fuse first, as an exchange will in Lecture 7.

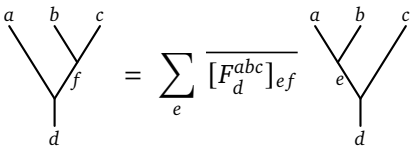
Proposition 6.2 (F is unitary). *Every F-matrix is unitary.*

Proof. By the bubble (5.1), applied twice,

$$L_e^\dagger L_{e'} = \delta_{ee'} \sqrt{\frac{d_a d_b}{d_e}} \sqrt{\frac{d_e d_c}{d_d}}\ \mathrm{id}_d$$
$$= \delta_{ee'} \sqrt{\frac{d_a d_b d_c}{d_d}}\ \mathrm{id}_d,$$

and likewise $R_f^\dagger R_{f'} = \delta_{ff'} \sqrt{d_a d_b d_c / d_d}\ \mathrm{id}_d$. So both families are orthogonal with one common norm, and a unitary map between the two spaces has a unitary matrix in these bases. ∎

So the move back is the conjugate transpose, a sum over the left channel:

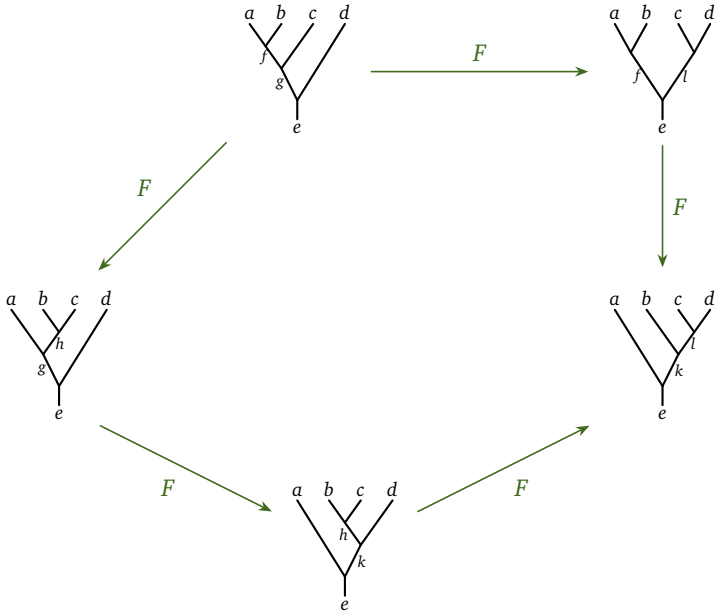


(6.3)

One more convention fixes the entries with a unit leg: $[F_d^{abc}]_{ef} = 1$ whenever a, b or c is 1 and the trees exist. Bonderson (p. 15) imposes it as the statement that fusing with the vacuum commutes with associativity, and notes that it can always be arranged by the choice of vertices.

6.2 The pentagon equation

Four particles a, b, c, d with total charge e have five bracketings, and the F-moves connect them. Labelling the inner edges as in the picture,



the two paths from the top-left tree to $a(b(cd))$ on the right must agree.

Theorem 6.3 (The pentagon equation). *The F-symbols of $\mathcal{C}$ satisfy, for all labels,*

$$[F_e^{fcd}]_{gl}\,[F_e^{abl}]_{fk} = \sum_h [F_g^{abc}]_{fh}\,[F_e^{ahd}]_{gk}\,[F_k^{bcd}]_{hl}.$$

Conversely · sketched, numbers $[F_d^{abc}]_{ef}$ for a fusion ring, invertible and equal to 1 on unit legs, that satisfy these equations define an associator which makes the skeletal category with these fusion rules monoidal.

Proof. Start from the tree $((ab)c)d$ with channels f for ab and g for $(ab)c$. The upper path first applies the F-move to f, c, d, which changes g into l, the channel of cd, with coefficient $[F_e^{fcd}]_{gl}$; then the F-move to a, b, l, which changes f into k, with coefficient $[F_e^{abl}]_{fk}$. This reaches the tree $a(b(cd))$ with channels l and k. The lower path applies F_g^{abc} inside, changing f into h, the channel of bc; then F_e^{ahd}, changing g into k; then F_k^{bcd}, changing h into l. It reaches the same tree after a sum over h. The two paths are the two sides of Mac Lane's pentagon (Definition 3.1), applied to the starting tree, so they are equal; the trees $a(b(cd))$ are a basis (Theorem 5.6, mirrored), so their coefficients are equal, which is the equation.

For the converse, define α on simple objects by the F-symbols and extend it by linearity. Mac Lane's pentagon then needs checking only on simple objects, where it is the equation, and the triangle holds by the unit legs. This is the argument of Theorem 3.8. ∎

For Vec_G^ω the equation at $(a, b, c, d) = (g, h, k, m)$ has $f = gh$, $l = km$ and one term on the right, and it reads

$$\omega(gh, k, m)\,\omega(g, h, km) = \omega(g, h, k)\,\omega(g, hk, m)\,\omega(h, k, m),$$

the cocycle condition of Theorem 3.8. In general it is a system of polynomial equations, one for each choice of the nine labels, in the unknown F-symbols; for Fibonacci it has twelve distinct non-trivial equations in five unknowns.

6.3 Gauge and Ocneanu

The vertices were chosen up to a phase (Lecture 5), and the F-symbols depend on the choice; any non-zero rescaling of the vertices changes them in the same way.

Theorem 6.4 (Gauge transformations). *Let $\psi_c^{ab} = u_c^{ab}\,\tilde{\psi}_c^{ab}$ with $u_c^{ab} \neq 0$ and $u_a^{1a} = u_a^{a1} = 1$. In the new vertices*

$$[\tilde{F}_d^{abc}]_{ef} = \frac{u_d^{af}\,u_f^{bc}}{u_e^{ab}\,u_d^{ec}}\,[F_d^{abc}]_{ef},$$

and the $\tilde{F}$ satisfy the pentagon equation and are 1 on unit legs. If every $\left|u_c^{ab}\right| = 1$, they are unitary.

Proof. Substitute into Definition 6.1: $L_e = u_e^{ab}\,u_d^{ec}\,\tilde{L}_e$ and $R_f = u_f^{bc}\,u_d^{af}\,\tilde{R}_f$. The pentagon is a statement about α, which has not changed, written in the new trees. If $|u| = 1$ the new vertices still satisfy (5.1), so Proposition 6.2 applies to them. With a unit leg the factor is 1. ∎

A number that does not change is a *gauge invariant*, and only those describe the category: two solutions related by a gauge transformation are the same category. An entry whose factor is 1 is invariant. For a self-dual a, the entry $[F_a^{aaa}]_{11}$ has the factor $u_a^{a1}\,u_1^{aa}/(u_1^{aa}\,u_a^{1a}) = 1$, and it is the Frobenius–Schur indicator of Lecture 4 divided by the dimension (Bonderson (2.23) · *cited*):

$$\kappa_a = d_a\,[F_a^{aaa}]_{11}. \tag{6.4}$$

For that twisted $\mathbb{Z}_2$ category, $\kappa_s = \omega(s,s,s) = -1$.

For Vec_G^{ω} the factor of Theorem 6.4 on F^{ghk} is $u(g,hk)\,u(h,k)/(u(g,h)\,u(gh,k))$, writing $u(g,h) = u_{gh}^{gh}$: the cocycle ω changes by a coboundary. With $j = u^{-1}$ this is exactly the change $\omega' = \omega\,(\delta j)^{-1}$ made by a monoidal functor that is the identity on objects (Definition 3.10). A gauge transformation is such a functor, and Proposition 3.11 found its invariant on $\mathbb{Z}_2$: $\omega(s,s,s)$, which is κ_s.

The pentagon solutions for fixed fusion rules are rigid.

Theorem 6.5 (Ocneanu rigidity · *cited*). *A fusion category admits no non-trivial deformations, and there are only finitely many fusion categories with a given fusion ring, up to equivalence (EGNO Thm. 9.1.4).*

In F-symbols: the pentagon for a given fusion ring has only finitely many solutions up to gauge transformations with $u \neq 0$ (Bonderson, p. 33).

6.4 Fibonacci solved

Fibonacci has one non-trivial fusion, $\tau \otimes \tau \cong 1 \oplus \tau$. With the unit legs fixed, the unknowns are the number $x = [F_1^{\tau\tau\tau}]_{\tau\tau}$ and the 2×2 matrix

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

rows and columns in the order $1, \tau$. By Theorem 6.4, with $u_1^{\tau\tau} = p$ and $u_\tau^{\tau\tau} = q$, the entries x, A and D are invariant, while

$$\tilde{B} = \frac{q^2}{p} B, \qquad \tilde{C} = \frac{p}{q^2} C,$$

so BC is invariant and the ratio B/C is not. Write the pentagon equation with $a = b = c = d = \tau$. To read an instance, take $e = f = 1$, $g = k = \tau$ and $l = 1$. On the left, $F_1^{1\tau\tau}$ and $F_1^{\tau\tau 1}$ have a unit leg, so the left side is 1. On the right, $h = 1$ gives $[F_\tau^{\tau\tau\tau}]_{11}[F_1^{\tau 1\tau}]_{\tau\tau}[F_\tau^{\tau\tau\tau}]_{11} = A^2$ and $h = \tau$ gives $B\,x\,C$: so $1 = A^2 + xBC$. The following instances list (e, f, g, k, l) beside each equation, in the order used below:

$$\begin{aligned}
(\tau,1,\tau,1,\tau): &\quad A = xBC,\\
(\tau,\tau,1,1,\tau): &\quad BC = x^2 A,\\
(1,1,\tau,\tau,1): &\quad xBC + A^2 = 1,\\
(\tau,1,\tau,\tau,1): &\quad A^2 + BCD = 0,\\
(\tau,\tau,1,\tau,\tau): &\quad BD = xBD,\\
(1,\tau,\tau,\tau,\tau): &\quad x^2 = xD^2 + BC.
\end{aligned}$$

Theorem 6.6 (Fibonacci's F-symbols). *Every solution has $x = 1$, $A \in \{\varphi^{-1}, -\varphi\}$, $D = -A$ and $BC = A$. The unitary ones have $A = \varphi^{-1}$, and in the gauge $B = C$*

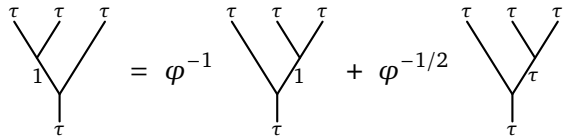
$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix},$$

unique up to the gauge $B, C \mapsto -B, -C$.

Proof. The first two equations give $A = x^3 A$. If $A = 0$, then $BC = 0$ and the third reads $0 = 1$; so $A \neq 0$, $x^3 = 1$, and B and C are non-zero. If $D = 0$, the fourth gives $A^2 = 0$; so $D \neq 0$, and the fifth gives $x = 1$. Now the first gives $BC = A$, the third $A^2 + A = 1$, and the fourth $A^2 + AD = 0$, so $D = -A$; the sixth then holds. The roots of $A^2 + A = 1$ are φ^{-1} and $-\varphi$, since $\varphi^2 = \varphi + 1$.

A unitary matrix has entries of modulus at most 1, which excludes $A = -\varphi$. For $A = \varphi^{-1}$, a unit first row needs $|B|^2 = 1 - \varphi^{-2} = \varphi^{-1}$. The gauge with $p = 1$ and $q^4 = C/B$ makes $B = C$, and then $B^2 = A$ gives $B = C = \pm\varphi^{-1/2}$; the sign is the gauge $q \mapsto iq$. The matrix is unitary, and every other pentagon equation holds (`scripts/check-l6-algebra.py` checks them all). ∎

In pictures, three τ's whose first two fuse to the vacuum are



(6.5)

The gauge-dependent entries are not physics: $B = \varphi^{-1/2}$ is a choice, while $x = 1$, A, D and BC are forced. By (6.4), $A = \kappa_\tau / d_\tau$; the unitary root $A = \varphi^{-1}$, with $d_\tau = \varphi$ from Lecture 5, gives $\kappa_\tau = 1$.

The rejected root is not an error. With $A = -\varphi$ the matrix, with $B = C = \sqrt{-\varphi}$, still solves every pentagon equation and squares to the identity. It is the *Yang–Lee* category, the Galois conjugate $\sqrt{5} \mapsto -\sqrt{5}$ of Fibonacci, with $d_\tau = 1/A = -\varphi^{-1}$, the other root of $d^2 = 1 + d$ that Lecture 5 set aside. It is a fusion category (EGNO Exer. 8.18.7 · *cited*), but not a unitary one, so it is no anyon model (Sheet 6). Up to gauge, then, the fusion rule $\tau \otimes \tau \cong 1 \oplus \tau$ allows exactly these two categories, as Ocneanu's theorem promised: finitely many.

6.5 Ising: κH verified; κ gauge invariant

Ising has eleven F-matrices with no unit leg: $F_\sigma^{\sigma\sigma\sigma}$, which is 2×2 in the channels $1,\psi$, and ten numbers. Let H be the Hadamard matrix of *Quantum Computing* (Eq. (2.1)),

$$H=\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}.$$

Theorem 6.7 (Ising's F-symbols · *proved by verification*)**.** *For $\kappa=\pm1$, the F-symbols*

$$F_\sigma^{\sigma\sigma\sigma}=\kappa H,$$
$$F_\psi^{\sigma\psi\sigma}=F_\sigma^{\psi\sigma\psi}=-1,$$

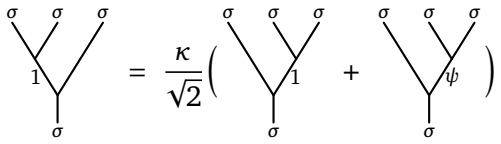
with every other one equal to 1, solve the pentagon equation. The two solutions are not related by any gauge transformation.

Proof. The pentagon equation is a finite list, and `scripts/check-l6-algebra.py` checks all 3^9 instances for both signs. One instance shows where the -1 comes from: take $a=b=c=d=\sigma$, $e=\psi$, $f=1$, $g=k=\sigma$ and $l=1$. The left side vanishes, since f and l would have to fuse to ψ, and $1\otimes 1\cong 1$. The right side has the two terms $h=1,\psi$:

$$0=\frac{\kappa^2}{2}\Big(1+[F_\psi^{\sigma\psi\sigma}]_{\sigma\sigma}\Big).$$

Since σ is self-dual, (6.4) gives the gauge invariant $d_\sigma[F_\sigma^{\sigma\sigma\sigma}]_{11}=\sqrt{2}\,\kappa/\sqrt{2}=\kappa$, which differs between the two solutions. ∎

In pictures, three σ's whose first two fuse to the vacuum are



(6.6)

A monoidal equivalence between the two would fix σ, the only simple object of dimension $\sqrt{2}$, and ψ. Identifying each image with the object, it carries each vertex to a multiple of a vertex, and its compatibility with the associators (Definition 3.10) becomes the formula of Theorem 6.4 with some $u\neq 0$ · *sketched*. That leaves κ alone.

So the Ising fusion rules carry two inequivalent fusion categories, told apart by the Frobenius–Schur indicator of σ. That there are no others is a theorem of Tambara and Yamagami · *cited* (EGNO Prop. 9.14.9). The course's *Ising* is $\kappa=+1$; the solution $\kappa=-1$ is the fusion category of $\mathrm{SU}(2)_2$, whose σ is spin-$\frac{1}{2}$ with $\kappa=-1$, as in Lecture 4. The two have the same fusion ring, as Lecture 5 found, and differ in their F-symbols (Simon, p. 98). As gates, H and $-H$ agree up to a phase; as F-symbols they give inequivalent categories. The printed entries are in this gauge: unlike κ, the off-diagonal entries of κH change under $u_\psi^{\sigma\sigma}\mapsto iu_\psi^{\sigma\sigma}$, and $F_\psi^{\sigma\psi\sigma}$, with the (ψ,ψ) entry of κH, under $u_\sigma^{\psi\sigma}\mapsto -u_\sigma^{\psi\sigma}$.

Summary.

- *Handed on:* the F-symbols are the associator on fusion trees, $\alpha\circ L_e=\sum_f[F_d^{abc}]_{ef}R_f$. For a unitary category with a norm-preserving vertex gauge, the admissible F-matrices are unitary. Entries on allowed unit legs are 1 (Definition 6.1 and Proposition 6.2).

- *The pentagon:* Mac Lane's pentagon is a polynomial system in the F-symbols (Theorem 6.3); for Vec_G^{ω} it is the cocycle condition.
- *Gauge:* changing vertices rescales F (Theorem 6.4); only invariants, such as $\kappa_a = d_a[F_a^{aaa}]_{11}$, are physics. Ocneanu: finitely many solutions up to gauge (Theorem 6.5).
- *Fibonacci:* $F_\tau^{\tau\tau\tau} = \left(\begin{smallmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{smallmatrix}\right)$, unique up to gauge among unitary solutions; Yang–Lee is its non-unitary conjugate (Theorem 6.6).
- *Ising:* $F_\sigma^{\sigma\sigma\sigma} = \kappa H$, two inequivalent solutions; the course's Ising has $\kappa = +1$ (Theorem 6.7).
- *Next:* Lecture 7 adds the braiding: R-symbols, the hexagon equations and the topological spin.

References

Kitaev, *Anyons in an exactly solved model and beyond*, cond-mat/0506438: the F-move, eq. (177), p. 68; the pentagon, eq. (179), p. 69; Ocneanu rigidity, p. 89, and gauge transformations, eqs. (239)–(242), p. 90; the Ising F-matrix κH, eq. (86), p. 38. Bonderson, PhD thesis (Caltech, 2007): F-moves and their unitarity, (2.6)–(2.7), pp. 14–15; unit legs, p. 15; κ_a and F, (2.23), p. 20; the multiplicity-free pentagon and gauge, (2.77) and (2.80)–(2.81), p. 33; Fibonacci and Ising data, pp. 82–84. Simon, *Topological Quantum*: F-matrices, (9.1), p. 113; Fibonacci, (9.3), p. 115; the pentagon and Ocneanu, (9.7), p. 117; gauge, (9.8), p. 118; Ising and $\mathrm{SU}(2)_2$, p. 98, footnote 7. Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: Ocneanu rigidity, Thm. 9.1.4, p. 276; Ising categories, Prop. 9.14.9, p. 315; the Yang–Lee categories, Exer. 8.18.7, p. 236, and Exer. 9.4.6(iii), p. 283.

Lecture 7

Braiding: hexagons, Yang–Baxter and topological spin

Objectives. *Lectures 5 and 6 built fusion categories, with their particles and associators. In three dimensions, exchanging two identical particles twice is the identity, and exchange statistics are those of the symmetric group: bosons and fermions. In a plane the two exchanges are not the same, and the symmetric group is replaced by the braid group. This lecture adds a braiding to a monoidal category, derives the braid relation from its axioms, writes the axioms as equations in F- and R-symbols, and solves them for Fibonacci. Closing a braided strand gives the topological spin.*

By the end of this lecture you will be able to:

- State the braid group and why it replaces the symmetric group in a plane.
- Define a braiding, prove the braid relation from the hexagons, and tell the braid form of the Yang–Baxter equation from *Integrability*'s form.
- Write the hexagon equations in F- and R-symbols and solve them for Fibonacci.
- Compute topological spins, and find the braidings of the pointed categories on $\mathbb{Z}_2$.

7.1 B_n versus S_n

The *braid group* B_n has generators $\sigma_1, \dots, \sigma_{n-1}$, with

$$\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1},$$
$$\sigma_i\sigma_j = \sigma_j\sigma_i \quad (|i-j| \geq 2).$$

Adding $\sigma_i^2 = 1$ gives the symmetric group S_n. The generator σ_i exchanges particles i and $i+1$, passing the left one over:

$$\sigma_i = \quad \begin{matrix} \vert & \cdots & \diagdown & & \cdots & \vert \\ 1 & & i & i{+}1 & & n \end{matrix}$$

A point of a configuration space specifies all particle positions at once. A path in that space is a possible motion without collisions. A closed path returns to the same unordered configuration, although the individual particles may have exchanged places. Choose one starting configuration. The fundamental group collects closed paths based there, identifying paths that can be continuously deformed into one another without collisions and with that configuration fixed. Its product follows one path by another. For these connected spaces, simply connected means that every closed path can shrink to a point. The codimension of an excluded set is the number of independent local conditions defining it. Requiring two positions to agree imposes one condition for each spatial coordinate. These meanings explain the geometric statement below; its planar case is cited.

Theorem 7.1 (Configuration spaces). *The fundamental group of the space of n unordered distinct points in $\mathbb{R}^2$ is B_n · cited (Kassel–Turaev §1.4.3); in $\mathbb{R}^d$ with $d \geq 3$ it is S_n · sketched.*

For $d \geq 3$: ordered configurations form $\mathbb{R}^{dn}$ with the diagonals $x_i = x_j$ removed, each of codimension $d \geq 3$, and removing subspaces of codimension at least 3 leaves a space simply connected. The unordered space is its quotient by the free action of S_n, so its fundamental group is S_n. In the plane the diagonals have codimension 2, like a point in a plane, and a loop taking one particle around another does not shrink. Quantum mechanics of identical particles in a plane therefore allows unitary representations of B_n, not only representations of S_n: *anyons* (*Symmetries* §15.4).

7.2 Braided categories and hexagons

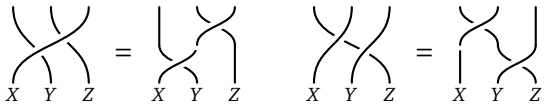
As in Lecture 4, we work as if the category were strict.

Definition 7.2 (Braiding). A *braiding* on a monoidal category is a natural isomorphism $c_{X,Y} : X \otimes Y \to Y \otimes X$, drawn with the strand from bottom left to top right over the other, such that the *hexagons* hold:

$$c_{X,Y\otimes Z} = (\mathrm{id}_Y \otimes c_{X,Z}) \circ (c_{X,Y} \otimes \mathrm{id}_Z),$$
$$c_{X\otimes Y,Z} = (c_{X,Z} \otimes \mathrm{id}_Y) \circ (\mathrm{id}_X \otimes c_{Y,Z}).$$

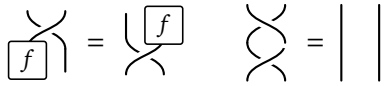
It is *symmetric* if $c_{Y,X} \circ c_{X,Y} = \mathrm{id}_{X\otimes Y}$ for all X, Y.

With the associators put back, each hexagon has six sides, hence the name (EGNO Def. 8.1.1). In pictures, the first says that a strand crossing over a pair crosses each in turn, and the second that a pair crossing over a strand does so one at a time:



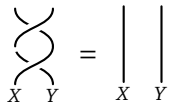
(7.1)

Naturality says that boxes slide through crossings, and invertibility that a crossing followed by its inverse is no crossing at all (Reidemeister II):



(7.2)

The *double braiding* $c_{Y,X} \circ c_{X,Y}$ takes one strand once around the other. A braiding is symmetric exactly when the double braiding is always trivial:



(7.3)

and then over- and under-crossings agree, and are drawn as a plain crossing.

Example 7.3 (Braidings).

- Vect and Rep(G) with the flip $v \otimes w \mapsto w \otimes v$, which is an intertwiner and natural: symmetric.

- *Super vector spaces* sVec: Vec_G on $\mathbb{Z}_2 = \{1, s\}$ with $c_{s,s} = -1$ (EGNO Ex. 8.2.2). It is symmetric, since $(-1)^2 = 1$, and the odd line is a fermion: exchanging two of them gives a sign.
- The same category with $c_{s,s} = +1$: symmetric again, but a different braiding, not equivalent to the first as a braided category: an equivalence fixes s, so it would preserve $c_{s,s} = \pm 1$. A braiding is structure, not a property.

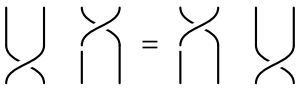
7.3 Yang–Baxter is Reidemeister III

Theorem 7.4 (The braid relation · *proved given coherence*)**.** *In a braided category, for all X, Y, Z,*

$$\begin{aligned}(c_{Y,Z} \otimes \mathrm{id}_X)(\mathrm{id}_Y \otimes c_{X,Z}) \\ \circ (c_{X,Y} \otimes \mathrm{id}_Z) \\ = (\mathrm{id}_Z \otimes c_{X,Y})(c_{X,Z} \otimes \mathrm{id}_Y) \\ \circ (\mathrm{id}_X \otimes c_{Y,Z}).\end{aligned}$$

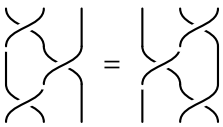
So $\sigma_i \mapsto \mathrm{id} \otimes c_{X,X} \otimes \mathrm{id}$, in positions $i, i+1$, is a representation of B_n on $X^{\otimes n}$.

Proof. By the first hexagon the last two factors on the left are $c_{X,Y\otimes Z}$. Naturality of $c_{X,-}$ applied to $c_{Y,Z} : Y \otimes Z \to Z \otimes Y$ gives $(c_{Y,Z} \otimes \mathrm{id}_X)\, c_{X,Y\otimes Z} = c_{X,Z\otimes Y}\, (\mathrm{id}_X \otimes c_{Y,Z})$, and the first hexagon again turns $c_{X,Z\otimes Y}$ into the first two factors on the right. The far commutation $\sigma_i \sigma_j = \sigma_j \sigma_i$ is the interchange law (Theorem 3.4):



■

In pictures this is Reidemeister III: a strand slides over a crossing.



(7.4)

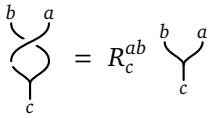
Let P be the flip, $P(v \otimes w) = w \otimes v$. For a map R on $V \otimes V$, a leg subscript specifies which ordered factors of $V^{\otimes 3}$ it acts on, leaving the other fixed:

$$\begin{aligned}R_{12} = R \otimes \mathrm{id}, \qquad R_{23} = \mathrm{id} \otimes R, \\ R_{13} = P_{23}(R \otimes \mathrm{id})P_{23}.\end{aligned}$$

Integrability's Yang–Baxter equation carries spectral parameters (its eq. (14.8)). For a constant matrix R on $V \otimes V$ it reads $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$, and then R satisfies it exactly when $c = P \circ R$ satisfies the braid form $c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23}$, which Kassel (Thm. XIII.1.3) and EGNO (Prop. 8.1.10) call the Yang–Baxter equation. Three things are called R, and must be kept apart: *Integrability*'s $R(\lambda)$, the braiding $c = P \circ R$, and the R-symbols of the next section.

7.4 R-symbols; Fibonacci solved

In a multiplicity-free fusion category with the vertices of Lecture 5, the braiding maps the line $\mathrm{Hom}(c, a\otimes b)$ to the line $\mathrm{Hom}(c, b\otimes a)$, so it acts on vertices by numbers, the *R-symbols*:



(7.5)

that is $c_{a,b}\circ\psi_c^{ab} = R_c^{ab}\,\psi_c^{ba}$, with $R_c^{ab}=1$ when a or b is 1: the first hexagon with $Y=Z=1$ gives $c_{X,1}=c_{X,1}^2$, so $c_{X,1}=\mathrm{id}$, and likewise $c_{1,X}=\mathrm{id}$. In a unitary category $\left|R_c^{ab}\right|=1$. A gauge transformation (Theorem 6.4) changes R_c^{ab} by u_c^{ba}/u_c^{ab}, so R_c^{aa} is a gauge invariant.

Theorem 7.5 (The hexagon equations · *proved given coherence*). *The F- and R-symbols of a braided multiplicity-free fusion category satisfy, for all labels,*

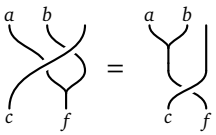
$$R_e^{ca}\,[F_d^{acb}]_{eg}\,R_g^{cb} = \sum_f [F_d^{cab}]_{ef}\,R_d^{cf}\,[F_d^{abc}]_{fg},$$

and, from the second hexagon,

$$(R_e^{ac})^{-1}[F_d^{acb}]_{eg}(R_g^{bc})^{-1} = \sum_f [F_d^{cab}]_{ef}\,(R_d^{fc})^{-1}[F_d^{abc}]_{fg}$$

(Bonderson (2.57)–(2.58)). Conversely, F- and R-symbols satisfying the pentagon and both families define a braiding · sketched.

Proof. The step that does the work is that a strand crosses a vertex as it crosses the two strands leaving it, by naturality and the first hexagon:



(7.6)

Apply both sides of the first hexagon, with $X=c$, $Y=a$, $Z=b$, to the tree in which a and c fuse first, through e, to total charge d. On the right, c crosses a, which multiplies by R_e^{ca}; an F-move brings c and b together, through g; c crosses b, which gives R_g^{cb}. On the left, an F-move brings a,b together, through f; c crosses the pair f at once, giving R_d^{cf}; a last F-move returns to the basis of trees fusing b and c first. Comparing coefficients of the basis trees gives the equation. The second family comes the same way from the second hexagon, written for the inverse braiding: $c_{b,a}^{-1}$ acts on ψ_c^{ab} by $(R_c^{ba})^{-1}$, so inverting a crossing also exchanges its labels. The converse is the argument of Theorem 6.3. ∎

For Fibonacci, with F from Theorem 6.6, the unknowns are $R_1=R_1^{\tau\tau}$ and $R_\tau=R_\tau^{\tau\tau}$. Two instances of the first family, with $a=b=c=\tau$, are enough. With $d=1$ and $e=g=\tau$, every F-symbol is 1 and the sum has only $f=\tau$:

$$R_\tau^2 = R_1.$$

With every label τ, the left side is $R_\tau(-\varphi^{-1})R_\tau$ and the right side $\varphi^{-1/2}\cdot 1\cdot\varphi^{-1/2} + (-\varphi^{-1})R_\tau(-\varphi^{-1})$, so

$$R_\tau^2 + \varphi^{-1}R_\tau + 1 = 0.$$

Theorem 7.6 (Fibonacci's R-symbols)**.** *The hexagons for Fibonacci have exactly two solutions,*

$$R_1^{\tau\tau} = e^{-4\pi i/5}, \quad R_\tau^{\tau\tau} = e^{3\pi i/5},$$

and its complex conjugate.

Proof. The roots of $x^2 + \varphi^{-1}x + 1 = 0$ have product 1 and sum $-\varphi^{-1} = 2\cos(3\pi/5)$, so they are $e^{\pm 3\pi i/5}$. Then $R_1 = R_\tau^2 = e^{\pm 6\pi i/5} = e^{\mp 4\pi i/5}$. Both pairs satisfy every instance of both families (`scripts/check-l7-algebra.py`). ∎

Bonderson (p. 82) and Simon (eq. (10.2)) print the first. The picture alone does not choose; the spin of the next section does.

Exchanging the last two particles of one state

Fix three Fibonacci particles with total charge τ. Use the channel order $1, \tau$ for both the left trees L_e and the right trees R_f of Lecture 6. The bubble calculation of Proposition 6.2 gives

$$L_e^\dagger L_{e'} = \delta_{ee'}\varphi\ \mathrm{id}_\tau, \qquad R_f^\dagger R_{f'} = \delta_{ff'}\varphi\ \mathrm{id}_\tau\,.$$

Thus the printed trees have squared norm φ. For normalized physical states use the orthonormal vectors

$$\ell_e = \varphi^{-1/2}L_e, \qquad r_f = \varphi^{-1/2}R_f.$$

The common rescaling leaves the F-matrix unchanged. A unit coefficient on the raw tree L_1 would instead have squared norm φ.

Write the coefficients of a state as a column c_L in the basis ℓ_1, ℓ_τ. The convention of Definition 6.1 has the input tree on a row: $\ell_e = \sum_f F_{ef}r_f$, with the associator suppressed. Consequently the column in the right basis is $c_R = F^T c_L$. For a unitary F-matrix, the inverse column change is $\overline{F}$. This transpose matters in general. For the real symmetric Fibonacci matrix of Theorem 6.6,

$$F^T = F, \qquad \overline{F} = F, \qquad F^{-1} = F.$$

The first exchange σ_1 acts diagonally on the left basis. The last-two exchange σ_2 acts diagonally on the right basis. Use the exchange phases selected by θ_τ in § 7.5. Set

$$D = \begin{pmatrix} R_1^{\tau\tau} & 0 \\ 0 & R_\tau^{\tau\tau} \end{pmatrix}, \qquad R_1^{\tau\tau} = e^{-4\pi i/5}, \quad R_\tau^{\tau\tau} = e^{3\pi i/5}.$$

To exchange the last two particles, expand in the right basis, apply D, then expand back in the left basis, as in Figure 7.1. The two changes of basis preserve the state. Only the crossing is a physical exchange; no intermediate channel is measured.

The column-action matrices are therefore

$$M_1 = D, \qquad M_2 = \overline{F}DF^T = FDF.$$

The last equality uses the displayed Fibonacci gauge. For a general unitary F-matrix the formula is $\overline{F}DF^T$.

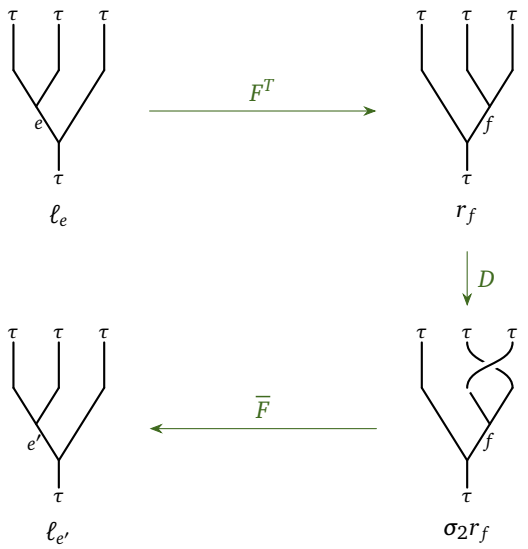


Figure 7.1: The bases used to calculate σ_2. Every tree includes the common factor $\varphi^{-1/2}$. The arrows give operations on coefficient columns; the crossing uses the left-over-right convention. The channels e, f and e' range over $1, \tau$.

Take the normalized initial state ℓ_1, whose first pair has vacuum charge. Its coefficient column passes through the three steps as follows:

$$\begin{aligned} c_L = \begin{pmatrix} 1 \\ 0 \end{pmatrix} &\longmapsto c_R = \begin{pmatrix} \varphi^{-1} \\ \varphi^{-1/2} \end{pmatrix} \\ &\longmapsto Dc_R = \begin{pmatrix} \varphi^{-1} R_1^{\tau\tau} \\ \varphi^{-1/2} R_\tau^{\tau\tau} \end{pmatrix} \\ &\longmapsto c'_L = \begin{pmatrix} \varphi^{-2} R_1^{\tau\tau} + \varphi^{-1} R_\tau^{\tau\tau} \\ \varphi^{-3/2} (R_1^{\tau\tau} - R_\tau^{\tau\tau}) \end{pmatrix}. \end{aligned}$$

Substituting the two exchange phases gives the output state

$$\sigma_2 \ell_1 = \varphi^{-1} e^{4\pi i/5} \ell_1 + \varphi^{-1/2} e^{-3\pi i/5} \ell_\tau .$$

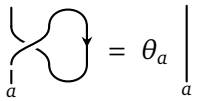
The first-pair charge is no longer definite. A subsequent measurement of that charge has probabilities

$$P(1) = \varphi^{-2}, \qquad P(\tau) = \varphi^{-1}, \qquad P(1) + P(\tau) = 1,$$

because $\varphi^{-2} + \varphi^{-1} = 1$. The relative phase remains part of the coherent state before that measurement. This exchange mixes two amplitudes while preserving the external particle types, the total charge and the norm. Sheet 7 checks the braid relation for M_1 and M_2.

7.5 Topological spin

Closing the right strand of $c_{a,a}$, its partial trace in the sense of (4.9), gives an endomorphism of the simple a, hence a number, the *topological spin* θ_a:



(7.7)

Theorem 7.7 (The spin from R · *proved given coherence*)**.** *In a unitary multiplicity-free fusion category with its canonical spherical structure,*

$$\theta_a = \frac{1}{d_a}\sum_c N_{aa}^{c}\, d_c\, R_c^{aa}.$$

Proof. Take the trace of both sides of (7.7): the left side becomes the trace of $c_{a,a}$, and the right side $\theta_a d_a$. On each channel c of $a\otimes a$, $c_{a,a}$ acts as R_c^{aa} by (7.5), so it is $\sum_c R_c^{aa}P_c$ with P_c the projections of (5.3), and the trace of P_c is $N_{aa}^{c}d_c$. ∎

For Fibonacci,

$$\theta_\tau = \frac{e^{-4\pi i/5} + \varphi\, e^{3\pi i/5}}{\varphi} = e^{4\pi i/5},$$

and the conjugate solution gives $e^{-4\pi i/5}$. The course takes $\theta_\tau = e^{4\pi i/5}$, which fixes the chirality. The spin is a gauge invariant, like the double braiding: a gauge changes R_c^{ab} by u_c^{ba}/u_c^{ab} and R_c^{ba} by the inverse. The two are related by the *ribbon relation* $R_c^{ab}R_c^{ba} = \theta_c/(\theta_a\theta_b)$ (Bonderson (2.60) · *cited*); Lecture 9 identifies θ_a with the twist of a ribbon category, the phase of rotating one particle by 2π.

7.6 Pointed braidings; Ising

For Vec_G^{ω} on $\mathbb{Z}_2 = \{1, s\}$, a braiding is determined by one number, written $c(s,s) = R_1^{ss}$. By Theorem 7.7, $\theta_s = c(s,s)$.

Theorem 7.8 (Braidings on $\mathbb{Z}_2$)**.** *The hexagons for* Vec_G^{ω} *on* $\mathbb{Z}_2$ *force* $c(s,s)^2 = \omega(s,s,s)$*, and each such* $c(s,s)$ *gives a braiding* · *sketched. So the trivial* ω *has two braidings,* $c(s,s) = \pm 1$*, the* boson *and the* fermion *(*sVec*); and* $\omega(s,s,s) = -1$ *has two,* $c(s,s) = \pm i$*, the* semion *and the* antisemion*.*

Proof. In the first family take $a = b = c = d = s$ and $e = g = 1$, so that $f = 1$: the left side is $c(s,s)\,\omega(s,s,s)\,c(s,s)$ and the right side $\omega(s,s,s)\cdot 1\cdot\omega(s,s,s)$. So $c(s,s)^2 = \omega(s,s,s)$. Conversely every instance of both families holds when it does (`scripts/check-l7-algebra.py`), and the converse of Theorem 7.5 gives the braiding. ∎

Here ω was fixed and the braiding varied. A *quadratic form* on a finite abelian group A is a function $q\colon A \to \mathbb{C}^\times$ with

$$q(-x) = q(x) \qquad (x \in A),$$

for which the ratio

$$\frac{q(x+y)}{q(x)q(y)} \qquad (x, y \in A)$$

is multiplicative in each argument. Pointed braided fusion categories are classified, up to braided equivalence, by these forms up to automorphisms of A · *cited*. Every form occurs, with $q(x) = \theta_x$, and determines ω up to a coboundary. For $\mathbb{Z}_5$, the five possible functions $q(x) = e^{2\pi i k x^2/5}$ give three classes: $k = 0$, $k \in \{1,4\}$ and $k \in \{2,3\}$ (EGNO Thms. 8.4.9 and 8.4.12, after Eilenberg–Mac Lane and Joyal–Street).

Theorem 7.9 (Braidings on Ising · *cited*)**.** *Each of the two Ising fusion categories has exactly four braidings (Siehler, Thm. 1.2). On the course's Ising* ($\kappa = +1$) *they have* $\theta_\sigma = e^{i\pi\nu/8}$ *with* $\nu \in \{1,7,9,15\}$*; the other odd* ν *belong to the* $\kappa = -1$ *category, which contains* $\mathrm{SU}(2)_2$ *(*$\nu = 3$*), and* $\kappa_\sigma = (-1)^{(\nu^2-1)/8}$ *(Kitaev, eqs. (91)–(96) and Table 1).*

The course's Ising is $\nu = 1$; in Lecture 6's gauge (Kitaev's Table 1, Bonderson p. 84),

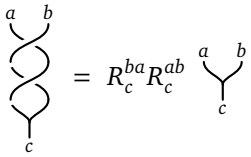
$$R_1^{\sigma\sigma} = e^{-i\pi/8}, \quad R_\psi^{\sigma\sigma} = e^{3i\pi/8},$$
$$R_\sigma^{\sigma\psi} = R_\sigma^{\psi\sigma} = -i, \quad R_1^{\psi\psi} = -1.$$

With the F of Lecture 6 these satisfy both hexagon families (the script checks them; Sheet 7 solves them). The spin formula gives

$$\theta_\sigma = \frac{e^{-i\pi/8} + e^{3i\pi/8}}{\sqrt{2}} = e^{i\pi/8}\,\frac{2\cos(\pi/4)}{\sqrt{2}} = e^{i\pi/8}.$$

The spin of ψ is $\theta_\psi = -1$, so ψ is a fermion.

On a vertex the double braiding is a number, since both crossings act by R-symbols:



(7.8)

Toric-code data

The *toric code* is an anyon model with four particle types $1, e, m, \varepsilon$. Its fusion is the group law of $\mathbb{Z}_2 \times \mathbb{Z}_2$. The quantum dimensions, topological spins and nontrivial products are

a	1	e	m	ε
d_a	1	1	1	1
θ_a	1	1	1	-1

$$e \otimes e = m \otimes m = \varepsilon \otimes \varepsilon = 1,$$
$$e \otimes m = \varepsilon, \quad e \otimes \varepsilon = m, \quad m \otimes \varepsilon = e.$$

Here the fusion equalities identify simple objects up to isomorphism. The rule $e \otimes m = \varepsilon$ identifies ε, the only fermion. These data do not distinguish e from m. Exchanging them is a symmetry (Lecture 12); the lattice convention in Lecture 13 fixes which is which. The double braiding of e with m is -1. It is the phase for taking e once around m; one exchange alone need not have this phase in every gauge. Sheet 7 constructs this data from a bicharacter, meaning a scalar braiding that is multiplicative in each argument. Lecture 12 recovers the same model as a centre.

Finally, an object X that braids trivially with everything, $c_{Y,X}c_{X,Y} = \mathrm{id}$ for all Y, is *transparent*; the transparent objects form the *Müger centre* (EGNO's centralizer C', §8.20). In sVec every object is transparent. In Fibonacci the only transparent simple object is 1, since $R_c^{\tau\tau}R_c^{\tau\tau} \neq 1$ for both channels. A *trivial* Müger centre consists of direct sums of the unit. Lecture 10 shows that this holds exactly when the S-matrix is invertible.

Summary.

- *Handed on:* in a plane, exchanges form the braid group B_n (Theorem 7.1); a braiding is a natural $c_{X,Y}$ with two hexagons (Definition 7.2), and it gives a representation of B_n (Theorem 7.4); Yang–Baxter in braid form is Reidemeister III.
- *R-symbols:* $c_{a,b}\psi_c^{ab} = R_c^{ab}\psi_c^{ba}$, constrained by the hexagon equations in F and R (Theorem 7.5); for Fibonacci $R_1^{\tau\tau} = e^{-4\pi i/5}$, $R_\tau^{\tau\tau} = e^{3\pi i/5}$ or the conjugate (Theorem 7.6).
- *Spin:* $\theta_a = d_a^{-1}\sum_c N_{aa}^c d_c R_c^{aa}$, the value of the closed curl (Theorem 7.7); $\theta_\tau = e^{4\pi i/5}$.

- *Pointed and Ising:* on $\mathbb{Z}_2$, $c(s,s)^2 = \omega(s,s,s)$: boson, fermion, semion, antisemion (Theorem 7.8); Ising has four braidings, and the course's has $\theta_\sigma = e^{i\pi/8}$ (Theorem 7.9).
- *Next:* Lecture 8 builds braidings from Hopf algebras: quantum groups.

References

Kassel and Turaev, *Braid Groups*: B_n as a fundamental group, §1.4.3, p. 29. Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: braidings and hexagons, Def. 8.1.1, p. 195; the Yang–Baxter equation, Prop. 8.1.10, p. 196; symmetric categories and sVec, Def. 8.1.12 and Ex. 8.2.2, p. 197; pointed braidings, Ex. 8.4.3, p. 204, and quadratic forms, Thms. 8.4.9 and 8.4.12, pp. 205–206; the centralizer, §8.20, p. 239. Kassel, *Quantum Groups*: braidings, Def. XIII.1.1, p. 315, and the Yang–Baxter equation, Thm. XIII.1.3, p. 317. Heunen and Vicary, *Categories for Quantum Theory*, Def. 1.17, p. 40. Kitaev, *Anyons in an exactly solved model and beyond*, cond-mat/0506438: Ising, eqs. (91)–(96), p. 40, and Table 1, p. 30; braiding, eqs. (204)–(209), pp. 80–81; spin, eqs. (211)–(215), p. 82. Bonderson, PhD thesis (Caltech, 2007): R-moves and hexagons, (2.53)–(2.58), pp. 25–26; the ribbon relation and spin, (2.60)–(2.61), p. 27; Fibonacci and Ising, pp. 82–84. Simon, *Topological Quantum*: braid groups, §3.3, pp. 28–31; Fibonacci, eq. (10.2), p. 126; hexagons, §13.3, p. 173. Siehler, *Braided near-group categories*, math/0011037, Thm. 1.2.

Block III

Quantum Groups and Knots

Lecture 8

Hopf algebras and quantum groups

Objectives. *Lecture 7 found braidings by solving equations. Where do they come from? For groups, the tensor product of representations comes from the diagonal $g \mapsto g \otimes g$, duals from $g \mapsto g^{-1}$, and the symmetry from the flip. This lecture asks which algebras H give a monoidal, rigid and braided* $\mathrm{Rep}(H)$*, and proves that bialgebras, Hopf algebras and quasitriangular Hopf algebras suffice. The quantum group $U_q(\mathfrak{sl}_2)$ then produces a braiding that is not a symmetry, and its matrix on two spins is, after a change of gauge, the braid limit of* Integrability*'s XXZ R-matrix. This is a side branch: Lecture 9 needs only that matrix.*

By the end of this lecture you will be able to:

- Define algebras, coalgebras, bialgebras and Hopf algebras in pictures, and prove that a bialgebra makes $\mathrm{Rep}(H)$ monoidal and an antipode makes it rigid.
- Prove that a universal R-matrix gives a braiding $c = P \circ R$, and tell R from c.
- State the reconstruction theorem, and show that Fibonacci and Ising are not representation categories of Hopf algebras.
- Compute with $U_q(\mathfrak{sl}_2)$ on $V_1 \otimes V_1$: the matrix of the braiding, the Hecke relation, and the dictionary to XXZ.

8.1 Bialgebras give monoidal representation categories

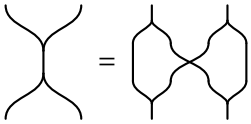
All modules here are finite-dimensional. The algebras acting on them may be infinite-dimensional, as for $U(\mathfrak{g})$ and U_q. The tensor product of linear maps is the ordinary one. An *algebra* is (H, μ, ι), with multiplication $\mu\colon H \otimes H \to H$ and unit $\iota\colon \mathbb{C} \to H$, associative and unital. A *coalgebra* is the same with arrows reversed: $(H, \Delta, \varepsilon_H)$, $\Delta\colon H \to H \otimes H$ coassociative and $\varepsilon_H\colon H \to \mathbb{C}$ a counit, $(\varepsilon_H \otimes \mathrm{id})\Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon_H)\Delta$. In pictures, read upward, μ and Δ are a fusion and a splitting vertex:

$\mu =$  $\Delta =$ 

Kassel writes η for the unit and plain ε for the counit; here η and ε are taken.

Definition 8.1 (Bialgebra)**.** A *bialgebra* is an algebra and a coalgebra on the same H such that Δ and ε_H are algebra maps: $\Delta(xy) = \Delta(x)\Delta(y)$, $\Delta(1) = 1 \otimes 1$, $\varepsilon_H(xy) = \varepsilon_H(x)\varepsilon_H(y)$, $\varepsilon_H(1) = 1$.

The first condition is the *bialgebra law*; with P the flip of Vect (a plain crossing, Lecture 7) it reads

 (8.1)

that is, $\Delta \circ \mu = (\mu \otimes \mu) \circ (\mathrm{id} \otimes P \otimes \mathrm{id}) \circ (\Delta \otimes \Delta)$.

Theorem 8.2 (Bialgebras and monoidal categories). *Let H be an algebra with algebra maps $\Delta\colon H \to H \otimes H$ and $\varepsilon_H\colon H \to \mathbb{C}$. If $(H, \Delta, \varepsilon_H)$ is a bialgebra, then* $\mathrm{Rep}(H)$ *is monoidal, with h acting on $V \otimes W$ by $\Delta(h)$ and on $\mathbb{C}$ by $\varepsilon_H(h)$, the associator and unitors those of* Vect*, and the forgetful functor to* Vect *strict monoidal. Conversely · sketched, for finite-dimensional H, if this recipe makes* $\mathrm{Rep}(H)$ *monoidal with these associators and unitors, then Δ is coassociative and ε_H a counit.*

Proof. Δ is an algebra map, so $\Delta(h)$ defines an action on $V \otimes W$; likewise ε_H on $\mathbb{C}$. If f and g intertwine, so does $f \otimes g$, since $(f \otimes g)\Delta(h) = \Delta(h)(f \otimes g)$ term by term. The associator of Vect, $(u \otimes v) \otimes w \mapsto u \otimes (v \otimes w)$, intertwines the actions $(\Delta \otimes \mathrm{id})\Delta(h)$ and $(\mathrm{id} \otimes \Delta)\Delta(h)$, which coassociativity makes equal; the unitors intertwine by the counit axiom. The pentagon and triangle hold because they hold in Vect. For the converse, apply the condition to $V = W = H$ itself, acting on itself, and evaluate at $1 \otimes 1 \otimes 1$ (Kassel Prop. XI.3.1). ∎

Example 8.3 (Three bialgebras).

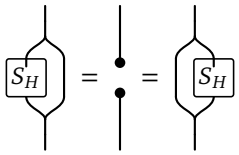
- The group algebra $\mathbb{C}[G]$ with $\Delta(g) = g \otimes g$, $\varepsilon_H(g) = 1$: g acts on $V \otimes W$ by $\rho_V(g) \otimes \rho_W(g)$, and $\mathrm{Rep}(\mathbb{C}[G])$ is $\mathrm{Rep}(G)$ with its tensor product.
- The functions $\mathbb{C}^G$, with basis δ_g, pointwise product and $\Delta(\delta_g) = \sum_{hk=g} \delta_h \otimes \delta_k$: a module is a G-graded vector space, and $\mathrm{Rep}(\mathbb{C}^G)$ is Vec_G; here $\varepsilon_H(\delta_g) = \delta_{g,1}$.
- The enveloping algebra $U(\mathfrak{g})$ with $\Delta(x) = x \otimes 1 + 1 \otimes x$ and $\varepsilon_H(x) = 0$ for $x \in \mathfrak{g}$: this is *QM-I*'s total angular momentum, $J_a = J_a^{(1)} \otimes 1 + 1 \otimes J_a^{(2)}$ (eq. (13.1)), adding angular momenta. (This J is angular momentum, not the structure map of a monoidal functor in Lecture 3.)

The coproduct is the rule for combining two systems. The diagonal $g \otimes g$ says a symmetry acts on both; the primitive $x \otimes 1 + 1 \otimes x$ says a generator, an infinitesimal symmetry, adds.

8.2 Antipodes give duals

Definition 8.4 (Hopf algebra). A *Hopf algebra* is a bialgebra with an invertible linear map $S_H\colon H \to H$, the *antipode*, such that $\mu(S_H \otimes \mathrm{id})\Delta = \iota\,\varepsilon_H = \mu(\mathrm{id} \otimes S_H)\Delta$.

In pictures, with the counit a strand that ends and the unit one that begins:



(8.2)

Proposition 8.5 (Antipodes give duals · *proved given Kassel Thm. III.3.4*). *If H is a Hopf algebra and V an H-module, then V^* with $(h \cdot f)(v) = f(S_H(h)v)$ is an H-module, and the evaluation and coevaluation of* Vect *are H-linear: a left dual. With S_H^{-1} in place of S_H, the same space is a right dual ${}^*V$. So* $\mathrm{Rep}(H)$ *is rigid.*

Proof. S_H reverses products, $S_H(xy) = S_H(y)S_H(x)$ (Kassel Thm. III.3.4), so the formula is an action. For $\mathrm{ev}(f \otimes v) = f(v)$, write $\Delta(h) = \sum h_{(1)} \otimes h_{(2)}$; then $\mathrm{ev}(\Delta(h)(f \otimes v)) = \sum f(S_H(h_{(1)})h_{(2)}v) = \varepsilon_H(h)f(v)$, by the antipode axiom: h acts on $\mathbb{C}$ by $\varepsilon_H(h)$, as it should. The coevaluation is checked in the same way with the other half of the axiom (Kassel

Props. III.5.2–5.3). The right dual uses S_H^{-1}, which satisfies the antipode identities for the opposite products, and is checked the same way with the evaluation $V \otimes {}^*V \to \mathbb{C}$ (EGNO Cor. 5.3.7; this is where invertibility is used). ∎

For $\mathbb{C}[G]$, $S_H(g) = g^{-1}$, and V^* is the contragredient representation of Lecture 4. For $U(\mathfrak{g})$, $S_H(x) = -x$: x acts on V^* by $-x^T$, which is why the dual of a spin has the opposite weights. For $\mathbb{C}^G$, $S_H(\delta_g) = \delta_{g^{-1}}$, and the dual of a vector of degree g has degree g^{-1}, as in Vec_G.

8.3 Quasitriangularity gives a braiding

The flip $P\colon v \otimes w \mapsto w \otimes v$ is H-linear for $\mathbb{C}[G]$ and $U(\mathfrak{g})$, because their coproducts are symmetric. In general P intertwines Δ with the opposite coproduct $\Delta^{\mathrm{op}} = P\Delta$, and a braiding needs a correction.

Definition 8.6 (Quasitriangular Hopf algebra)**.** A *universal R-matrix* is an invertible $R \in H \otimes H$ with

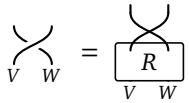
$$R\,\Delta(x)\,R^{-1} = \Delta^{\mathrm{op}}(x) \quad (x \in H),$$
$$(\Delta \otimes \mathrm{id})(R) = R_{13}R_{23},$$
$$(\mathrm{id} \otimes \Delta)(R) = R_{13}R_{12},$$

where $R_{12} = R \otimes 1$, and so on. Kassel calls (H, R) *braided* (Def. VIII.2.2).

Theorem 8.7 (R gives a braiding)**.** *For a universal R-matrix, $c_{V,W} = P \circ R_{V,W}\colon V \otimes W \to W \otimes V$ is a braiding on* $\mathrm{Rep}(H)$*, and R satisfies*

$$R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}.$$

In pictures, the crossing is a box followed by the plain flip:



(8.3)

Proof. c is H-linear: $c\,\Delta(x) = P\,R\,\Delta(x) = P\,\Delta^{\mathrm{op}}(x)\,R = \Delta(x)\,P\,R$, since $P\Delta^{\mathrm{op}}(x) = \Delta(x)P$. It is natural because R acts through the actions, which intertwiners commute with. For the second hexagon of Definition 7.2, R acts on $(U \otimes V) \otimes W$ by $(\Delta \otimes \mathrm{id})(R) = R_{13}R_{23}$, and $P_{U\otimes V,W} = (P_{U,W} \otimes \mathrm{id})(\mathrm{id} \otimes P_{V,W})$; moving $\mathrm{id} \otimes P_{V,W}$ past R_{13} turns it into R_{12}, so $c_{U\otimes V,W} = (c_{U,W} \otimes \mathrm{id})(\mathrm{id} \otimes c_{V,W})$. The first follows in the same way from $(\mathrm{id} \otimes \Delta)(R) = R_{13}R_{12}$. Finally the braid relation of Theorem 7.4 for $c = P \circ R$ is, after moving the flips to one side, the displayed equation (Kassel Prop. VIII.3.1, Thm. VIII.2.4). ∎

So R is not the braiding: $c = P \circ R$, Kassel's $\tau \circ R$. This is the *Integrability* equation $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$, now with no spectral parameter, and c satisfies the braid form of Lecture 7. For $\mathbb{C}[G]$ and $U(\mathfrak{g})$, $R = 1 \otimes 1$ and $c = P$: symmetric. But the same Hopf algebra can carry another R. On $\mathbb{C}[\mathbb{Z}_2] = \mathrm{span}(1, g)$,

$$R = \tfrac{1}{2}\,(1 \otimes 1 + 1 \otimes g + g \otimes 1 - g \otimes g)$$

satisfies the axioms and acts on $V_a \otimes V_b$, where g acts by $(-1)^a$, by $(-1)^{ab}$: the braiding of sVec (Lecture 7). A braiding is structure, and here it is the choice of R.

The algebraic maps have distinct jobs in the representation category. The formulas above give the following dictionary.

algebraic data	operation on modules
μ, ι	Specify the unital and associative laws respected by a module action.
Δ	Define the action on a tensor product of two modules.
ε_H	Define the action on the tensor unit $\mathbb{C}$.
S_H, S_H^{-1}	Define the actions on left and right duals.
R	Correct the flip to the braiding $c = P \circ R$.

Coassociativity makes the ordinary associator a module map. The antipode and universal R-matrix add duality and braiding to this tensor product; neither replaces the coproduct.

8.4 Reconstruction, and what it forbids

A *fibre functor* is a faithful $\mathbb{C}$-linear monoidal functor to Vect, like the forgetful functor of $\mathrm{Rep}(H)$ (EGNO Def. 5.1.1; on a semisimple category the exactness EGNO asks for is automatic).

Theorem 8.8 (Reconstruction · *cited*)**.** *A fusion category with a fibre functor is equivalent to* $\mathrm{Rep}(H)$ *for a finite-dimensional semisimple Hopf algebra* H*, built from the endomorphisms of the functor (EGNO Thm. 5.3.12, applied to fusion categories).*

A fibre functor preserves Frobenius–Perron dimensions (EGNO Prop. 4.5.7), and in Vect they are dimensions, integers. Fibonacci has $\mathrm{FPdim}(\tau) = \varphi$ and Ising $\mathrm{FPdim}(\sigma) = \sqrt{2}$, so neither has a fibre functor: no Hopf algebra has Fibonacci or Ising as its representations. Integer FPdims are necessary, not sufficient: Vec_G^{ω} with ω not a coboundary has every FPdim 1 and no fibre functor (EGNO Ex. 5.1.3), since a fibre functor would trivialise ω. The obstruction is FPdim, not the pivotal d: Vec_G for $G = \mathbb{Z}_3$ has the fibre functor that forgets the grading, yet a pivotal structure with $d = e^{2\pi i/3}$ (Lecture 4). The Fibonacci category does come from a quantum group, but as a subcategory of a quotient of its representations at a root of unity, the subject of Lecture 9.

8.5 $U_q(\mathfrak{sl}_2)$ and the XXZ dictionary

Fix $q \in \mathbb{C}^\times$ that is not a root of unity. The algebra $U_q(\mathfrak{sl}_2)$ is generated by E, F, $K^{\pm 1}$ with

$$KEK^{-1} = q^2 E, \qquad KFK^{-1} = q^{-2} F,$$
$$EF - FE = \frac{K - K^{-1}}{q - q^{-1}},$$

and it is a Hopf algebra with

$$\Delta(E) = K \otimes E + E \otimes 1,$$
$$\Delta(F) = 1 \otimes F + F \otimes K^{-1},$$
$$\Delta(K) = K \otimes K,$$
$$S_H(E) = -K^{-1}E, \quad S_H(F) = -FK,$$
$$S_H(K) = K^{-1},$$

and $\varepsilon_H(E) = \varepsilon_H(F) = 0$, $\varepsilon_H(K) = 1$ · *cited* (Kassel Prop. VII.1.1). This Δ is the opposite of Kassel's and EGNO's (Thm. 5.6.2), and S_H is the inverse of Kassel's antipode, as it must be for the opposite coproduct; the course uses it throughout. We use only the modules $V_1^{\otimes n}$ and their direct summands: on them K acts by integer powers of q, and the braiding below exists. (On the one-dimensional module with $K = -1$ and $E = F = 0$, which also satisfies the relations, no braiding is compatible with V_1.) On $V_1 = \mathbb{C}^2$, with basis $e_1 = \uparrow$, $e_2 = \downarrow$,

$$K = \begin{pmatrix} q & 0 \\ 0 & q^{-1} \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

As $q \to 1$, E, F and $(K - K^{-1})/(q - q^{-1})$ become the e, f, h of $\mathfrak{sl}_2$ (the E, F, H of *Symmetries*, L12), and Δ becomes the primitive coproduct. This is a limit, not substitution at $q = 1$: the presentation above has denominator $q - q^{-1}$.

The construction of the universal R-matrix uses a formal parameter. The ring $\mathbb{C}[[\hbar]]$ consists of formal power series $\sum_{n \geq 0} a_n \hbar^n$, with $a_n \in \mathbb{C}$. Here $\hbar$ is an indeterminate: no numerical value or convergence is assumed. In particular, $q = e^{\hbar/2}$ means its formal exponential series. The numerical parameter q fixed above serves a different role. On the two-spin space the sum involving powers of E and F truncates. The Cartan factor gives diagonal powers of a chosen root $A^2 = q$, yielding the finite matrix below. We use that matrix directly, rather than substituting a complex number into an arbitrary formal power series.

Theorem 8.9 (The braiding on two spins · *cited; proved on* $V_1 \otimes V_1$)**.** *The formal quantum group $U_\hbar(\mathfrak{sl}_2)$, over $\mathbb{C}[[\hbar]]$ with $q = e^{\hbar/2}$, has a universal R-matrix (Kassel Thm. XVII.4.2, his h), and with its legs exchanged it serves our Δ (his Lemma VIII.2.3); it acts on the modules $V_1^{\otimes n}$ and their summands, giving a braiding there (EGNO Ex. 8.3.13). On $V_1 \otimes V_1$, with $q = A^2$, it is $R = \operatorname{diag}(A, A^{-1}, A^{-1}, A)\,(1 + (q - q^{-1})F \otimes E)$, and the braiding is $c = P \circ R = A^{-1} B_q$, with*

$$B_q = \begin{pmatrix} q & 0 & 0 & 0 \\ 0 & q - q^{-1} & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & q \end{pmatrix}$$

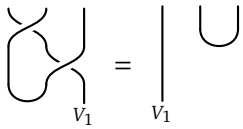
in the basis $\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow$.

Proof on $V_1 \otimes V_1$. Kassel's series, legs exchanged, is $q^{h \otimes h/2} \sum_n c_n F^n \otimes E^n$ with $c_0 = 1$, $c_1 = q - q^{-1}$; on V_1, $E^2 = F^2 = 0$, so only $n \leq 1$ survives, and $q^{h \otimes h/2} = \operatorname{diag}(A, A^{-1}, A^{-1}, A)$ since $h = \operatorname{diag}(1, -1)$. Direct computation then gives $R\,\Delta(x)R^{-1} = \Delta^{\mathrm{op}}(x)$ for $x = E, F, K$, the two coproduct conditions on $V_1^{\otimes 3}$, and $PR = A^{-1}B_q$ (`scripts/check-l8-algebra.py`). ∎

B_q satisfies the braid relation and the *Hecke relation* $(B_q - q)(B_q + q^{-1}) = 0$: its eigenvalues are q on the triplet and $-q^{-1}$ on the *q-singlet* $v = e_1 \otimes e_2 - q\, e_2 \otimes e_1$, the cup of Lecture 9. So c has eigenvalues A and $-A^{-3}$, and $c^2 \neq 1$: a braiding that is not a symmetry. At $q = 1$, $B_q = P$. The q-singlet is invariant: $\Delta(E)v = \Delta(F)v = 0$ and $\Delta(K)v = v$, so $v \colon \mathbb{C} \to V_1 \otimes V_1$ is a module map, and V_1 is its own dual. The operator $U = B_q - q$, the projection onto the singlet up to a factor, satisfies $U^2 = -(q + q^{-1})U$ (Sheet 8); it is the generator of the Temperley–Lieb algebra of Lecture 9.

Naturality of the braiding with the cup $v \colon \mathbb{C} \to V_1 \otimes V_1$ demands that a strand pass the

cup as if it were not there, $c_1 c_2(v \otimes \mathrm{id}) = \mathrm{id} \otimes v$:



(8.4)

Write $e_{ijk} = e_i \otimes e_j \otimes e_k$. Applying the displayed matrix B_q on the adjacent factors gives

$$\begin{aligned} B_1 B_2(v \otimes e_1) &= q e_{112} - q^2 e_{121} \\ &= q(e_1 \otimes v), \\ B_1 B_2(v \otimes e_2) &= q e_{212} - q^2 e_{221} \\ &= q(e_2 \otimes v). \end{aligned}$$

Thus for a scalar $z \in \mathbb{C}^\times$ and a crossing $c = zB_q$, cup naturality requires $z^2 q = 1$. Choosing a root A with $A^2 = q$ gives $z = A^{-1}$; the other sign corresponds to the other root. The Hecke-normalised B_q solves the braid relation, but it fails this naturality condition.

The following XXZ comparison is optional context for readers of *Integrability*. No later argument uses the spectral parameter. That course's eq. (15.22) gives a trigonometric R-matrix with spectral parameter λ and anisotropy $\Delta_{\mathrm{XXZ}} = \cos\eta$. The subscript distinguishes this number from the coproduct map Δ. Its spectral limits give B_q and its inverse:

Integrability	this course
anisotropy $\Delta_{\mathrm{XXZ}} = \cos\eta$	$q = e^{i\eta} = A^2$, $\Delta_{\mathrm{XXZ}} = (q + q^{-1})/2$
$R(\lambda)$, after the gauge $\mathrm{diag}(e^{\lambda/2}, e^{-\lambda/2})$ on factor 1	the same, $\tilde{R}(\lambda)$
$\lim_{\lambda\to-\infty}(-2e^{\lambda})P\tilde{R}(\lambda)$	B_q, the over-crossing up to A
$\lim_{\lambda\to+\infty}(2e^{-\lambda})P\tilde{R}(\lambda)$	B_q^{-1}, the under-crossing up to A^{-1}
$\eta \to 0$	$q \to 1$, $B_q \to P$

Sheet 8 derives the displayed spectral limits, including the gauge change and scalar factors. The two limits give the two crossings, with the categorical normalisation $c = A^{-1}B_q$ fixed above.

Summary.

- *Handed on:* a coproduct combines two systems; a bialgebra makes $\mathrm{Rep}(H)$ monoidal (Theorem 8.2), an antipode makes it rigid (Proposition 8.5), a universal R-matrix makes it braided, with $c = P \circ R$ (Theorem 8.7).
- *Reconstruction:* a fibre functor makes a fusion category $\mathrm{Rep}(H)$ (Theorem 8.8); integer FPdims are necessary, so Fibonacci and Ising are not $\mathrm{Rep}(H)$.
- *The quantum group:* on two spins, $c = A^{-1}B_q$ with $(B_q - q)(B_q + q^{-1}) = 0$ (Theorem 8.9); B_q is, after a change of gauge, the braid limit of the XXZ R-matrix.
- *Next:* Lecture 9 turns B_q into the Temperley–Lieb and Kauffman-bracket calculus, adds the twist, and builds $\mathrm{SU}(2)_k$.

References

Kassel, *Quantum Groups*: algebras, coalgebras, bialgebras and antipodes (Thm. III.3.4), §§III.1–III.3, pp. 39–52; modules and their duals, Props. III.5.1–III.5.3, pp. 58–60; $U(\mathfrak{g})$,

Prop. V.2.4, p. 97; $U_q(\mathfrak{sl}_2)$, Def. VI.1.1, p. 122, and its Hopf structure, §VII.1, p. 140; braided bialgebras, Def. VIII.2.2, Lemma VIII.2.3 and Thm. VIII.2.4, pp. 173–175, and their braidings, Prop. VIII.3.1, p. 178; Prop. XI.3.1, p. 285; the universal R, Thm. XVII.4.2, p. 413. Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: fibre functors, Def. 5.1.1, p. 91; reconstruction, Thm. 5.3.12, p. 96; duals, Cor. 5.3.7, p. 95; $U_q(\mathfrak{sl}_2)$, §5.6, p. 101; quasitriangular Hopf algebras, Def. 8.3.1, p. 198; the $U_q(\mathfrak{sl}_2)$ braiding, Ex. 8.3.13, p. 200; FPdim preserved, Prop. 4.5.7, p. 73. Heunen and Vicary, *Categories for Quantum Theory*: monoids and comonoids, Defs. 4.1, 4.3, pp. 128–129; Hopf algebras, Defs. 6.20, 6.31, pp. 205–210. Majid, *A Quantum Groups Primer*, Defs. 1.1–1.3 and 5.1, pp. 2–4, 29.

Lecture 9

Ribbon categories, Temperley–Lieb and $\mathrm{SU}(2)_k$

Objectives. *Lecture 7 found the topological spin as a closed curl, and Lecture 8 the matrix $c = A^{-1}B_q$ of the braiding on two spins. This lecture puts the two together. A twist makes a braided category draw ribbons, not strings, and then every framed knot has a value. On tensor powers of one spin the calculus is the Temperley–Lieb algebra, the value is the Kauffman bracket, and its projectors are the higher spins. At a root of unity one projector vanishes in every closed picture, and discarding it leaves a finite category: $\mathrm{SU}(2)_k$, the source of the Ising-like and Fibonacci-like anyons of Lecture 10.*

By the end of this lecture you will be able to:

- Define a twist and a ribbon category, and show that the curl of Lecture 7 satisfies the twist axiom.
- Compute with the Temperley–Lieb skein relation, and prove that the Kauffman bracket is invariant under Reidemeister II and changes by $-A^{\pm 3}$ under Reidemeister I.
- Construct the Jones–Wenzl projector p_2 and compute traces and twists of coloured strands.
- Explain how $\mathrm{SU}(2)_k$ arises at $A = e^{i\pi/(2(k+2))}$ by discarding negligible morphisms and flipping the pivotal structure.

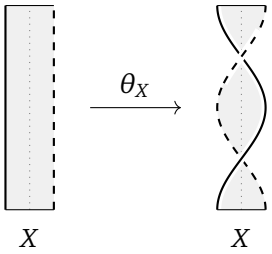
9.1 Twists and ribbon categories

Definition 9.1 (Twist, ribbon category)**.** A *twist* on a braided rigid category is a natural isomorphism $\theta_X : X \to X$ with $\theta_1 = \mathrm{id}$ and

$$\theta_{X\otimes Y} = c_{Y,X}\, c_{X,Y}\, (\theta_X \otimes \theta_Y).$$

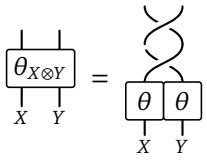
It is a *ribbon structure* if moreover $\theta_{X^*} = (\theta_X)^*$ (EGNO Def. 8.10.1; Kassel Def. XIV.3.2).

A ribbon carries a transverse direction as well as a centreline. The following band has one solid edge and one dashed edge, so that a full turn can be followed even when its centreline has not moved.



The untwisted band and the band with one full twist have the same dotted centreline. The marked edges distinguish their framings. The two edge crossings in the projection record one full turn of the band; they are not crossings of two independent object strands. The morphism θ_X records this turn, using the positive-twist convention of (9.2). Its mirror represents θ_X^{-1}.

The axiom is the picture of a ribbon, or a belt: a full turn of a belt holding two strands turns each strand and also wraps them once around each other,



(9.1)

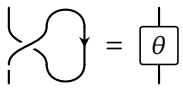
For a simple object θ_a is a number, and $\theta_{a\otimes b}$ acts on the channel c by θ_c; so the axiom is the ribbon relation $R_c^{ab}R_c^{ba} = \theta_c/(\theta_a\theta_b)$ that Lecture 7 cited.

Theorem 9.2 (The curl is a twist · *proved given coherence*)**.** *In a braided pivotal category, the curl of Lecture 7, the right partial trace of $c_{X,X}$, is a twist. In a fusion category with its canonical spherical structure, its value on a simple object is the topological spin θ_a of Theorem 7.7.*

Proof. Naturality: boxes slide through crossings, cups and caps. Invertibility: the curl drawn with the inverse crossing and the cap on the other side is its inverse, by RII and a snake. For $X \otimes Y$, draw the curl of the pair as a belt: the crossing $c_{X\otimes Y,X\otimes Y}$ is four crossings by the hexagons, and closing the right pair of strands with the nested cup and cap of Proposition 4.3 leaves a curl on each strand and the two crossings of $c_{Y,X}c_{X,Y}$ between them, which is the axiom. The value on a simple object is Theorem 7.7, which assumes the canonical structure. ∎

In another pivotal structure the curl can differ by a sign (§ 9.5). In a braided fusion category the curl is a ribbon structure exactly when the pivotal structure is spherical (EGNO Prop. 8.10.12); beyond fusion categories this can fail.

So framing matters. A closed curl is not a straight strand, as it would be in a plane drawing of a knot (Reidemeister I); with a twist, *framed* RI holds:



(9.2)

The strands of a ribbon category are thin bands lying flat in the page (blackboard framing), and a curl puts a full twist into the band.

9.2 Tangles and the Reshetikhin–Turaev functor

A *framed tangle* is a piece of a framed oriented link in a slab, with endpoints on the bottom and top; up to isotopy, tangles form a ribbon category under stacking and juxtaposition.

Theorem 9.3 (Shum; Reshetikhin–Turaev · *cited*)**.** *Framed oriented tangles form the free ribbon category on one object: for every object V of a strict ribbon category $\mathcal{C}$ there is a unique strict ribbon functor from tangles to $\mathcal{C}$ sending an upward strand to V (Kassel Thm. XIV.5.1).*

A framed link is a tangle from nothing to nothing, so its image is an endomorphism of 1, a number when $\mathrm{End}(1) = \mathbb{C}$, as in a fusion category: a link invariant for every object of such a ribbon category. The pictures of these lectures are honest because of this theorem.

9.3 Temperley–Lieb and the Kauffman bracket

Take $V = V_1$ of $U_q(\mathfrak{sl}_2)$, $q = A^2$, with the cup $v = e_1 \otimes e_2 - q\, e_2 \otimes e_1$ of Lecture 8. The cap $u\colon V \otimes V \to \mathbb{C}$ is given by

$$u(e_1 \otimes e_1) = u(e_2 \otimes e_2) = 0,$$
$$u(e_1 \otimes e_2) = -q^{-1}, \qquad u(e_2 \otimes e_1) = 1.$$

Both snakes are straight, as in Lecture 4. The contraction $u(v) = -q^{-1} - q$ gives the closed loop:

$$\bigcirc \;=\; \delta = -A^2 - A^{-2} = -[2]_q, \tag{9.3}$$

where $[n]_q = (q^n - q^{-n})/(q - q^{-1})$; at $A = \pm 1$ it is the Penrose loop -2 of Lecture 4. Write $U = v \circ u$, a cap followed by a cup. Then $U^2 = \delta U$, and on three strands $U_1U_2U_1 = U_1$, $U_2U_1U_2 = U_2$: the *Temperley–Lieb* relations. The *Temperley–Lieb algebra* $\mathrm{TL}_n(\delta)$ has generators $U_1, \dots, U_{n-1}$, these relations, and $U_iU_j = U_jU_i$ for $|i - j| \geq 2$; by Kauffman's theorem it is the algebra of the diagrams $n \to n$ below (Kauffman–Lins p. 9).

Definition 9.4 (The Temperley–Lieb category)**.** $\mathrm{TL}(\delta)$ has objects $n = 0, 1, 2, \dots$ and as morphisms $n \to m$ the linear combinations of planar diagrams of non-crossing strands joining n points below to m above; composition stacks diagrams and replaces each closed loop by δ.

Theorem 9.5 (Temperley–Lieb acts on spin $\frac{1}{2}$)**.** *With $\delta = -[2]_q$, $U_i \mapsto \mathrm{id} \otimes U \otimes \mathrm{id}$ (acting on strands $i, i+1$) defines a representation of $\mathrm{TL}_n(\delta)$ on $V_1^{\otimes n}$ by U_q-module maps.*

Proof. $U = v \circ u = B_q - q$ (multiply out) is a module map because $B_q = Ac$ is (Lecture 8). Far-apart generators act on different factors and commute. The relation $U^2 = \delta U$ follows from $u \circ v = \delta$. For the adjacent relation, expand each U as $v \circ u$. The two middle composites are the following snakes:



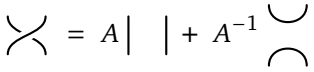
$S_L = (u \otimes \mathrm{id})(\mathrm{id} \otimes v)$ $\qquad$ $S_R = (\mathrm{id} \otimes u)(v \otimes \mathrm{id})$

Both S_L and S_R equal id_V, by the straight-snake identities for u and v. Associativity of composition and interchange give

$$\begin{aligned} U_1U_2U_1 &= (v\otimes \mathrm{id})\, S_L\, S_R\, (u\otimes \mathrm{id}) \\ &= (v\otimes \mathrm{id})(u\otimes \mathrm{id}) = U_1. \end{aligned}$$

Reflecting this calculation left to right gives $U_2U_1U_2 = U_2$. Thus all the defining TL relations hold locally on the tensor factors. ∎

The braiding is then a combination of planar diagrams: from $c = A^{-1}B_q$ and $B_q = q + U$, the *skein relation*



(9.4)

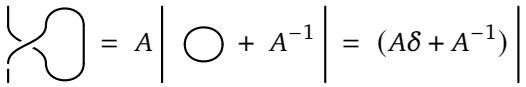
and the inverse crossing exchanges A and A^{-1}. This is the Kauffman bracket's rule as in Simon (Fig. 2.3) and Kauffman–Lins (p. 5); many references, and mirror images, exchange A and A^{-1}.

Theorem 9.6 (The Kauffman bracket · *sketched; proved for RI and RII*)**.** *Evaluating a link diagram by* (9.4) *and* (9.3)*, the value $\langle L\rangle$ is unchanged by Reidemeister II and III, and a curl multiplies it by $-A^3$, or $-A^{-3}$ for the opposite curl.*

Proof. RII: a crossing followed by its inverse is

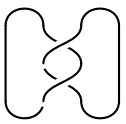
$$\begin{aligned} &(A + A^{-1}U)(A^{-1} + AU) \\ &= 1 + (A^2 + A^{-2})U + U^2 \\ &= 1 + (A^2 + A^{-2} + \delta)U = 1. \end{aligned}$$

RI: expanding the crossing of a curl gives



and $A\delta + A^{-1} = -A^3$. RIII is the braid relation of Theorem 7.4 for $c = A + A^{-1}U$, which follows from the TL relations alone (Sheet 9). ∎

So $\langle L\rangle$ is an invariant of framed links, and the factor $-A^3$ is the twist θ_{V_1}. For an oriented link, the *writhe* $w(L)$ is the sum of crossing signs. Our /-over crossing of two upward strands has sign +1; reversing one strand reverses the sign. Let O denote the unknot. Then $f_L(A) = (-A^3)^{-w(L)}\langle L\rangle/\langle O\rangle$ is an invariant of unframed links, and at $A = t^{-1/4}$ it is the *Jones polynomial* $V_L(t)$ (Kauffman–Lins Thm. 1). For the Hopf link with two positive crossings,



(9.5)

Indeed, squaring $c = A + A^{-1}U$ and using $U^2 = \delta U$ gives

$$c^2 = A^2 + (2 + A^{-2}\delta)U.$$

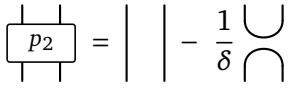
Closing the identity gives two loops. Closing U gives one. Hence

$$\begin{aligned} \langle L\rangle &= A^2\delta^2 + (2 + A^{-2}\delta)\delta \\ &= \delta(-A^4 - A^{-4}). \end{aligned}$$

Thus $\langle L\rangle/\langle O\rangle = -A^4 - A^{-4}$. Sheet 9 does the trefoil.

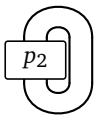
9.4 p_2 and coloured strands

Spin 1 sits inside $V_1 \otimes V_1$ as the image of a projection that kills the cup:



(9.6)

Then $p_2^2 = p_2$ and $Up_2 = 0 = p_2U$, whenever $\delta \neq 0$. Its trace closes both strands:



$$= \delta^2 - \frac{\delta}{\delta} = \delta^2 - 1 = [3]_q, \tag{9.7}$$

the dimension of spin 1, deformed; at $q = 1$, p_2 is the symmetriser $\frac{1}{2}(1 + P)$ of the triplet, the symmetric part of $\mathbb{C}^2 \otimes \mathbb{C}^2$ (*Symmetries* Ex. 14.13).

The image of p_2 is an ordinary subspace in the representation model. For example, at $q = 1$ the matrix acts by

$$p_2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} x_1 \\ (x_2 + x_3)/2 \\ (x_2 + x_3)/2 \\ x_4 \end{pmatrix}.$$

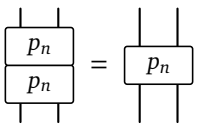
Its image is the triplet, spanned by $e_1 \otimes e_1$, $e_1 \otimes e_2 + e_2 \otimes e_1$ and $e_2 \otimes e_2$. Applying p_2 again fixes every vector in this image. Thus the path from projector to object is: the endomorphism p_2 of two strands selects its image, and the coloured strand $(2, p_2)$ names that image as an object in its own right. The original TL category has only the objects n; its idempotent completion adds a name for each such selected part.

Theorem 9.7 (Jones–Wenzl projectors · *cited; proved for p_2*)**.** *On $n+1$ strands, starting from $p_1 = \mathrm{id}$,*

$$p_{n+1} = p_n \otimes 1 + \frac{[n]_q}{[n+1]_q} (p_n \otimes 1)\, U_n\, (p_n \otimes 1)$$

defines idempotents killed by every U_i, for as long as $[m]_q \neq 0$ for $m \leq n+1$ (Kauffman–Lins Def. 1 and Lemma 1; for p_2 it is (9.6)). Their traces are $\operatorname{tr} p_n = (-1)^n [n+1]_q$ (ibid., p. 13).

In pictures, a projector absorbs a second copy of itself,



(9.8)

and is killed by any cap across two of its strands:



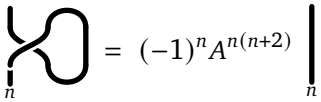
(9.9)

A pair (n, p_n), n strands carrying the idempotent p_n, is a *coloured strand*, drawn as one strand labelled n. It is an object of the *idempotent completion*: objects (X, p) with $p^2 = p$, morphisms $f\colon X \to Y$ with $f = p_Y f p_X$. Adding formal direct sums of such pairs gives

the *additive* idempotent completion. This is how spin $n/2$ enters the diagram calculus. A coloured strand has a twist, from the full twist of its n strands,

$$\theta_n = (-1)^n A^{n(n+2)}$$

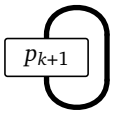
(Kauffman–Lins Prop. 6), and $\theta_1 = -A^3$. In pictures, a curl on a coloured strand is that number:



(9.10)

9.5 Roots of unity, the negligible quotient, the pivotal flip

Fix $k \geq 1$ and $A = e^{i\pi/(2(k+2))}$, so $q = A^2 = e^{i\pi/(k+2)}$. Then $[k+2]_q = 0$ while $[m]_q \neq 0$ for $m \leq k+1$: the recursion reaches p_{k+1}, and $\operatorname{tr} p_{k+1} = (-1)^{k+1}[k+2]_q = 0$:



(9.11)

Definition 9.8 (Negligible). A morphism $f\colon X \to Y$ is *negligible* if $\operatorname{tr}(gf) = 0$ for every $g\colon Y \to X$.

The negligible morphisms form an ideal: closed under composition and $\otimes$ with anything (EGNO Exer. 8.18.9(ii)). By Kauffman–Lins (Lemma 5), p_{k+1} closes to 0 in every network: it is negligible.

A zero trace is only one closure test. For instance, in $\mathrm{Vect}_{\mathrm{fd}}$ the map $f = \operatorname{diag}(1,-1)$ has trace zero, but $g = \operatorname{diag}(1,0)$ gives $\operatorname{tr}(gf) = 1$. So this f is not negligible. The cited result about p_{k+1} tests every closure, not just $\operatorname{tr}(p_{k+1})$.

The quotient at $k = 1$. Here $A = e^{i\pi/6}$, $q = e^{i\pi/3}$ and $\delta = -1$. The projector is $p_2 = 1 + U$, a nonzero diagrammatic endomorphism before the quotient. Every endomorphism of two strands is $a\,1 + bU$. The relations $Up_2 = 0$ and $\operatorname{tr}(p_2) = \delta^2 - 1 = 0$ give

$$\operatorname{tr}\big((a\,1 + bU)p_2\big) = 0 \qquad (a, b \in \mathbb{C}).$$

Thus p_2 is negligible in this case by the definition itself. In the idempotent completion it is the identity of the object $(2, p_2)$. Setting that identity to zero makes this object a zero object in the quotient. It does not assert that the matrix of p_2 was zero before quotienting.

The complementary projector $1 - p_2 = -U$ remains. The cup $v\colon 0 \to 2$ and the cap $-u\colon 2 \to 0$ identify this surviving part with the tensor unit, because in the quotient

$$(-u)v = 1_0, \qquad v(-u) = -U = 1_2.$$

Write s for the surviving one-strand colour 1. There are two simple objects, the unit and s, with $s \otimes s \cong 1$, as the general theorem below confirms. The quotient removes a channel. A subsequent pivotal change alters its closing and twist conventions: the raw values $d_s = -1$ and $\theta_s = -A^3 = -i$ become $d_s = 1$ and $\theta_s = i$ after multiplication by $(-1)^1$. These are different operations.

Theorem 9.9 (SU(2)$_k$ · *cited*)**.** *The additive idempotent completion of* TL(δ) *at* $A = e^{i\pi/(2(k+2))}$*, modulo negligible morphisms, is a semisimple ribbon category with simple objects the coloured strands* $0, 1, \dots, k$*, fusing by the truncated Clebsch–Gordan rule of Lecture 5 (Kauffman–Lins Lemmas 4–5; Simon Ch. 22, Tables 22.1–22.3). As a tensor category it is EGNO's* $C_k(q)$ *(Ex. 8.18.5): the quotient of the tilting* U_q*-modules by the negligible ones (Bakalov–Kirillov Thm. 3.3.20). Here "tilting modules" names that cited alternative construction; our construction uses the TL diagrams and projectors above. In the pivotal structure of* TL*,*

$$d_n = (-1)^n [n+1]_q,$$
$$\theta_n = (-1)^n A^{n(n+2)}.$$

These d_n are negative for odd n: the Penrose sign of Lecture 4, now on every half-integer spin. The cure is the one Lecture 4 found, a different pivotal structure. Multiplying the pivotal structure by $(-1)^n$ on the colour n is allowed, since $(-1)^n$ is a monoidal function of n (the truncated rule preserves parity); it multiplies d_n and θ_n by $(-1)^n$ (Bakalov–Kirillov Ex. 2.2.8; Simon (15.1)). With $j = n/2$:

$$d_j = [2j+1]_q > 0,$$
$$[2j+1]_q = \frac{\sin\big((2j+1)\pi/(k+2)\big)}{\sin\big(\pi/(k+2)\big)},$$
$$\theta_j = e^{2\pi i\, j(j+1)/(k+2)}.$$

At $k = 2$ the simple objects are 0, $\frac{1}{2}$ and 1, the coloured strands $0, 1, 2$; the truncated rule gives $\frac{1}{2} \otimes \frac{1}{2} \cong 0 \oplus 1$, $\frac{1}{2} \otimes 1 \cong \frac{1}{2}$ (since p_3, which would carry spin $\frac{3}{2}$, is negligible) and $1 \otimes 1 \cong 0$. This is the unitary SU(2)$_k$ of Lecture 5, with the dimensions found there as Frobenius–Perron dimensions (Bonderson §5.4). At $k = 1$, $\theta_{1/2} = -A^3 = -i$ in the TL structure and $+i$ after the flip: the semion of Lecture 7. At $k = 2$, $d_{1/2} = \sqrt{2}$ and $\theta_{1/2} = e^{3\pi i/8}$, Kitaev's $\nu = 3$: the $\kappa = -1$ category of Lecture 6, not the course's Ising. At $k = 3$, the integer spins 0 and 1 close under fusion, $1 \otimes 1 = 0 \oplus 1$, with $d_1 = [3]_q = \varphi$ and $\theta_1 = e^{4\pi i/5}$: Fibonacci, with the chirality chosen in Lecture 7. The other choice $-A$ of the root multiplies θ_j by $(-1)^{2j}$, and reverses the sign of the braiding $c = A^{-1}B_q$ on two spins.

Why is the quotient semisimple? Each coloured strand $n \le k$ has non-zero trace, so its identity is not negligible and it stays a simple object; the morphisms that would make the category non-semisimple, through p_{k+1} and beyond, are negligible and are set to zero. The traces that survive are the dimensions above.

Summary.

- *Handed on:* a twist satisfies $\theta_{X\otimes Y} = c_{Y,X}c_{X,Y}(\theta_X \otimes \theta_Y)$, and the curl is one (Theorem 9.2); framed tangles form the free ribbon category (Theorem 9.3).
- *Temperley–Lieb:* on $V_1^{\otimes n}$ the crossing is $A \cdot \mathrm{id} + A^{-1} \cdot (\mathrm{cup} \circ \mathrm{cap})$, the loop is $-A^2 - A^{-2}$, and the Kauffman bracket is a framed link invariant (Theorem 9.6).
- *Projectors:* for $\delta \neq 0$, $p_2 = 1 - U/\delta$. The Jones–Wenzl recursion reaches p_n while $[m]_q \neq 0$ for $1 \le m \le n$, with $\mathrm{tr}\, p_n = (-1)^n[n+1]_q$ and $\theta_n = (-1)^n A^{n(n+2)}$ (Theorem 9.7).
- SU(2)$_k$*:* at $A = e^{i\pi/(2(k+2))}$, p_{k+1} is negligible; the quotient, with the pivotal flip, has $d_j = [2j+1]_q$ and $\theta_j = e^{2\pi i j(j+1)/(k+2)}$ (Theorem 9.9).
- *Next:* Lecture 10 adds the S-matrix and makes these categories modular.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: twists and ribbon structures, Def. 8.10.1 and Rem. 8.10.3, p. 216; ribbon iff spherical, Prop. 8.10.12, p. 220; the pivotal flip, Rem. 8.10.16, p. 221; the categories $C_k(q)$ and negligible morphisms, Ex. 8.18.5 and Exer. 8.18.9, pp. 235–236 (their twist $q^{-m(m+2)/2}$ is the inverse of ours). Kassel, *Quantum Groups*: framed tangles, §X.8, pp. 270–272; ribbon categories, Def. XIV.3.2, p. 349; the universal ribbon category, Thm. XIV.5.1, p. 360. Bakalov and Kirillov, *Lectures on Tensor Categories and Modular Functors*: ribbon categories, Def. 2.2.1 and Ex. 2.2.8, pp. 33–35; Reshetikhin–Turaev, Thm. 2.3.10, p. 42; $\mathrm{SU}(2)_k$, Thm. 3.3.20, p. 67. Kauffman and Lins, *Temperley–Lieb Recoupling Theory*: the bracket, pp. 5–9 and Thm. 1; the Jones–Wenzl recursion, Def. 1 and Lemma 1, pp. 13–14; the root of unity, Lemmas 4–5, pp. 37–38; twisted projectors, Prop. 6, p. 43. Simon, *Topological Quantum*: the skein relation, Fig. 2.3, p. 4; twists, Ch. 15, pp. 199–200; Temperley–Lieb–Jones, Ch. 22, pp. 297–310, and the theories at $A = \pm e^{i\pi/(2(k+2))}$, Tables 22.1–22.3. Bonderson, PhD thesis, §5.4, p. 81.

Block IV

Topological Order

Lecture 10

Modular tensor categories and anyon models

Objectives. *Lectures 5 to 7 and 9 built braided fusion categories with a twist: fusion rules, F- and R-symbols, spins. Which of them describe anyons in a two-dimensional medium? Physically, in a bosonic phase every anyon other than the vacuum must be detectable by braiding something around it. Categorically, this is the statement that one matrix, the Hopf link S, is invertible. From S and the spins come the Verlinde formula, which recovers the fusion rules, a projective representation of $\mathrm{SL}(2,\mathbb{Z})$, and a central charge c modulo 8. This lecture computes all of it for Fibonacci and tabulates it for the other models of the course.*

By the end of this lecture you will be able to:

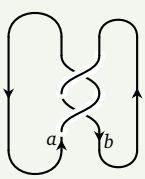
- Compute S and T of a premodular category from its fusion rules, dimensions and spins, and remove a loop encircling a strand.
- Prove that the killing loop projects onto the vacuum when the Müger centre is trivial, and deduce $S^2 = C$ and the criterion for modularity.
- Prove the Verlinde formula and recover fusion rules from S.
- Compute Fibonacci's modular data and central charge, and read c from the Gauss sum for the semion, Ising and the toric code.

10.1 S and T

Definition 10.1 (Premodular category)**.** A *premodular* category is a ribbon fusion category (EGNO Def. 8.13.1): a fusion category with a braiding and a ribbon structure, as in Lecture 9.

As everywhere in the course, the category is unitary with its canonical spherical structure, so $d_a = \mathrm{FPdim}(a) > 0$, $|\theta_a| = 1$, and $\dim \mathcal{C} = \sum_a d_a^2$ as in Lecture 5. A ribbon structure has $\theta_{\bar{a}} = \theta_a$, since $\theta_{a^*} = (\theta_a)^*$ and a scalar is its own transpose.

Definition 10.2 (S- and T-matrices)**.** For simple objects a, b, let $\tilde{S}_{ab}$ be the Hopf link of a and b, oriented oppositely; equivalently of a and $\bar{b}$, both pointing up:

$$\tilde{S}_{ab} \;=\; \text{  } \;=\; \mathrm{Tr}\big(c_{\bar{b},a}\, c_{a,\bar{b}}\big). \tag{10.1}$$

The *S-matrix* is $S_{ab} = (\dim \mathcal{C})^{-1/2}\, \tilde{S}_{ab}$, and the *T-matrix* is $T = \mathrm{diag}(\theta_a)$ (Kitaev eq. (223)).

Proposition 10.3 (S from the data · *proved given coherence*)**.**

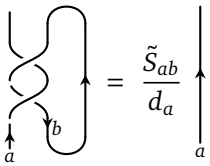
$$\tilde{S}_{ab} = \sum_c N_{a\bar{b}}^{c}\, d_c\, \frac{\theta_c}{\theta_a \theta_b}.$$

In particular $\tilde{S}_{1a} = \tilde{S}_{a1} = d_a$*,* $\tilde{S}$ *is symmetric, and* S *depends only on* N*,* d *and* θ*.*

Proof. Insert completeness (Equation (5.3)) on $a \otimes \bar{b}$ under the double braiding. On the channel c the double braiding is the scalar $R_c^{\bar{b}a} R_c^{a\bar{b}} = \theta_c/(\theta_a\theta_b)$ (Equation (7.8) and the ribbon relation of Lecture 9). Closing $\psi_c^{a\bar{b}} \psi_c^{a\bar{b}\dagger}$ gives the theta graph $\sqrt{d_a d_b d_c}$ (Equation (5.4)), and the completeness coefficient $\sqrt{d_c/(d_a d_b)}$ turns it into d_c. For the symmetry, dualising $a \otimes \bar{b} \to c$ gives $b \otimes \bar{a} \to \bar{c}$, so $N_{a\bar{b}}^{c} = N_{b\bar{a}}^{\bar{c}}$, while $d_{\bar{c}} = d_c$ and $\theta_{\bar{c}} = \theta_c$. ∎

Two conventions differ from this one. EGNO, Bakalov–Kirillov, Müger and Rowell–Wang use the unnormalised $\tilde{S}$. EGNO and Müger also take the Hopf link of a and b with the same orientation, and Bonderson's (2.63) sums over N_{ab}^{c}: either way the result is our $\bar{S}$, the complex conjugate. The two agree for self-dual models, but with $T = \mathrm{diag}(\theta)$ only our S satisfies the relations of § 10.4 in general.

A loop around a strand is a number times the strand, since $\mathrm{End}(a) = \mathbb{C}$; closing the strand shows which number:



(10.2)

Write $\lambda_a(b) = \tilde{S}_{ab}/d_a$ for this number. Loops can carry combinations of labels, linearly.

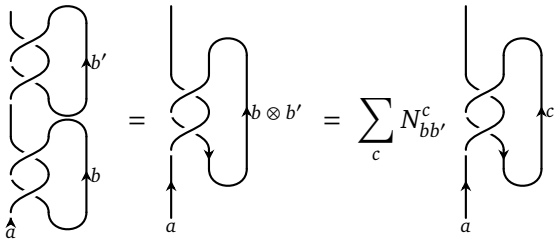
For a numerical preview, Fibonacci gives the following matrix, derived in Example 10.11:

$$\tilde{S} = \begin{pmatrix} 1 & \varphi \\ \varphi & -1 \end{pmatrix}, \qquad S = \frac{\tilde{S}}{\sqrt{2+\varphi}}.$$

An encircling τ-loop therefore acts on the vacuum strand by $\lambda_1(\tau) = \varphi$, but on a τ-strand by $\lambda_\tau(\tau) = -\varphi^{-1}$. These distinct scalars explain how a loop can distinguish an anyon from the vacuum. The entry $S_{\tau\tau}$ itself is not the loop scalar: one must undo the normalisation of S and divide by the dimension of the encircled strand.

Lemma 10.4 (Loops fuse · *proved given coherence*)**.** *For every* a*,* $\lambda_a(1) = 1$ *and* $\lambda_a(b)\,\lambda_a(b') = \sum_c N_{bb'}^{c}\, \lambda_a(c)$*: the map* $b \mapsto \lambda_a(b)$ *is a character of the fusion ring.*

Proof. Two loops b and b' around a are scalars, so their product is the picture with both. Their fusion is shown with the loop orientations of (10.2):



Slide one loop along the strand a until the two run parallel. By the hexagons and the nested cup and cap of Proposition 4.3 they become one loop labelled $b \otimes b'$. Decompose $b \otimes b'$

by completeness. Each vertex slides around the loop, since the double braiding is natural, until it meets its partner; the bubble $\sqrt{d_b d_{b'}/d_c}$ cancels the coefficient, leaving $\sum_c N_{bb'}^c$ loops labelled c. A loop labelled 1 is empty. ∎

10.2 Premodular, modularity, the killing loop

Lecture 7 called x *transparent* if $c_{y,x}c_{x,y} = \mathrm{id}$ for every y, and the transparent objects the *Müger centre*. A transparent x has $\lambda_x(b) = d_b$ for every b: its loops are those of the vacuum, so no loop around it detects it.

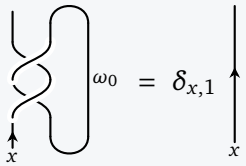
Definition 10.5 (Modular category, anyon model)**.** A premodular category is *modular* if S is invertible (EGNO Def. 8.13.4). A unitary modular category is an *anyon model*: its simple objects are the anyon types, N the fusion rules, and F, R the recoupling and exchange data (Kitaev App. E).

Every gapped phase of bosons in two dimensions, with no symmetry imposed, carries an anyon model, in which only the vacuum is transparent · *physical principle*; this is Kitaev's "braiding nondegeneracy" (pp. 66 and 86). Fermions are excluded: a transparent fermion braids trivially with everything, yet its spin $\theta = -1$ tells it from the vacuum.

The *killing loop* is the loop labelled

$$\omega_0 = \frac{1}{\dim C} \sum_b d_b \, b.$$

Lemma 10.6 (The killing loop · *proved given coherence*)**.** *If the Müger centre of a unitary premodular C is trivial, then for every simple x*



(10.3)

Proof. The loop is the scalar $\Omega_x/\dim C$ with $\Omega_x = \sum_b d_b \lambda_x(b)$. By Lemma 10.4,

$$\begin{aligned}\Omega_x \lambda_x(b') &= \sum_c \Big(\sum_b d_b N_{bb'}^c \Big) \lambda_x(c) \\ &= d_{b'}\, \Omega_x,\end{aligned}$$

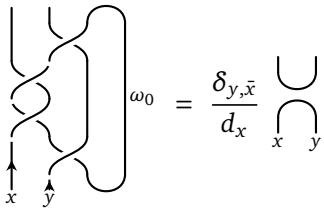
because $N_{bb'}^c = N_{c\bar{b}'}^b$ and $\sum_b N_{c\bar{b}'}^b d_b = d_c d_{b'}$. If $\Omega_x \neq 0$, then $\lambda_x(b') = d_{b'}$ for all b', that is,

$$\begin{aligned}&\sum_c N_{x\bar{b}'}^c d_c \, \frac{\theta_c}{\theta_x \theta_{b'}} \\ &= \sum_c N_{x\bar{b}'}^c d_c .\end{aligned}$$

Each phase has modulus 1 and each $d_c > 0$, so every phase is 1: the double braiding of x with every $\bar{b}'$ is the identity, and x is transparent, hence $x = 1$. Finally $\Omega_1 = \sum_b d_b^2 = \dim C$. ∎

The loop is natural in what it encircles, so around $x \otimes y$ it acts on each channel c of the completeness relation by $\delta_{c,1}$. Only the vacuum channel survives, with coefficient $1/d_x$ and

$\psi_1^{x\bar{x}}$ the cup:



(10.4)

The killing loop cuts two strands and reconnects them through the vacuum (Kitaev eqs. (230)–(231); Simon Fig. 17.10).

Theorem 10.7 (Modularity · *cited; proved for unitary C given coherence*)**.** *For a premodular C, S is invertible if and only if the Müger centre is trivial (EGNO Prop. 8.20.12 and Thm. 8.20.7). Then $S^2 = C$, the charge conjugation $C_{ab} = \delta_{\bar{a},b}$.*

Proof. If $x \neq 1$ is transparent, then $\tilde{S}_{xb} = d_x d_b = d_x \tilde{S}_{1b}$: two rows of S are proportional, and S is singular. Conversely, take the trace of (10.4). The right side closes to one loop, $\delta_{y,\bar{x}}$. On the left, the loop b acts on the channel c of $x \otimes y$ by $\lambda_c(b)$, so its trace is

$$\sum_c N_{xy}^c d_c \lambda_c(b) = \sum_c N_{xy}^c \tilde{S}_{bc} = d_b \lambda_b(x)\lambda_b(y),$$

by symmetry and Lemma 10.4. Summing with weights $d_b/\dim C$,

$$\frac{1}{\dim C}\sum_b \tilde{S}_{xb}\tilde{S}_{by} = \delta_{y,\bar{x}}.$$

So $\tilde{S}^2 = \dim C \cdot C$ and $S^2 = C$. ∎

For example, sVec with odd simple s and $\theta_s = -1$ is premodular but every object is transparent (Lecture 7): $S = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ is singular. A fermion alone is not an anyon model.

10.3 Verlinde

Theorem 10.8 (Verlinde formula · *cited; first form proved given modularity and coherence*)**.** *In a unitary modular category,*

$$N_{ab}^c = \sum_x \frac{S_{ax}\,S_{bx}\,S_{x\bar{c}}}{S_{1x}} = \sum_x \frac{S_{ax}\,S_{bx}\,\overline{S_{cx}}}{S_{1x}}.$$

Proof. By the symmetry of S, $S_{ax} = S_{1x}\lambda_x(a)$, with $S_{1x} = d_x/\sqrt{\dim C} > 0$. Lemma 10.4 becomes

$$\frac{S_{ax}S_{bx}}{S_{1x}} = \sum_e N_{ab}^e S_{ex}.$$

Multiply by $(S^{-1})_{xc}$ and sum over x. By Theorem 10.7, $S^{-1} = SC$, so $(S^{-1})_{xc} = S_{x\bar{c}}$, which gives the first form. For the second, $\overline{S_{cx}} = S_{c\bar{x}} = S_{x\bar{c}}$: reversing every crossing of the Hopf link and the orientation of one component is an isotopy, and in a unitary category the inverse of the double braiding is its adjoint (Kitaev eq. (224)). ∎

With the fusion matrices of Lecture 5, $(N_a)_{cb} = N_{ab}^{c}$, S diagonalises fusion:

$$N_a = S^{-1}\,\mathrm{diag}\Big(\frac{S_{ax}}{S_{1x}}\Big)_x\,S.$$

Its eigenvalues $S_{ax}/S_{1x} = \lambda_x(a)$ are the characters of Lemma 10.4, one for each simple x. The fusion ring is commutative, and S alone determines it. The eigenvalue at $x = 1$ is $\lambda_1(a) = d_a$, the Frobenius–Perron eigenvalue of Lecture 5.

The proof of the first form is graphical at every step: a loop is a number (10.2), loops fuse (Lemma 10.4), and the killing loop exists (Lemma 10.6), which forces $S^2 = C$. Modularity is used once, to make ω_0 kill every non-vacuum strand.

10.4 $\mathrm{SL}(2,\mathbb{Z})$ and the central charge

Theorem 10.9 (Gauss sums · *cited*). *In a modular category, the Gauss sums $p_\pm = \sum_a d_a^2\theta_a^{\pm 1}$ satisfy $p_+p_- = \dim C$ (Bakalov–Kirillov Cor. 3.1.11, (3.1.20); EGNO Prop. 8.15.4). In the unitary case $p_- = \overline{p_+}$, so $|p_+| = \sqrt{\dim C}$, and*

$$\sum_a d_a^2\theta_a = \sqrt{\dim C}\;e^{2\pi i c/8}$$

defines the central charge *c modulo* 8.

Theorem 10.10 (Modular data · *cited; $S^2 = C$ proved for unitary C given coherence*). *In a modular category, $S^2 = C$ and*

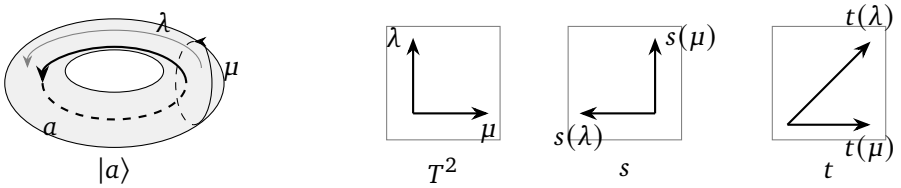
$$(ST)^3 = e^{2\pi i c/8}\,S^2$$

(Kitaev eqs. (232)–(233); Bakalov–Kirillov (3.1.17)). Since $C^2 = 1$ and C commutes with T, the matrices S and T give a projective representation of

$$\mathrm{SL}(2,\mathbb{Z}) = \langle s, t \mid (st)^3 = s^2,\ s^4 = 1\rangle.$$

EGNO writes $T = \mathrm{diag}(\theta^{-1})$ (Thm. 8.16.1), and with it the relation reads $(ST)^3 = p_-S^2$ for its unnormalised S: the same content in another convention. The number c is defined only modulo 8: the modular data see it only through $e^{2\pi i c/8}$. Physically it is the chiral central charge of the edge of the phase · *physical principle* (Kitaev eq. (172)).

The geometric interpretation is a cited statement of the Reshetikhin–Turaev theory, rather than an additional consequence proved here (Simon §17.3.2; Kitaev App. E.5, eqs. (232)–(233)). A solid torus is a filled doughnut, whose boundary is a torus T^2. A core ribbon labelled by a simple object a prepares a boundary state $|a\rangle$. These states form a basis of the torus state space. In particular the toric code has four such basis states. The drawing shows the core and the two boundary cycles, followed by their images in a square with opposite sides identified.



The meridian μ goes around the tube and bounds a disk in the solid torus. The longitude λ goes around the hole, parallel to the core. Orient them so that $\mu \cdot \lambda = +1$. The arrows in each square represent homology classes; they are not anyon worldlines or braidings. Fix the torus generators by

$$s(\mu) = \lambda, \qquad s(\lambda) = -\mu,$$
$$t(\mu) = \mu, \qquad t(\lambda) = \lambda + \mu.$$

The map s exchanges the two cycles with one orientation reversed. The map t is a Dehn twist about the meridian: cutting along that cycle, turning once and regluing adds a meridian to the longitude.

We choose the boundary and core orientations so that the basis matrix of s is precisely the opposite-orientation Hopf-link S of (10.1), and the matrix of t is $T = \mathrm{diag}(\theta_a)$, up to the common scalar phases of the projective action. Thus t twists the core ribbon, while s changes which cycle is filled by the meridian disk. Reversing the cycle exchange replaces s by s^{-1} and the matrix by $S^{-1} = \bar{S}$ in these unitary conventions. The figure uses the displayed s, not that inverse. This distinction is invisible in self-dual examples such as Fibonacci but matters in general. The mapping class group consists of orientation-preserving diffeomorphisms up to deformation. For a torus it is $\mathrm{SL}(2,\mathbb{Z})$. The geometric construction and its anomaly are cited here; Lecture 14 returns to the state-space and gluing interpretation.

10.5 Fibonacci live; tables of the others

Example 10.11 (Fibonacci's S)**.** Fibonacci has $d_\tau = \varphi$ and $\theta_\tau = e^{4\pi i/5}$ (Lecture 7), so $\dim C = 1 + \varphi^2 = 2 + \varphi$. By Proposition 10.3, with $\bar{\tau} = \tau$ and $\tau \otimes \tau = 1 \oplus \tau$,

$$\tilde{S}_{\tau\tau} = \theta_\tau^{-2} + \varphi\, \theta_\tau^{-1} = e^{2\pi i/5} + \varphi\, e^{-4\pi i/5}.$$

With $\cos(2\pi/5) = 1/(2\varphi)$, $\cos(4\pi/5) = -\varphi/2$ and $\sin(2\pi/5) = \varphi \sin(4\pi/5)$, the real part is $\frac{1}{2}(\varphi^{-1} - \varphi^2) = -1$ and the imaginary part is 0. So

$$S = \frac{1}{\sqrt{2+\varphi}} \begin{pmatrix} 1 & \varphi \\ \varphi & -1 \end{pmatrix}.$$

Example 10.12 (Fibonacci's modular data)**.** The determinant is $\det S = -1$, so Fibonacci is modular, as Lecture 7 found: the only transparent simple object is 1. And, since τ is self-dual,

$$S^2 = \frac{1+\varphi^2}{2+\varphi} \cdot 1 = 1 = C.$$

Verlinde (Theorem 10.8) returns the fusion rule:

$$N_{\tau\tau}^{\tau} = \frac{\varphi^3 - \varphi^{-1}}{2+\varphi} = 1,$$
$$N_{\tau\tau}^{1} = \frac{\varphi^2 + 1}{2+\varphi} = 1,$$

using $\varphi^3 = 2\varphi + 1$ and $\varphi^{-1} = \varphi - 1$. Sheet 10 checks $(ST)^3 = e^{2\pi i c/8} S^2$ with the c of the next example.

Example 10.13 (Fibonacci's central charge)**.** The Gauss sum is $1+\varphi^2 e^{4\pi i/5}$. Its real part is $1-\varphi^3/2=\frac{1}{2}-\varphi$, and its imaginary part $\varphi^2\sin(\pi/5)>0$; its modulus is $\sqrt{2+\varphi}$, as Theorem 10.9 requires. Dividing by this modulus gives

$$\frac{1/2-\varphi}{\sqrt{2+\varphi}}=-\sin\frac{\pi}{5}=\cos\frac{7\pi}{10},$$
$$\frac{\varphi^2\sin(\pi/5)}{\sqrt{2+\varphi}}=\cos\frac{\pi}{5}=\sin\frac{7\pi}{10}.$$

Thus the argument is $7\pi/10=2\pi c/8$, so

$$c=\tfrac{14}{5}\pmod 8.$$

The conjugate chirality of Lecture 7 has $c=-\frac{14}{5}$.

The other models of the course are tabulated, with c from the Gauss sum; each row is recomputed by `scripts/check-l10-algebra.py`, and Sheet 10 recomputes the toric code, $\mathrm{SU}(2)_k$ and the $\nu/2$ (Kitaev Table 1 and App. E; Bonderson §§5.1–5.6; Simon Table 17.1).

model	d	θ	c
semion $1,s$	$1,1$	$1,i$	1
Ising $1,\sigma,\psi$	$1,\sqrt{2},1$	$1,e^{i\pi/8},-1$	$\frac{1}{2}$
toric code $1,e,m,\varepsilon$	$1,1,1,1$	$1,1,1,-1$	0
Fibonacci $1,\tau$	$1,\varphi$	$1,e^{4\pi i/5}$	$\frac{14}{5}$

Example 10.14 (The toric code's S)**.** For an abelian model every d is 1 and each fusion has one channel, so $\tilde S_{ab}$ is the double braiding of a and $\bar b$, a sign here. With $\theta=(1,1,1,-1)$ on $(1,e,m,\varepsilon)$ and $e\otimes m=\varepsilon$,

$$S=\frac{1}{2}\begin{pmatrix}1&1&1&1\\1&1&-1&-1\\1&-1&1&-1\\1&-1&-1&1\end{pmatrix}.$$

The normalised entry is $S_{em}=-\frac{1}{2}$. The unnormalised entry $\tilde S_{em}=-1$ is the double-braiding phase *Quantum Computing* measured by taking e around m.

For $\mathrm{SU}(2)_k$ (Theorem 9.9), with labels $j=0,\frac{1}{2},\dots,\frac{k}{2}$,

$$S_{jj'}=\sqrt{\frac{2}{k+2}}\,\sin\frac{(2j+1)(2j'+1)\pi}{k+2},$$
$$c=\frac{3k}{k+2}$$

(Bakalov–Kirillov Thm. 3.3.20 and Ex. 3.3.22 · *cited*; EGNO Ex. 8.18.5). At $k=1$ it is the semion row. At $k=2$, $c=\frac{3}{2}$.

Remark 10.15 (The sixteen-fold way)**.** The braidings with Ising's fusion rules (Theorem 7.9) have $\theta_\sigma=e^{i\pi\nu/8}$ for odd ν, and all eight are modular with $c=\nu/2$ modulo 8 (Kitaev Table 1 and eq. (58); EGNO Ex. 8.18.4). With the eight abelian theories of even ν they form Kitaev's sixteen-fold way (§9).

Remark 10.16 (Ising is not $\mathrm{SU}(2)_2$)**.** Ising and $\mathrm{SU}(2)_2$ have the same fusion rules but different modular data: $\theta_\sigma = e^{i\pi/8}$ and $c = \frac{1}{2}$ for Ising ($\nu = 1$, $\kappa_\sigma = +1$), $\theta_{1/2} = e^{3\pi i/8}$ and $c = \frac{3}{2}$ for $\mathrm{SU}(2)_2$ ($\nu = 3$, $\kappa = -1$). The fusion rules do not determine the anyon model.

Fibonacci's braiding is universal for quantum computation (Freedman, Larsen and Wang, Commun. Math. Phys. 227 (2002) 605–622), beyond this course.

Summary.

- *Handed on:* $\tilde{S}_{ab}$ is the Hopf link, $\sum_c N_{ab}^c d_c \theta_c/(\theta_a\theta_b)$, $S = \tilde{S}/\sqrt{\dim C}$ and $T = \mathrm{diag}(\theta)$ (Definition 10.2 and Proposition 10.3); a loop around a is $\tilde{S}_{ab}/d_a$, and loops fuse (Lemma 10.4).
- *Modularity:* with trivial Müger centre the killing loop projects onto the vacuum (Lemma 10.6), so $S^2 = C$; S is invertible exactly when the Müger centre is trivial (Theorem 10.7).
- *Verlinde:* $N_{ab}^c = \sum_x S_{ax}S_{bx}\overline{S_{cx}}/S_{1x}$, and S diagonalises fusion (Theorem 10.8).
- *Central charge:* $\sum_a d_a^2\theta_a = \sqrt{\dim C}\, e^{2\pi i c/8}$ and $(ST)^3 = e^{2\pi i c/8}S^2$ (Theorems 10.9 and 10.10); Fibonacci has $c = \frac{14}{5}$, Ising $\frac{1}{2}$, the semion 1, the toric code 0.
- *Next:* Lecture 11 studies algebras and module categories, the systems on which a fusion category acts.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: premodular and modular categories and the S-matrix, Defs. 8.13.1–8.13.4 and (8.46), p. 224, and Prop. 8.13.8, p. 225; S^2 and the Verlinde formula, Prop. 8.14.2 and Cor. 8.14.4, p. 227; Gauss sums, Def. 8.15.1 and Prop. 8.15.4, pp. 229–230; $(ST)^3$ with $T = \mathrm{diag}(\theta^{-1})$, Thm. 8.16.1, p. 231; Ising, Ex. 8.18.4, pp. 234–235; c for $C(\mathfrak{g}, q)$, Ex. 8.18.5, p. 235; non-degeneracy and the centralizer, Def. 8.19.1, p. 237, Def. 8.20.1, p. 239, Thm. 8.20.7, p. 241, and Prop. 8.20.12, p. 243. Bakalov and Kirillov, *Lectures on Tensor Categories and Modular Functors*: $\tilde{s}$, (3.1.1)–(3.1.2), p. 47; modular categories, Def. 3.1.1 and Lemma 3.1.4, p. 48; Gauss sums and $(\tilde{s}t)^3$, Lemma 3.1.5 and Thm. 3.1.7, pp. 49–50; (3.1.15)–(3.1.17), p. 51; the killing property, Cor. 3.1.11, p. 52; Verlinde, Thm. 3.1.14, p. 54; $\mathrm{SU}(2)_k$, Thm. 3.3.20 and Ex. 3.3.22, pp. 67–68. Kitaev, *Anyons in an exactly solved model and beyond*, cond-mat/0506438 (v3): Ising-type theories, Table 1, p. 30, and eq. (58), p. 27; the sixteen-fold way, §9, pp. 42–43; the chiral central charge, eq. (172), p. 66; S and Verlinde, eqs. (223)–(228), pp. 84–85; braiding nondegeneracy, §E.5, Def. E.11 and Thm. E.12, p. 86; the killing loop, eqs. (230)–(231), p. 87; $(ST)^3$, eqs. (232)–(233), p. 88. Simon, *Topological Quantum*: S and T, (17.5)–(17.17), pp. 224–226, and the torus, §17.3.2, p. 226; the table of models, Table 17.1, p. 231; the Ω loop, Figs. 17.7–17.11, pp. 230–232. Müger, *Modular categories* (2012), Def. 5.1, p. 13, and S, T, p. 14. Bonderson, PhD thesis (Caltech, 2007): S and T, (2.63)–(2.64), pp. 27–28 (his (2.63) is our $\bar{S}$), and (A.6)–(A.7), p. 96 (his (A.7) is our S); the models, §§5.1–5.6, pp. 77–84. Rowell and Wang, *Mathematics of topological quantum computing*, arXiv:1705.06206, §2.4.2, pp. 11–12.

Block V

Boundaries and Defects

Lecture 11

Algebras in tensor categories and module categories

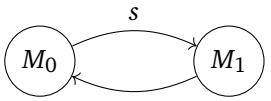
Objectives. *A fusion category acts on a category by functors $X \otimes -$, one for each object X, with coherent associativity and unit isomorphisms. A category with such an action is a* module category. *This lecture builds module categories from algebras and computes a rank-one module associator. Ostrik's theorem says that every finite semisimple indecomposable module category comes from an algebra.*

By the end of this lecture you will be able to:

- Define algebras in C and their modules, and prove that the modules over an algebra form a C-module category.
- Write the module pentagon for a rank-one module category over the part of Vec_G^ω on a subgroup H, and show that it exists exactly when $\omega|_H$ is a coboundary.
- Compute internal Homs in a semisimple module category, and state Ostrik's theorem.
- Classify the indecomposable semisimple module categories over Vec_G by pairs (H, ψ), and list them for $\mathbb{Z}_2$.

Two actions of the same symmetry

An action need not act on the objects of the symmetry category itself. Take $C = \mathrm{Vec}_{\mathbb{Z}_2}$ with simple objects 1 and s and $s \otimes s = 1$. We seek categories on which this same s acts. In the regular action it exchanges the simple objects $M_0 = 1$ and $M_1 = s$. In the action on $\mathrm{Vect}_{\mathrm{fd}}$ obtained by forgetting the grading, it fixes the single simple object $M = \mathbb{C}$. The graphs show the action of s on simple objects, rather than morphisms between those objects.



$$Q_s = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$



$$Q_s = (1)$$

The column indexed by M_j records the multiplicities in $s \otimes M_j$. Both matrices square to the identity, as the fusion rule requires. Each graph is connected under the action, whereas their disjoint union would give a decomposable action.

The classification problem is now concrete: are these all the indecomposable actions, and what extra data must accompany their graphs? The associativity of successive actions must also be specified. The twisted $\mathbb{Z}_2$ example below shows that the one-vertex graph can fail to admit that extra data. For the untwisted category the two graphs will give the

two toric-code boundaries in Lecture 13. There the question becomes which bulk charge a boundary can absorb. First we construct the actions from algebras in C.

11.1 Algebras and modules in C

Throughout, C is a fusion category, as in Lecture 5, and pictures suppress associators by coherence (Theorem 3.3).

Definition 11.1 (Algebra, module)**.** An *algebra* in C is (A, μ, ι) with $\mu\colon A \otimes A \to A$ and $\iota\colon 1 \to A$ such that

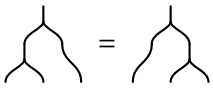
$$\begin{aligned} &\mu \circ (\mu \otimes \mathrm{id}) \\ &= \mu \circ (\mathrm{id} \otimes \mu) \circ \alpha_{A,A,A}, \\ &\mu \circ (\iota \otimes \mathrm{id}) = l_A, \\ &\mu \circ (\mathrm{id} \otimes \iota) = r_A \end{aligned}$$

(EGNO Def. 7.8.1, which writes m, u). A *right A-module* is (M, p) with $p\colon M \otimes A \to M$,

$$\begin{aligned} &p \circ (p \otimes \mathrm{id}) \\ &= p \circ (\mathrm{id} \otimes \mu) \circ \alpha_{M,A,A}, \\ &p \circ (\mathrm{id} \otimes \iota) = r_M. \end{aligned}$$

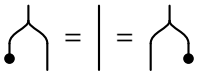
A module map is a morphism f of C with $f \circ p = p' \circ (f \otimes \mathrm{id})$; they form $\mathrm{Mod}_C(A)$ (EGNO Def. 7.8.5, Rem. 7.8.6). Left modules and *bimodules*, with commuting left and right actions, are defined alike.

With μ drawn as a fusion vertex and ι as a dot at the bottom of a strand, associativity is



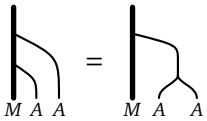
(11.1)

and the unit laws are



(11.2)

A right module is a thick strand, the *wall*, which A-strands join from the right; its axiom says that two A-strands may be multiplied before or after they join it:



(11.3)

Example 11.2 (Algebras)**.**

- In $\mathrm{Vect}_{\mathrm{fd}}$, an algebra is an ordinary finite-dimensional algebra, and $\mathrm{Mod}(A)$ its right modules.
- The unit 1, with $\mu = l_1$: its modules are all of C.
- For X in C, $X \otimes X^*$, with $\mu = \mathrm{id}_X \otimes \mathrm{ev}_X \otimes \mathrm{id}_{X^*}$ and $\iota = \mathrm{coev}_X$ (the duals of Lecture 4).
- In Vec_G^{ω}, for a subgroup $H \leq G$ and a function $\psi\colon H \times H \to \mathbb{C}^\times$: $A = \bigoplus_{h \in H} \delta_h$ with $\mu = \psi(g,h)$ on $\delta_g \otimes \delta_h \to \delta_{gh}$, the *twisted group algebra* $\mathbb{C}_\psi[H]$. Composing with $\alpha_{g,h,k} = \omega(g,h,k)$, associativity reads

$$\begin{aligned} &\psi(g,h)\,\psi(gh,k) \\ &= \omega(g,h,k)\,\psi(h,k)\,\psi(g,hk) \end{aligned}$$

for $g, h, k \in H$, and the unit is δ_1 when $\psi(1,h) = \psi(h,1) = 1$.

For $\omega = 1$ the displayed equation says that ψ is a 2-*cocycle*; rescaling the basis of each δ_h by $\phi(h)$ changes ψ by the *coboundary* $(\delta\phi)(g,h) = \phi(g)\phi(h)/\phi(gh)$, and the classes form $H^2(H, \mathbb{C}^\times)$. These are exactly the cocycles of projective representations, $U_g U_h = \psi(g,h) U_{gh}$ (*Symmetries* Def. 13.19, which writes ω).

Every object X gives a *free* module $X \otimes A$, with $p = (\mathrm{id} \otimes \mu) \circ \alpha_{X,A,A}$, and module maps out of it are morphisms out of X: $\mathrm{Hom}_A(X \otimes A, M) \cong \mathrm{Hom}_C(X, M)$ (EGNO Lemma 7.8.12). An algebra is *separable* if μ has a right inverse $A \to A \otimes A$ that is a map of A-bimodules (EGNO Def. 7.8.29); then $\mathrm{Mod}_C(A)$ is semisimple (EGNO Prop. 7.8.30).

11.2 Module categories; the explicit module associator

Definition 11.3 (Module category). A *left C-module category* is a $\mathbb{C}$-linear category $\mathcal{M}$ with a bilinear functor $\otimes : C \times \mathcal{M} \to \mathcal{M}$, a natural isomorphism, the *module associator*,

$$\alpha^{\mathcal{M}}_{X,Y,M} : (X \otimes Y) \otimes M \to X \otimes (Y \otimes M),$$

and a unit isomorphism $l^{\mathcal{M}}_M : 1 \otimes M \to M$, such that the *module pentagon*

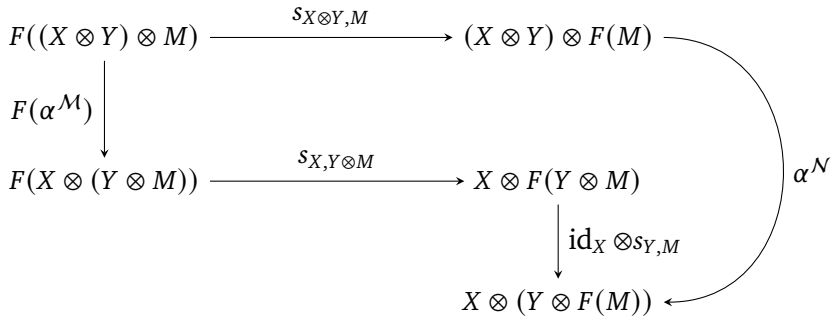
$$\begin{aligned} &\alpha^{\mathcal{M}}_{X,Y,Z\otimes M} \circ \alpha^{\mathcal{M}}_{X\otimes Y,Z,M} \\ &\quad = (\mathrm{id}_X \otimes \alpha^{\mathcal{M}}_{Y,Z,M}) \circ \alpha^{\mathcal{M}}_{X,Y\otimes Z,M} \\ &\qquad \circ (\alpha_{X,Y,Z} \otimes \mathrm{id}_M) \end{aligned}$$

and the triangle $(\mathrm{id}_X \otimes l^{\mathcal{M}}_M) \circ \alpha^{\mathcal{M}}_{X,1,M} = r_X \otimes \mathrm{id}_M$ hold (EGNO Defs. 7.1.1–7.1.2).

Example 11.4 (Module categories). C acts on itself, with $\alpha^C = \alpha$: the *regular* module. A monoidal functor $(F, J, J_0) : C \to \mathrm{Vect}_{\mathrm{fd}}$, such as the forgetful functor of $\mathrm{Rep}(G)$, makes $\mathrm{Vect}_{\mathrm{fd}}$ a C-module category, with $X \otimes V = F(X) \otimes V$ and $\alpha^{\mathcal{M}}$ built from J^{-1} (EGNO Ex. 7.4.6).

A *module functor* $(F, s) : \mathcal{M} \to \mathcal{N}$ is a functor with natural isomorphisms $s_{X,M} : F(X \otimes M) \to X \otimes F(M)$ compatible with the associators and units (EGNO Def. 7.2.1). The module category $\mathcal{M}$ is *indecomposable* if it is not equivalent to a direct sum $\mathcal{M}_1 \oplus \mathcal{M}_2$ of two nonzero module categories (EGNO Def. 7.3.6).

The compatibility of a module functor has a useful typed form. For $X, Y \in C$ and $M \in \mathcal{M}$ the two paths below must agree. Every vertex is an object of $\mathcal{N}$.



Thus compatibility means

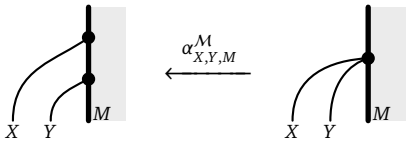
$$\begin{aligned}&\alpha^{\mathcal{N}}_{X,Y,F(M)} \circ s_{X\otimes Y,M}\\ &\quad = (\mathrm{id}_X \otimes s_{Y,M}) \circ s_{X,Y\otimes M} \circ F(\alpha^{\mathcal{M}}_{X,Y,M}).\end{aligned}$$

The unit condition is an equality of maps $F(1 \otimes M) \to F(M)$:

$$F(l^{\mathcal{M}}_M) = l^{\mathcal{N}}_{F(M)} \circ s_{1,M}.$$

The long path first reassociates inside F, then takes X outside F, then takes Y outside it. This order will determine the multiplication order in Theorem 12.2.

In pictures $\mathcal{M}$ is a shaded region: an object M is a wall bounding it, and an object X of C acts by a strand that joins the wall from the unshaded side at a marked point. The module associator compares joining Y then X at two points with joining $X \otimes Y$ at one:



A module category is not a category of modules: its objects need not be modules over anything. Ostrik's theorem (§ 11.3) relates the two.

Theorem 11.5 (Modules form a module category · *proved given coherence*)**.** *For an algebra A in C, $\mathrm{Mod}_C(A)$ is a left C-module category: $X \otimes (M,p) = (X \otimes M, (\mathrm{id}_X \otimes p) \circ \alpha_{X,M,A})$, with module associator $\alpha_{X,Y,M}$ and unit l_M (EGNO Prop. 7.8.10).*

Proof. In pictures $X \otimes M$ is the strand X beside the wall M; A-strands join the wall from the right, past nothing, so the module axioms of $X \otimes M$ are those of M. The associator and unitors of C are module maps, by their naturality in the last variable and the pentagon of C; the module pentagon and triangle are the pentagon and triangle of C, with M as the fourth object. ∎

On a module category with one simple object M the invertible δ_h act by $h \otimes M \cong M$, and the module associator is a number.

Theorem 11.6 (A rank-one module category)**.** *Let ω be a normalised 3-cocycle on G, $H \le G$, and C_H the full subcategory of Vec^{ω}_G on the δ_h, $h \in H$. Make $\mathrm{Vect}_{\mathrm{fd}}$ a C_H-module category with $\delta_h \otimes M = M$, $l^{\mathcal{M}} = \mathrm{id}$ and $\alpha^{\mathcal{M}}_{g,h,M} = \psi(g,h)\,\mathrm{id}_M$. The module pentagon and triangle hold if and only if $\psi(g,1) = \psi(1,g) = 1$ and, for $g,h,k \in H$,*

$$\begin{aligned}&\psi(g,h)\,\psi(gh,k)\\ &= \omega(g,h,k)\,\psi(h,k)\,\psi(g,hk),\end{aligned}$$

that is, $\omega|_H = (\delta\psi)^{-1}$ in the notation of Lecture 3. So such a module category exists exactly when $\omega|_H$ is a coboundary.

Proof. On simple objects every arrow is a scalar, as in Theorem 3.8. In the module pentagon for g,h,k the left side is $\psi(g,h)\,\psi(gh,k)$ and the right side $\psi(h,k)\,\psi(g,hk)\,\omega(g,h,k)$. The triangle for g gives $\psi(g,1) = 1$, and the pentagon at $(1,1,k)$ then gives $\psi(1,k) = \psi(1,1) = 1$. Conversely, if $\omega|_H = (\delta\psi)^{-1}$, the equation at $(1,1,k)$ and $(g,1,1)$ gives $\psi(1,k) = \psi(g,1) = \psi(1,1)$, and $\psi/\psi(1,1)$ is normalised and still solves it. ∎

The same equation gives associativity of $\mathbb{C}_\psi[H]$ in Example 11.2. Sheet 11 reconstructs this algebra. EGNO prints this condition as $\omega = d\psi$ (Ex. 7.8.3, (9.13)); its own coboundary (Ex. 1.7.1), associator ((2.19)) and module pentagon ((7.2)) give the one above.

For $\mathbb{Z}_2$ with $\omega(s,s,s) = -1$, the semion's category of Lecture 7, the equation at (s,s,s) reads $\psi(s,s) = -\psi(s,s)$: no module category $\mathrm{Vect}_{\mathrm{fd}}$ with s acting, and no fibre functor, as Lecture 8 stated (EGNO Ex. 5.1.3). For $H = G$ the module category $\mathrm{Vect}_{\mathrm{fd}}$ needs $[\omega] = 0$ (Lecture 13).

Induced to all of G, the algebra $\mathbb{C}_\psi[H]$ in Vec_G^ω gives the semisimple module category $\mathrm{Mod}(\mathbb{C}_\psi[H])$, whose simple objects are the free modules $\delta_g \otimes \mathbb{C}_\psi[H]$, one for each coset gH, permuted by G · *cited* (EGNO Ex. 9.7.2, for right module categories, with the sign corrected as above). The coset is visible in $\mathrm{Hom}_A(\delta_g \otimes A, \delta_{g'} \otimes A) \cong \mathrm{Hom}(\delta_g, \delta_{g'} \otimes A)$, which is nonzero exactly when $g \in g'H$.

11.3 Internal Hom; Ostrik

An internal Hom reconstructs an algebra from a module category.

Definition 11.7 (Internal Hom)**.** For objects M, N of a C-module category $\mathcal{M}$, the *internal Hom* $[M,N]$ is an object of C with isomorphisms

$$\begin{aligned}&\mathrm{Hom}_C(X, [M,N])\\ &\cong \mathrm{Hom}_{\mathcal{M}}(X \otimes M, N),\end{aligned}$$

natural in X (EGNO §7.9, (7.21)). By Yoneda (Theorem 2.6) it is unique up to unique isomorphism.

Theorem 11.8 (The semisimple internal Hom)**.** *If $\mathcal{M}$ is semisimple with finitely many simple objects, then*

$$\begin{aligned}&[M,N]\\ &= \bigoplus_c \mathrm{Hom}_{\mathcal{M}}(c \otimes M, N) \otimes c,\end{aligned}$$

the sum over the simple objects c of C, where $V \otimes c$ means $c^{\oplus \dim V}$ with multiplicity space $V = \mathrm{Hom}_{\mathcal{M}}(c \otimes M, N)$, and no dual. The evaluation $\mathrm{ev}_{M,N} \colon [M,N] \otimes M \to N$ is, on each summand, the canonical map $\mathrm{Hom}_{\mathcal{M}}(c \otimes M, N) \otimes (c \otimes M) \to N$, and the composite of evaluations makes $[M,M]$ an algebra, with unit from $l_M^{\mathcal{M}}$.

Proof. For a simple d, Schur's lemma (Theorem 2.13) gives $\mathrm{Hom}(d, \bigoplus_c V_c \otimes c) = V_d$, which with $V_c = \mathrm{Hom}_{\mathcal{M}}(c \otimes M, N)$ is $\mathrm{Hom}_{\mathcal{M}}(d \otimes M, N)$; both sides are additive in X, and every X is a sum of simples. Under the adjunction, $\mathrm{id}_{[M,N]}$ corresponds to $\mathrm{ev}_{M,N}$, which is the stated map. The multiplication $[M,M] \otimes [M,M] \to [M,M]$ is the morphism corresponding to $\mathrm{ev} \circ (\mathrm{id} \otimes \mathrm{ev}) \circ \alpha^{\mathcal{M}}$; associativity and the unit laws follow from the module pentagon and triangle, by the uniqueness in the adjunction. The unit corresponds to $l_M^{\mathcal{M}} \colon 1 \otimes M \to M$. ∎

Example 11.9 (Internal Homs)**.** For the regular module, duality (Lecture 4) gives $\mathrm{Hom}(c \otimes x, y) \cong \mathrm{Hom}(c, y \otimes x^*)$, so $[x,y] = y \otimes x^*$ and $[1,1] = 1$; in Vec_G, $[\delta_x, \delta_y] = \delta_{yx^{-1}}$.

For $G = \mathbb{Z}_2$ and $A = \mathbb{C}[\mathbb{Z}_2]$, the module category $\mathrm{Mod}(A)$ has one simple object $M = A$. Both $1 \otimes M$ and $s \otimes M$ are isomorphic to M, so both multiplicity spaces $\mathrm{Hom}(M,M)$ and $\mathrm{Hom}(s \otimes M, M)$ are $\mathbb{C}$. Hence $[M,M] = 1 \oplus s$. Fix a basis vector v_s of the line s. Let a_1 and

a_s be the homogeneous basis of A. Choose $t_1 = \mathrm{id}_M$ and the evaluation map $t_s : s \otimes M \to M$ given by

$$t_s(v_s \otimes a_1) = a_s, \qquad t_s(v_s \otimes a_s) = a_1.$$

With $s \otimes s = 1$, two evaluations give

$$t_s \circ (\mathrm{id}_s \otimes t_s) = t_1.$$

Thus the product on $[M, M]$ reconstructs $\mathbb{C}[\mathbb{Z}_2]$. Sheet 11 checks the general product.

Theorem 11.10 (Ostrik · *cited*)**.** *Let C be a fusion category over $\mathbb{C}$ and $\mathcal{M}$ a semisimple indecomposable C-module category with finitely many simple objects. For any nonzero M in $\mathcal{M}$, $A = [M, M]$ is a separable algebra, and $N \mapsto [M, N]$ is an equivalence of module categories $\mathcal{M} \simeq \mathrm{Mod}_C(A)$.*

This is Ostrik's Theorem 1 (math/0111139, p. 10), who writes "semisimple" algebra; over $\mathbb{C}$ this is separable (Müger 2012, Lemma 3.14 and Thm. 3.21). Every finite semisimple module category is a direct sum of indecomposable ones (EGNO Prop. 7.6.7), so classifying them means classifying algebras up to equivalence of their module categories (*Morita equivalence* of algebras, EGNO Def. 7.8.17).

11.4 Vec_G: (H, ψ), worked for $\mathbb{Z}_2$; $\mathrm{Rep}(G)$

Theorem 11.11 (Module categories over Vec_G · *sketched given Ostrik's theorem*)**.** *Indecomposable semisimple Vec_G-module categories with finitely many simple objects, up to equivalence, correspond to pairs $(H, [\psi])$ with $H \leq G$ and $[\psi] \in H^2(H, \mathbb{C}^\times)$, up to conjugating both, through $(H, \psi) \mapsto \mathrm{Mod}(\mathbb{C}_\psi[H])$ (EGNO Ex. 7.4.10–7.4.11 and 9.7.2).*

Sketch. Each δ_g is invertible, so $\delta_g \otimes -$ permutes the simple objects, transitively since $\mathcal{M}$ is indecomposable. Fix a simple M and let H be its stabiliser. By Theorem 11.8, $A = [M, M] = \bigoplus_g \mathrm{Hom}(\delta_g \otimes M, M) \otimes \delta_g$ has a one-dimensional summand for each $h \in H$ and none elsewhere; choosing a basis vector in each gives $\mu = \psi(g, h)$, and associativity makes ψ a 2-cocycle (Example 11.2). Other basis vectors change ψ by a coboundary, and another simple $\delta_g \otimes M$ conjugates H and ψ. Ostrik's theorem gives $\mathcal{M} \simeq \mathrm{Mod}(A)$, and Theorem 11.5 gives a module category for every pair. ∎

Example 11.12 ($\mathbb{Z}_2$)**.** On $\mathbb{Z}_2$ a normalised 2-cocycle has one value, $\psi(s, s)$, and the cocycle condition at (s, s, s), $\psi(s, s)\psi(1, s) = \psi(s, s)\psi(s, 1)$, is automatic. Every value is a coboundary: $\psi = \delta\phi$ with $\phi(s)^2 = \psi(s, s)$. So $H^2(\mathbb{Z}_2, \mathbb{C}^\times) = 0$, and the subgroups 1 and $\mathbb{Z}_2$ give the two indecomposable module categories over Vec_G, $G = \mathbb{Z}_2$:

- $H = 1$, $A = \delta_1 = 1$: $\mathrm{Mod}(1) = \mathrm{Vec}_G$, the regular module, with two simple objects that s exchanges;
- $H = \mathbb{Z}_2$, $A = \mathbb{C}[\mathbb{Z}_2] = \delta_1 \oplus \delta_s$: one simple object $\delta_1 \otimes A \cong \delta_s \otimes A$, the module category $\mathrm{Vect}_{\mathrm{fd}}$ on which s acts trivially; it comes from the fibre functor that forgets the grading.

Over Vec_G^ω with $\omega(s, s, s) = -1$ only $H = 1$ survives, since $\omega|_{\mathbb{Z}_2}$ is not a coboundary (Theorem 11.6; EGNO Ex. 9.7.2): the regular module is the only one.

Lecture 13 meets these two as the two gapped boundaries of the toric code. Sheet 11 counts the four module categories over Vec_{S_3}.

Over $\mathrm{Rep}(G)$ the same labels classify: the indecomposable semisimple module categories are $\mathrm{Rep}^{\psi}(H)$, the ψ-projective representations of H, on which $\mathrm{Rep}(G)$ acts by restriction and tensor product $\cdot$ *cited* (Ostrik Thm. 2, p. 12; EGNO Cor. 7.12.20). The lists agree because $\mathrm{Rep}(G)$ is the dual of Vec_G with respect to $\mathrm{Vect}_{\mathrm{fd}}$ (Lecture 12) and module categories over a category and its dual correspond (EGNO Thm. 7.12.16).

Summary.

- *Handed on:* an algebra in C is (A, μ, ι); its right modules form a left C-module category $\mathrm{Mod}_C(A)$ (Definitions 11.1 and 11.3 and Theorem 11.5).
- *The module associator:* a rank-one module over the part of Vec_G^{ω} on H is a normalised ψ with $\psi(g,h)\psi(gh,k) = \omega(g,h,k)\psi(h,k)\psi(g,hk)$, which exists exactly when $\omega|_H$ is a coboundary (Theorem 11.6).
- *Ostrik:* in the semisimple case $[M,N] = \bigoplus_c \mathrm{Hom}(c\otimes M, N)\otimes c$, and every finite semisimple indecomposable module category is $\mathrm{Mod}([M,M])$ for any chosen nonzero object M (Theorems 11.8 and 11.10).
- Vec_G*:* indecomposable semisimple module categories are pairs $(H, [\psi])$ up to conjugacy; $\mathbb{Z}_2$ has two (Theorem 11.11 and Example 11.12).
- *Next:* Lecture 12 uses module categories to compare fusion categories (Morita equivalence) and builds the Drinfeld centre.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: cochains, Ex. 1.7.1, p. 8; no fibre functor on Vec_G^{ω}, Ex. 5.1.3, p. 91; module categories, Defs. 7.1.1–7.1.2, pp. 131–132; module functors, Def. 7.2.1, p. 134; indecomposable, Def. 7.3.6, p. 135; fibre functors as module structures, Ex. 7.4.6, p. 136; Vec_G and $\mathrm{Rep}(G)$, Ex. 7.4.9–7.4.11 and Rem. 7.4.12, pp. 136–137; direct sums, Prop. 7.6.7, p. 140; algebras and modules, Def. 7.8.1, Ex. 7.8.3, Def. 7.8.5, Rem. 7.8.6, Prop. 7.8.10 and Lemma 7.8.12, pp. 141–144; Morita equivalence of algebras, Def. 7.8.17, p. 145; separable algebras, Def. 7.8.29 and Prop. 7.8.30, p. 146; internal Hom, §7.9, pp. 147–150; Ostrik's theorem, Thm. 7.10.1, pp. 150–152; duals and module categories, Thm. 7.12.16, and $\mathrm{Rep}(G)$, Cor. 7.12.20, pp. 157–158; Vec_G^{ω}, Ex. 9.7.2 and Rem. 9.7.3, p. 287 (their $d\psi = \omega$ is our $\omega = (\delta\psi)^{-1}$). Ostrik, *Module categories, weak Hopf algebras and modular invariants*, math/0111139: module categories, Def. 6, pp. 5–6; internal Hom, Def. 11, p. 9; Theorem 1, §3.3, p. 10; $\mathrm{Rep}(G)$, Thm. 2, p. 12. Müger, *Modular categories* (2012): separable algebras, Def. 3.11 and Lemma 3.14, pp. 5–6, and Ostrik's theorem, Thm. 3.21, p. 6. Gruber, lecture notes: module categories (right-handed), Def. 2.1, pp. 9–10.

Lecture 12

Morita equivalence and the Drinfeld centre

Objectives. *A fusion category C acting on a module category $\mathcal{M}$ (Lecture 11) has a second symmetry category, its* dual*, whose objects are the module endofunctors of $\mathcal{M}$. For* Vec_G *acting on* $\mathrm{Vect}_{\mathrm{fd}}$ *the dual is* $\mathrm{Rep}(G)$*. These categories have equivalent* Drinfeld centres*. An object of the centre $\mathcal{Z}(C)$ is an object of C equipped with a compatible family of crossings with every other object. The chosen crossings are additional data. The centre is braided, and modular when C is spherical. For* Vec_G *with $G=\mathbb{Z}_2$, its anyons are those of the toric code of* Quantum Computing*.*

By the end of this lecture you will be able to:

- Define the dual category of a module category, and identify $(\mathrm{Vec}_G)^*_{\mathrm{Vect}_{\mathrm{fd}}}$ with $\mathrm{Rep}(G)$.
- Define half-braidings and the Drinfeld centre, and prove that it is braided.
- Compute $\mathcal{Z}(\mathrm{Vec}_G)$ for $G=\mathbb{Z}_2$ by half-braidings, and identify it with the toric code, braiding and spins included.
- State Müger's theorem on the centre and the theorem of Etingof, Nikshych and Ostrik on Morita equivalence, and use them on Vec_G, $\mathrm{Rep}(G)$ and Fibonacci.

12.1 Module functors; the dual category; Morita equivalence

Module functors $(F,s)\colon \mathcal{M}\to\mathcal{N}$ were defined in Lecture 11 (§ 11.2). A *module natural transformation* $\xi\colon F\to F'$ is a natural transformation with

$$s'_{X,M}\circ \xi_{X\otimes M} = (\mathrm{id}_X\otimes \xi_M)\circ s_{X,M}.$$

Definition 12.1 (Dual category)**.** For a C-module category $\mathcal{M}$, the *dual category* $C^*_{\mathcal{M}} = \mathrm{Fun}_C(\mathcal{M},\mathcal{M})$ has the module endofunctors as objects and the module natural transformations as morphisms, with tensor product the composition, $F\otimes G = F\circ G$ (EGNO Def. 7.12.2).

Theorem 12.2 (Two duals · *cited; (ii) proved*)**.** *(i) For the regular module, $X\mapsto(-\otimes X)$ is a tensor equivalence $C^{\otimes\mathrm{op}}\simeq C^*_C$, where $C^{\otimes\mathrm{op}}$ has the reversed tensor product (EGNO Ex. 7.12.3). For* Vec_G*, $g\mapsto(-\otimes\delta_{g^{-1}})$ identifies* Vec_G *with* $(\mathrm{Vec}_G)^*_{\mathrm{Vec}_G}$*. (ii) For* Vec_G *acting on* $\mathrm{Vect}_{\mathrm{fd}}$ *by $\delta_g\otimes V=V$,* $(\mathrm{Vec}_G)^*_{\mathrm{Vect}_{\mathrm{fd}}}\simeq\mathrm{Rep}(G)$ *(EGNO Ex. 7.12.19).*

Proof of (ii), and the order in (i). In (i), $(-\otimes\delta_{g^{-1}})\circ(-\otimes\delta_{h^{-1}}) = -\otimes\delta_{h^{-1}g^{-1}} = -\otimes\delta_{(gh)^{-1}}$, so the assignment respects the tensor product, while $g\mapsto(-\otimes\delta_g)$ reverses it for non-abelian G; that it is an equivalence is EGNO's.

For the second dual, put $V = F(\mathbb{C})$. For $w \in W$, let $f_w \colon \mathbb{C} \to W$ send 1 to w. The natural map $V \otimes W \to F(W)$ is given by

$$v \otimes w \longmapsto F(f_w)(v).$$

It is an isomorphism because F is linear and preserves finite direct sums. The module structure is determined by the maps $s_g \colon F(\delta_g \otimes \mathbb{C}) \to \delta_g \otimes F(\mathbb{C})$, hence by $s_g \in \mathrm{GL}(V)$. With trivial module associators, compatibility (EGNO (7.6)) reads $s_{gh} = s_h\, s_g$. Thus $\rho(g) = s_g^{-1}$ is a representation.

More explicitly, put $X = \delta_g$, $Y = \delta_h$ and $M = \mathbb{C}$ in the compatibility diagram of Lecture 11. Identify all the action spaces with $V = F(\mathbb{C})$. Both module associators become identities. The short path is s_{gh}, while the long path is

$$V \xrightarrow{s_g} V \xrightarrow{s_h} V, \qquad s_{gh} = s_h \circ s_g.$$

Composition applies the rightmost map first, so taking inverses gives

$$\rho(gh) = s_{gh}^{-1} = s_g^{-1} s_h^{-1} = \rho(g)\rho(h).$$

The unit condition gives $s_1 = \mathrm{id}_V$. For a non-abelian group, replacing the long path by $s_g s_h$ would change the answer; the $\mathbb{Z}_2$ example alone cannot detect that error.

Conversely, a representation (V, ρ) gives the functor $F_V(W) = V \otimes W$, with $F_V(f) = \mathrm{id}_V \otimes f$ and $s_g = \rho(g)^{-1}$ on V. Extend the module structure to graded direct sums using the ordinary flip of vector spaces. The representation law gives the required compatibility. These constructions are inverse up to the natural isomorphism above. A module natural transformation is determined by an intertwiner on V. Composing F_V with F_U gives $V \otimes U \otimes W$, with representation $\rho_V \otimes \rho_U$. ∎

So one symmetry, Vec_G, has two duals: itself (from the regular module) and $\mathrm{Rep}(G)$ (from $\mathrm{Vect}_{\mathrm{fd}}$).

Example 12.3 (The duals of $\mathrm{Vec}_{\mathbb{Z}_2}$)**.** Lecture 11 found two indecomposable module categories over $\mathrm{Vec}_{\mathbb{Z}_2}$: the regular one and $\mathrm{Vect}_{\mathrm{fd}}$. Their duals are $\mathrm{Vec}_{\mathbb{Z}_2}$ (the group is abelian, so $\otimes$op changes nothing) and $\mathrm{Rep}(\mathbb{Z}_2)$, whose two simple objects, the trivial and the sign representation, fuse like $\mathbb{Z}_2$ with the trivial associator of vector spaces: $\mathrm{Rep}(\mathbb{Z}_2) \simeq \mathrm{Vec}_{\mathbb{Z}_2}$ as fusion categories ($\omega = 1$, Lecture 3). So $\mathrm{Vec}_{\mathbb{Z}_2}$ is dual to itself in two different ways. In the toric code of § 12.3 this is the exchange of e and m, and in Lecture 13 the Kramers–Wannier duality of the Ising chain.

To compare categories through module categories we need products of categories.

Definition 12.4 (Deligne product, bimodule category)**.** For fusion categories $\mathcal{C}$ and $\mathcal{D}$, the *Deligne product* $\mathcal{C} \boxtimes \mathcal{D}$ is the fusion category with simple objects $a \boxtimes b$, for simple a in $\mathcal{C}$ and b in $\mathcal{D}$, and $(a \boxtimes b) \otimes (a' \boxtimes b') = (a \otimes a') \boxtimes (b \otimes b')$. Its morphism spaces are

$$\begin{aligned} &\mathrm{Hom}(X \boxtimes Y, X' \boxtimes Y') \\ &= \mathrm{Hom}_{\mathcal{C}}(X, X') \otimes \mathrm{Hom}_{\mathcal{D}}(Y, Y'). \end{aligned}$$

Composition, associators and unitors are taken componentwise, and objects are finite direct sums of the simple pairs (EGNO §1.11, §4.6). A $(\mathcal{C}, \mathcal{D})$-*bimodule category* is a module category over $\mathcal{C} \boxtimes \mathcal{D}^{\otimes \mathrm{op}}$ (EGNO Def. 7.1.7, Ex. 7.4.3).

Definition 12.5 (Morita equivalence)**.** Fusion categories $\mathcal{C}$ and $\mathcal{D}$ are *Morita equivalent* if there is an indecomposable semisimple $\mathcal{C}$-module category $\mathcal{M}$ with $\mathcal{D}^{\otimes\mathrm{op}} \simeq \mathcal{C}^*_{\mathcal{M}}$ as tensor categories (EGNO Def. 7.12.17). Then $\mathcal{M}$ is a $(\mathcal{C},\mathcal{D})$-bimodule category.

Without $\otimes$op the centres of § 12.4 would match only up to reversing the braiding. By Theorem 12.2, Vec_G and $\mathrm{Rep}(G)$ are Morita equivalent, since $\mathrm{Rep}(G)^{\otimes\mathrm{op}} \simeq \mathrm{Rep}(G)$ by the flip. Physically, Morita equivalent symmetry categories describe the same two-dimensional bulk, seen through different boundaries · *physical principle* (Kitaev and Kong; Lecture 13).

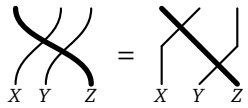
12.2 The Drinfeld centre

Definition 12.6 (Half-braiding, Drinfeld centre)**.** A *half-braiding* on an object Z of $\mathcal{C}$ is a natural isomorphism $\gamma_X : X \otimes Z \to Z \otimes X$ such that

$$\gamma_{X\otimes Y} = \alpha_{Z,X,Y} \circ (\gamma_X \otimes \mathrm{id}_Y) \circ \alpha^{-1}_{X,Z,Y} \circ (\mathrm{id}_X \otimes \gamma_Y) \circ \alpha_{X,Y,Z}$$

for all X, Y. The *Drinfeld centre* $\mathcal{Z}(\mathcal{C})$ has objects the pairs (Z,γ) and morphisms the $f : Z \to Z'$ with $(f \otimes \mathrm{id}_X) \circ \gamma_X = \gamma'_X \circ (\mathrm{id}_X \otimes f)$ (EGNO Def. 7.13.1).

A half-braiding is a strand Z, drawn thick, that every strand of $\mathcal{C}$ can cross. It is not a braiding of $\mathcal{C}$: only Z carries one. We initially draw this crossing without over or under; the braiding of the centre below will use the over-crossing convention of Lecture 7. The condition says that crossing $X \otimes Y$ is crossing Y, then X:



(12.1)

The tensor product of (Z,γ) and (Z',γ') is $Z \otimes Z'$, with X crossing Z and then Z', and the unit is 1 with the trivial half-braiding (EGNO (7.42)).

Theorem 12.7 (The centre is braided · *proved given coherence*)**.** *$\mathcal{Z}(\mathcal{C})$ is a monoidal category, and*

$$c_{(Z,\gamma),(Z',\gamma')} = \gamma'_Z : Z \otimes Z' \to Z' \otimes Z$$

is a braiding on it (EGNO Prop. 8.5.1). In pictures, the strand Z crosses the thick strand Z', over it as in Lecture 7, by Z''s half-braiding:

$$c_{(Z,\gamma),(Z',\gamma')} \;=\; \gamma'_Z. \tag{12.2}$$



Proof. γ'_Z is a morphism of $\mathcal{Z}(\mathcal{C})$: for any X, pulling X across $Z \otimes Z'$ and then exchanging Z past Z' equals exchanging first, by naturality of γ' in its argument (applied to γ_X) and the condition (12.1) for γ'. It is natural in the first variable because γ' is natural in its argument, and in the second because morphisms of the centre commute with half-braidings. That the tensor product of two half-braidings is again one is the same kind of check, with the pentagon (EGNO (7.42)). The two hexagons of Definition 7.2: $c_{(Z,\gamma),(Z',\gamma')\otimes(Z'',\gamma'')}$ is $\gamma' \otimes \gamma''$ evaluated on Z, which by the definition of the tensor product is Z crossing Z' then Z''; and $c_{(Z,\gamma)\otimes(Z',\gamma'),(Z'',\gamma'')} = \gamma''_{Z\otimes Z'}$ is Z' then Z crossing Z'', which is (12.1) for γ''. In pictures both are strands sliding through crossings. ∎

The forgetful functor $\mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ keeps Z and forgets γ. If $\mathcal{C}$ is already braided, $X \mapsto (X, c_{-,X})$ is a braided functor $\mathcal{C} \to \mathcal{Z}(\mathcal{C})$ (EGNO Prop. 8.6.1). Two centres must not be

confused: the Drinfeld centre $\mathcal{Z}(C)$ is a new category built from any monoidal C; the Müger centre of Lecture 7 is a subcategory of a braided one, its transparent objects.

12.3 $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$ in full; $D(G)$ and $\mathcal{Z}(\mathrm{Vec}_G)$

Theorem 12.8 (The centre of Vec_A · *proved given coherence*). *For an abelian group A, the simple objects of $\mathcal{Z}(\mathrm{Vec}_A)$ are the pairs (g,χ), $g \in A$, χ a character of A, with $\gamma_{\delta_h} = \chi(h)\,\mathrm{id}$ on $\delta_h \otimes \delta_g$. Their braiding and spin are*

$$c_{(g,\chi),(g',\chi')} = \chi'(g),$$
$$\theta_{(g,\chi)} = \chi(g).$$

For $A = \mathbb{Z}_2$, write the group additively as $\{0,1\}$: the generator s of Lecture 3 is now 1, and its identity is 0. The four simple objects $1 = (0,1)$, $e = (0,\chi_-)$, $m = (1,1)$ and $\varepsilon = (1,\chi_-)$ have $\theta = 1,1,1,-1$, $e \otimes m = \varepsilon$ and double braiding $c_{m,e}c_{e,m} = -1$.

Proof. Every morphism of Vec_A preserves the grading, so an object of the centre is a sum of pieces $\delta_g \otimes V$, and γ_{δ_h} maps $\delta_h \otimes (\delta_g \otimes V)$ to $(\delta_g \otimes V) \otimes \delta_h$. Since A is abelian both are $\delta_{hg} \otimes V$, so γ_{δ_h} is a map $\gamma_h \in \mathrm{GL}(V)$. The associators are trivial, so (12.1) reads $\gamma_{hk} = \gamma_h\gamma_k$: a representation of A, which splits into characters. So the simple objects are the (g,χ), and their tensor product multiplies both, $(g,\chi)\otimes(g',\chi') = (gg',\chi\chi')$; in particular $e \otimes m = (1,\chi_-) = \varepsilon$. By Theorem 12.7 $c_{(g,\chi),(g',\chi')} = \gamma'_{\delta_g} = \chi'(g)$, and for an invertible object the spin is its self-braiding, $\chi(g)$ (Theorem 7.7). For $\mathbb{Z}_2$: $\theta_\varepsilon = \chi_-(1) = -1$, $c_{e,m} = \chi_m(0) = 1$ and $c_{m,e} = \chi_-(1) = -1$. ∎

These are the anyons of the toric code (Kong–Zhang (4.1.3), Ex. 4.2.17). In full, with rows X and columns Y in the order $1, e, m, \varepsilon$, the braiding $c_{X,Y} = \chi_Y(g_X)$ is

$$\left(c_{X,Y}\right) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix}.$$

It is not symmetric: $c_{e,m} = 1$ but $c_{m,e} = -1$. For $X \neq Y$, neither of them 1, a single entry can depend on the gauge, but the double braiding $c_{Y,X}c_{X,Y}$, the product of the matrix and its transpose entry by entry, does not, nor do the self-braidings on the diagonal; the double braiding is -1 for e and m, e and ε, m and ε. The double braiding -1 of e and m is the phase *Quantum Computing* measured by taking e around m, and the S-matrix of Example 10.14 follows from these numbers (Sheet 12). The anyons of the toric code are the centre of its symmetry $\mathbb{Z}_2$. Exchanging e and m preserves the fusion rules, the spins and the double braidings; it is a braided autoequivalence (Kong–Zhang §4.3.2; Sheet 12), the shadow of the two duals of Example 12.3. Twisting by $\omega(s,s,s) = -1$, on the generator s of $\mathbb{Z}_2$, changes the centre: the same computation, with associators, gives spins $1, 1, i, -i$, the *double semion* (`scripts/check-l12-algebra.py`).

For a non-abelian G a simple δ_g need not carry a half-braiding at all: γ_{δ_h} must map $\delta_h \otimes \delta_g = \delta_{hg}$ to $\delta_g \otimes \delta_h = \delta_{gh}$, which is zero unless $hg = gh$. So δ_g alone works only for central g; in general the object must be a sum $\bigoplus_{g' \in [g]} \delta_{g'} \otimes V_{g'}$ over a whole conjugacy class, and the half-braiding is then fixed by a representation of the centraliser $C_G(g)$ on V_g. This is why the labels below are classes and centraliser representations. The centre is the representation category of a Hopf algebra.

Theorem 12.9 (The Drinfeld double · *cited*)**.** *The* Drinfeld double $D(G) = \mathbb{C}^G \rtimes \mathbb{C}[G]$ *has basis* $\delta_g \otimes h$ *and product* $(\delta_g \otimes h)(\delta_{g'} \otimes h') = \delta_{g,hg'h^{-1}}\, \delta_g \otimes hh'$*; it is quasitriangular (EGNO Ex. 8.3.9). There are braided equivalences* $\mathrm{Rep}D(G) \simeq \mathcal{Z}(\mathrm{Vec}_G) \simeq \mathcal{Z}(\mathrm{Rep}G)$ *(EGNO Prop. 7.14.6, Ex. 8.5.4 and 8.5.6). The simple objects are the pairs* $([g], \pi)$ *of a conjugacy class and an irreducible representation of the centraliser* $C_G(g)$ *(*Symmetries *Defs. 2.4, 2.11), with* $\theta_{([g],\pi)} = \chi_\pi(g)/\dim \pi$ *(Bakalov–Kirillov (3.2.8)).*

The labels are not pairs (g, an irrep of G): for S_3 the classes of 1, the transpositions and the 3-cycles have centralisers S_3, $\mathbb{Z}_2$ and $\mathbb{Z}_3$, with $3 + 2 + 3 = 8$ simple objects (Sheet 12), and $\dim \mathcal{Z}(\mathrm{Vec}_G) = |G|^2$.

12.4 Müger and ENO (statements)

For a braided C, the *reverse* C^{rev} has the braiding $c^{\mathrm{rev}}_{X,Y} = c^{-1}_{Y,X}$ (EGNO Def. 8.1.4), and conjugate spins.

Theorem 12.10 (Müger · *cited*)**.** *Let C be a fusion category over $\mathbb{C}$. (i) $\mathcal{Z}(C)$ is non-degenerate, that is, its Müger centre is trivial (EGNO Cor. 8.20.13), and* $\dim \mathcal{Z}(C) = (\dim C)^2$ *(EGNO Prop. 9.3.4). (ii) If C is spherical, $\mathcal{Z}(C)$ is modular (EGNO Cor. 8.20.14). (iii) If C is modular, $\mathcal{Z}(C) \simeq C \boxtimes C^{\mathrm{rev}}$ as braided categories (Müger 2012, Thm. 7.1(vi)).*

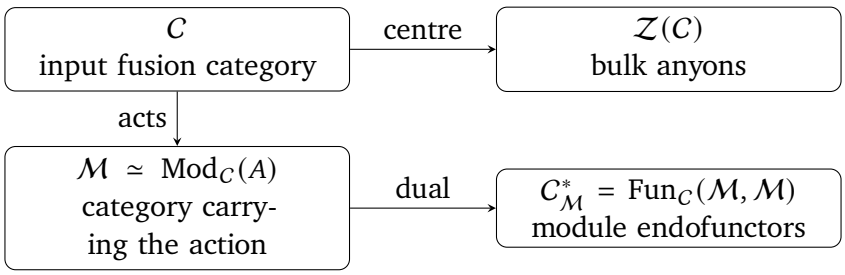
"Modular" needs the spherical structure of Lecture 10. For Fibonacci, $\mathcal{Z}(\mathrm{Fib}) \simeq \mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ has dimension $(2+\varphi)^2$ and central charge $\frac{14}{5} - \frac{14}{5} = 0$ (the script checks the Gauss sum). The toric code, too, has $c = 0$: for spherical C the Gauss sums of $\mathcal{Z}(C)$ are $\dim C$, so $c = 0$ (Müger 2012, Thm. 7.1(iv)).

Theorem 12.11 (Etingof, Nikshych and Ostrik · *cited*)**.** *Fusion categories C and $\mathcal{D}$ are Morita equivalent if and only if $\mathcal{Z}(C) \simeq \mathcal{Z}(\mathcal{D})$ as braided categories (EGNO Thm. 8.12.3).*

So $\mathcal{Z}(\mathrm{Vec}_G) \simeq \mathcal{Z}(\mathrm{Rep}G)$ is Morita equivalence seen from the centre, and $\mathrm{Vec}_{\mathbb{Z}_2}$ and its twist by ω are *not* Morita equivalent: the double semion has spins $\pm i$, the toric code none. Physically, a gapped two-dimensional phase whose boundary carries the symmetry C has the anyons of $\mathcal{Z}(C)$ in its bulk, and every indecomposable module category is a gapped boundary of it · *physical principle* (Kitaev and Kong, §4; the bulk of a string net is the centre, Simon p. 485): the toric code is the bulk of $\mathrm{Vec}_{\mathbb{Z}_2}$, whose two module categories are its two boundaries, and Morita equivalent symmetries share a bulk. Lecture 13 makes this precise on the lattice.

The action, its endofunctors and the bulk

The same module category now has several associated objects of different types. The diagram records their roles. The physical interpretation uses the unitary setting of Theorem 13.5; the algebraic definitions above do not need that interpretation.



An object of $\mathcal{M}$ is an object being acted on. An object of $C^*_{\mathcal{M}}$ is a functor on that whole category. In particular, one simple object in $\mathcal{M} = \mathrm{Vect}_{\mathrm{fd}}$ does not imply one simple boundary excitation: its dual $\mathrm{Rep}(\mathbb{Z}_2)$ has two. The algebra $A = [M, M]$ lies in C and reconstructs the action from a chosen nonzero object M. The algebra of absorbed *bulk* charges will instead be denoted L; it lies in $\mathcal{Z}(C)$ and is introduced in Lecture 13. Thus A and L belong to different categories.

For the two $\mathbb{Z}_2$ action graphs, the data obtained so far are

A in $C = \mathrm{Vec}_{\mathbb{Z}_2}$	$\mathcal{M}$	$C^*_{\mathcal{M}}$
1	$\mathrm{Vec}_{\mathbb{Z}_2}$	$\mathrm{Vec}_{\mathbb{Z}_2}$
$1 \oplus s$	$\mathrm{Vect}_{\mathrm{fd}}$	$\mathrm{Rep}(\mathbb{Z}_2)$

For the regular graph choose $M = M_0$; only 1 fixes it, so $[M_0, M_0] = 1$. For the fixed graph both 1 and s fix M, so $[M, M] = 1 \oplus s$, with the product computed in Example 11.9. The next lecture completes both rows with their absorbed bulk charges. With our composition convention the boundary excitation tensor product is that of $(C^*_{\mathcal{M}})^{\otimes\mathrm{op}}$ in the cited bulk–boundary correspondence. The reversal is invisible up to tensor equivalence in these symmetric $\mathbb{Z}_2$ examples; it remains part of the general convention.

Summary.

- *Handed on:* the dual $C^*_{\mathcal{M}} = \mathrm{Fun}_C(\mathcal{M},\mathcal{M})$, with $(\mathrm{Vec}_G)^*_{\mathrm{Vect}_{\mathrm{fd}}} \simeq \mathrm{Rep}(G)$; Morita equivalence $\mathcal{D}^{\otimes\mathrm{op}} \simeq C^*_{\mathcal{M}}$ (Definitions 12.1 and 12.5 and Theorem 12.2).
- *The centre:* half-braidings $\gamma_X : X \otimes Z \to Z \otimes X$; $\mathcal{Z}(C)$ is braided by $c = \gamma'_Z$ (Definition 12.6 and Theorem 12.7).
- *The toric code:* $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$ has $1, e, m, \varepsilon$ with $\theta = \chi(g)$ and double braiding -1 for e, m (Theorem 12.8); in general $\mathcal{Z}(\mathrm{Vec}_G) \simeq \mathrm{Rep}D(G)$, labelled by classes and centraliser irreps (Theorem 12.9).
- *Theorems:* the centre is non-degenerate, of dimension $(\dim C)^2$, modular if C is spherical, and $C \boxtimes C^{\mathrm{rev}}$ if C is modular; Morita equivalence is braided equivalence of centres (Theorems 12.10 and 12.11).
- *Next:* Lecture 13 reads module categories as boundaries and defects, and the dual category as the symmetry on them.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: the Deligne product, §1.11, p. 15, and §4.6, p. 73; bimodule categories, Def. 7.1.7, p. 133, and Ex. 7.4.3, p. 136; the dual category, §7.11, p. 154, Def. 7.12.2 and Ex. 7.12.3, p. 155; Morita equivalence, Def. 7.12.17, p. 157; $(\mathrm{Vec}_G)^*_{\mathrm{Vect}_{\mathrm{fd}}} \simeq \mathrm{Rep}(G)$, Ex. 7.12.19, p. 158; the centre, Def. 7.13.1, p. 162, and (7.42); $\mathrm{Rep}D(H)$, Prop. 7.14.6, p. 165; the reverse braiding, Def. 8.1.4, pp. 195–196; $D(G)$,

Ex. 8.3.9, p. 199; the braiding of the centre, Prop. 8.5.1, p. 207; $\mathcal{Z}(\mathrm{Vec}_G)$ and Rep$D(H)$, Ex. 8.5.4 and 8.5.6, pp. 207–208; braided $C \to \mathcal{Z}(C)$, Prop. 8.6.1, p. 208; Morita and centres, Thm. 8.12.3, p. 222; non-degenerate and modular centres, Cor. 8.20.13–8.20.14, p. 243; $\dim \mathcal{Z}(C)$, Prop. 9.3.4, p. 280. Kitaev and Kong, *Models for gapped boundaries and domain walls*, arXiv:1104.5047, §§4–5, pp. 8–17. Müger, *Modular categories* (2012): the centre, Def./Prop. 4.5, p. 9 (his half-braiding sits on the other side, so his centre is our $\mathcal{Z}(C)^{\mathrm{rev}}$), and Thm. 7.1, p. 21. Kong and Zhang, arXiv:2205.05565: the toric code from half-braidings, (4.1.3), p. 77 (their fermion is f); Ex. 4.2.17, p. 84; the $e \leftrightarrow m$ autoequivalence, §4.3.2, p. 91. Bakalov and Kirillov, *Lectures on Tensor Categories and Modular Functors*: $D(G)$, §3.2, pp. 59–61. Simon, *Topological Quantum*: quantum doubles, pp. 455–458; Morita equivalence and bulks, p. 485.

Lecture 13

Boundaries, defects and non-invertible symmetries

Objectives. *Lectures 11 and 12 built module categories, dual categories and the centre as mathematics. This lecture interprets them physically. The toric code has rough and smooth lattice boundaries, corresponding to the two module categories over* $\mathrm{Vec}_{\mathbb{Z}_2}$. *The anyons absorbed at each boundary form a* Lagrangian algebra. *In the critical Ising field theory, an interface between the theory and its gauged version becomes a topological line* N. *Its fusion rule* $N \otimes N = 1 \oplus \eta$ *makes this symmetry non-invertible. Gauging a group symmetry means summing over networks of its symmetry lines. Full gauging uses a module category with one simple object, which exists only when the symmetry has no anomaly.*

By the end of this lecture you will be able to:

- Find the anyons that condense on the rough and smooth boundaries of the toric code, from the lattice.
- Define commutative, connected, étale and Lagrangian algebras, and find the Lagrangian algebras of the toric code; state their correspondence with module categories.
- State the Kramers–Wannier line of the critical Ising model, its fusion rule and its category, and explain why the lattice operator is not a fusion category.
- State full gauging of Vec_G^{ω} symmetry, its anomaly and its dual symmetry $\mathrm{Rep}(G)$.

13.1 The toric-code boundaries

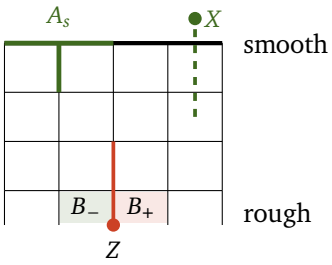
Recall the toric code of *Quantum Computing* (Lecture 14): qubits on the edges of a square lattice, the star term $A_s = \prod X$ on the four edges at a vertex and the plaquette term $B_p = \prod Z$ around a face. The Hamiltonian is

$$H = -\sum_s A_s - \sum_p B_p.$$

The commuting terms have eigenvalues $+1$ and -1. A ground state has every $A_s = B_p = +1$. A term with eigenvalue -1 is *violated*; with these coefficients each violated term raises the energy by 2. A string of Z's on a path violates the stars at its two ends, creating e's; a string of X's crossing a path on the dual lattice violates the plaquettes at its ends, creating m's. In Lecture 12 these are the simple objects $e = (0, \chi_-)$ and $m = (1, 1)$ of $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$. These pairs use additive group labels: 0 is the identity and 1 is the generator previously called s.

A Hamiltonian is *gapped* if the energy gap above its ground states is bounded below by a constant independent of the system size (Kitaev and Kong, fn. 1). A boundary ends the

lattice along a line, and its terms are the truncated stars and plaquettes that remain; it is a *gapped boundary* if the Hamiltonian with it stays gapped. There are two natural ways to cut:



The thick green edges support one boundary star. The shaded three-edge plaquettes B_- and B_+ share the dangling edge carrying the Z-string endpoint. The dashed X-string exits through the smooth boundary. At the *smooth* boundary (top) the lattice ends along a row of edges, each in one plaquette, and the boundary stars have three edges. At the *rough* boundary (bottom) edges stick out, each meeting one vertex, and the boundary plaquettes have three edges (Kitaev–Kong §2; Bravyi and Kitaev call the rough one an x-boundary and the smooth one a z-boundary, and the names vary between sources).

Theorem 13.1 (The toric-code boundaries). *At the smooth boundary m condenses: an X-string ending on the boundary leaves no excitation there, while e cannot be absorbed. At the rough boundary e condenses and m cannot be absorbed. The excitations that remain are $\{1, e\}$ on the smooth boundary and $\{1, m\}$ on the rough one.*

Proof. Absorption is local: consider operators supported near one long boundary, away from its ends and from the other boundary. An X on an edge flips the plaquettes containing it, a Z flips the stars at its ends, and every operator is a sum of products of these. A smooth-boundary edge lies in one plaquette, so an X-string on the dual lattice that ends by crossing it violates only the plaquette at its other end: m is absorbed. But every edge there still has two ends, each with a star, so every product of Paulis supported there flips an even number of stars: the number of e's keeps its parity, and a single e cannot disappear. At the rough boundary the outer end of a dangling edge has no star, so a Z-string ending there violates nothing at that end: e is absorbed. Every edge, the dangling ones included, lies in two plaquettes, so the same parity count keeps a single m. ∎

So the two boundaries are told apart by which anyon disappears into them: m on the smooth one, e on the rough one (Kong–Zhang Rem. 4.1.1 calls them the m- and e-condensed boundaries). Bringing any bulk anyon to the boundary gives a boundary excitation, and fusion decides which. At the smooth boundary m becomes 1, so $\varepsilon = e \otimes m$ becomes e: the map is $1, m \mapsto 1$ and $e, \varepsilon \mapsto e$. At the rough boundary it is $1, e \mapsto 1$ and $m, \varepsilon \mapsto m$ (Kitaev–Kong, maps (3)–(4)). The boundary excitations fuse like $\mathbb{Z}_2$, but they are not anyons: on a line they cannot braid.

13.2 Module categories and boundaries; Lagrangian algebras

The anyons that condense on a boundary form an algebra in the braided centre, and the algebra must be of a special kind.

Definition 13.2 (Commutative, connected, étale, Lagrangian)**.** An algebra (L, μ, ι) in a braided category is *commutative* if $\mu \circ c_{L,L} = \mu$, and *connected* if $\dim \mathrm{Hom}(1, L) = 1$; it is *étale* if it is commutative and separable (Lecture 11). In a non-degenerate braided fusion category $\mathcal{D}$, a connected étale algebra L is *Lagrangian* if $\mathrm{FPdim}(L)^2 = \mathrm{FPdim}(\mathcal{D})$ (DMNO Defs. 3.1 and 4.6). For unitary categories, FPdim = d, so this is $d(L)^2 = \dim \mathcal{D}$.

Theorem 13.3 (Lagrangian algebras of the toric code · *proved given coherence and EGNO Prop. 7.8.30*)**.** *In $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$ the Lagrangian algebras are exactly $1 \oplus e$ and $1 \oplus m$, up to isomorphism. The square-zero algebras on $1 \oplus e$, $1 \oplus m$ and $1 \oplus \varepsilon$ are connected, commutative and two-dimensional, but not separable.*

Proof. $\dim \mathcal{Z} = 4$, so $d(L) = 2$, and connectedness leaves $L = 1 \oplus X$ with $X \neq 1$ one of e, m, ε, each with $X \otimes X = 1$. The unit is 1; the product on $X \otimes X \to 1$ is a scalar λ, and associativity holds for every λ. If $\lambda = 0$, the copy of X is a submodule of the right module L with no complement, since $1 \cdot X = X$: the modules are not semisimple, so L is not separable (Lecture 11, EGNO Prop. 7.8.30); commutativity holds, both sides being 0. If $\lambda \neq 0$, rescaling X makes $\lambda = 1$, a unique algebra. It is separable: the map $\Delta : L \to L \otimes L$ sending 1 to $\frac{1}{2}(1 \otimes 1 + X \otimes X)$ and X to $\frac{1}{2}(X \otimes 1 + 1 \otimes X)$ is a bimodule section of μ, as for $\mathbb{C}[\mathbb{Z}_2]$, and a morphism of $\mathcal{Z}$, since each component maps between copies of one simple object. On $X \otimes X \cong 1$ the braiding $c_{X,X}$ acts by θ_X (Theorem 12.8), so $\mu \circ c = \mu$ forces $\theta_X = 1$: $X = e$ or m, not ε, whose $\theta = -1$. ∎

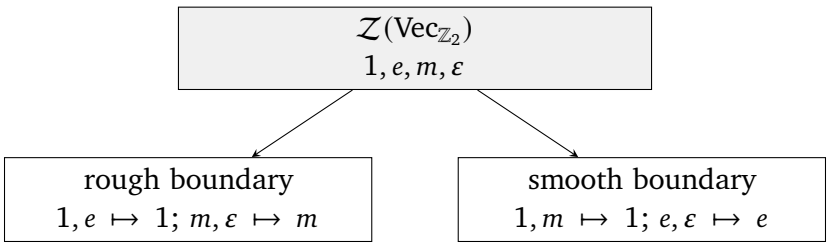
Two Lagrangian algebras and, in Lecture 11, two indecomposable module categories over $\mathrm{Vec}_{\mathbb{Z}_2}$ are related by the following correspondence.

Theorem 13.4 (Davydov, Müger, Nikshych, Ostrik · *cited*)**.** *For a fusion category C, Lagrangian algebras in $\mathcal{Z}(C)$ correspond bijectively to indecomposable C-module categories $\mathcal{M}$; the category of L-modules in $\mathcal{Z}(C)$ is then $(C_{\mathcal{M}}^*)^{\otimes \mathrm{op}}$, the dual with its tensor product reversed (DMNO Prop. 4.8 and Rem. 4.9(i), for right module categories; Kong 2014, (4.3)).*

For $\mathrm{Vec}_{\mathbb{Z}_2}$, the regular module ($H = 1$) has dual $\mathrm{Vec}_{\mathbb{Z}_2}$, and the module $\mathrm{Vect}_{\mathrm{fd}}$ ($H = \mathbb{Z}_2$) has dual $\mathrm{Rep}(\mathbb{Z}_2)$ (Theorem 12.2). The algebra $1 \oplus e$ condenses pure charges, the $H = 1$ boundary, with excitations $\{1, m\} \simeq \mathrm{Vec}_{\mathbb{Z}_2}$: the rough boundary. The algebra $1 \oplus m$ is the $H = \mathbb{Z}_2$ boundary, with $\{1, e\} \simeq \mathrm{Rep}(\mathbb{Z}_2)$: the smooth one (Beigi, Shor and Whalen, p. 13; Kong–Zhang Ex. 4.2.17). So the Morita equivalent symmetries $\mathrm{Vec}_{\mathbb{Z}_2}$ and $\mathrm{Rep}(\mathbb{Z}_2)$ of Lecture 12 live on two boundaries of one bulk. The toric code is the simplest *Levin–Wen* (string-net) model, the one built from $\mathrm{Vec}_{\mathbb{Z}_2}$ (Kitaev and Kong, §3). Its edges carry simple-object labels, and its local terms enforce fusion rules and F-moves. § 14.4 gives the construction.

The two action graphs completed

The data of Lectures 11 and 12 combine with § 13.1 to give two different restrictions from the same bulk. The arrows below send a bulk anyon to its boundary excitation.



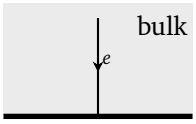
The following table keeps the algebra that presents the action separate from the algebra of absorbed bulk anyons. In both columns the input category is $C = \mathrm{Vec}_{\mathbb{Z}_2}$.

	rough	smooth
Algebra in C	$A = 1$	$A = 1 \oplus s$
Action category $\mathcal{M}$	$\mathrm{Vec}_{\mathbb{Z}_2}$	$\mathrm{Vect}_{\mathrm{fd}}$
Action of s	exchanges M_0, M_1	fixes M
Dual $C^*_{\mathcal{M}}$	$\mathrm{Vec}_{\mathbb{Z}_2}$	$\mathrm{Rep}(\mathbb{Z}_2)$
Algebra in $\mathcal{Z}(C)$	$L = 1 \oplus e$	$L = 1 \oplus m$
Boundary labels	$\{1, m\}$	$\{1, e\}$

For example, ε becomes m on the rough boundary because e is absorbed; it becomes e on the smooth boundary because m is absorbed. The boundary tensor category in the correspondence is the dual with reversed tensor order, $(C^*_{\mathcal{M}})^{\otimes\mathrm{op}}$. Here the two duals are symmetric and tensor equivalent to $\mathrm{Vec}_{\mathbb{Z}_2}$, so that reversal does not distinguish them. Their restrictions from the fixed bulk do distinguish the boundaries. Neither the one simple object of $\mathrm{Vect}_{\mathrm{fd}}$ nor the constituents of L are the list of surviving boundary excitations.

Theorem 13.5 (Kitaev and Kong · *cited*)**.** *For a unitary fusion category C and an indecomposable C-module category $\mathcal{M}$, the Levin–Wen model built from C has a gapped boundary determined by $\mathcal{M}$, whose excitations form* $\mathrm{Fun}_C(\mathcal{M}, \mathcal{M})$ *(Kitaev and Kong, §4, with the tensor product in the reversed order; Kong 2014, §4.2).*

That every gapped boundary arises this way is a physical principle · *physical principle*: Kitaev and Kong "cannot readily defend the word 'all'" (p. 2), and a gapped edge needs a condensable set of bosons (Simon §25.8, pp. 359–360). With it, the gapped boundaries of the toric code are exactly the rough and the smooth one. The same count applies to any centre: $\mathcal{Z}(\mathrm{Fib})$ has exactly one gapped boundary, and $\mathcal{Z}(\mathrm{Vec}_G)$ for $G = S_3$ has four (Sheet 13). A boundary is drawn as a line ending the bulk, and an anyon in L ends on it:



rough boundary, $L = 1 \oplus e$

13.3 TFIM recap; KW at criticality; the continuum line and the lattice caveat

The transverse-field Ising chain on n sites, with periodic boundary conditions, is

$$H_{\mathrm{chain}} = -\sum_j \left(Z_j Z_{j+1} + g\, X_j\right)$$

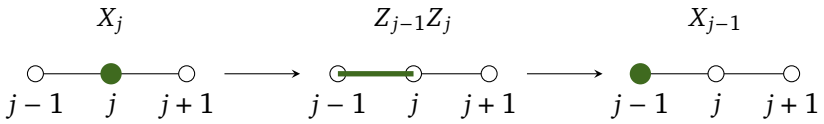
(Shao (3.41)). Its $\mathbb{Z}_2$ symmetry is $\eta = \prod_j X_j$. In *Integrability* (Lecture 18, eq. (18.1)) the same chain is written with $\sigma^x\sigma^x$ and σ^z, so $X \leftrightarrow \sigma^z$, $Z \leftrightarrow \sigma^x$, and η is its parity $\Pi = \prod \sigma^z$. For $g > 0$, in the thermodynamic limit, the ground state is unique and symmetric for $g > 1$, the symmetry is broken for $0 < g < 1$, and $g = 1$ is the critical point.

The *Kramers–Wannier* map sends $X_j \mapsto Z_{j-1}Z_j$ and $Z_{j-1}Z_j \mapsto X_{j-1}$: a site term becomes a bond term and a bond term a site term. Applied to H_{chain} it gives

$$-\sum_j \left(X_j + g\, Z_{j-1}Z_j\right) = g\, H_{\text{chain}}(1/g),$$

so $g \leftrightarrow 1/g$: the spectra at g and $1/g$ agree up to the scale g in the η-even sector of the periodic chain (*Integrability* Prop. 18.9). Away from $g = 1$ it maps the disordered phase to the ordered one: it is a duality between two Hamiltonians, not a symmetry of one (Shao, fn. 33). At $g = 1$ it commutes with H_{chain}, but it is not invertible. The product of all X_j is η, while the product of all $Z_{j-1}Z_j$ around the ring is 1; so the operator D realising the map has $D\eta = D$, and it kills the η-odd states.

The site shift can be read directly from the local assignments. Filled circles mark the site carrying X; a thick segment marks the bond carrying ZZ. Each arrow applies the same KW assignment once.



Two applications therefore shift X_j to X_{j-1}. With the translation convention below, this is $\mathcal{T}^{-1}$. This is an identity of operator assignments, before any continuum limit.

A continuum limit describes distances much larger than the lattice spacing and excitations with energies small compared with the microscopic scale. It does not identify every lattice state with a continuum state. A correlation function is an expectation value of a product of local operator insertions; it tests how those insertions are related. These meanings suffice for the topological-line statement below, whose Ising field-theory content is cited.

A *topological line* in a field theory is an operator supported on a curve in spacetime whose small deformations change no correlation function away from the other insertions; along a spatial circle it commutes with the Hamiltonian (Shao §2.2).

To *gauge* a finite symmetry, we sum over insertions of networks of its lines, joined by the multiplication of an algebra in the symmetry category. The next section states when this construction is possible for the whole group.

Definition 13.6 (Tambara–Yamagami categories)**.** For a finite abelian group A, the fusion category $\mathrm{TY}(A, \chi, \pm 1/\sqrt{|A|})$ has simple objects $a \in A$ and one more, N, with $a \otimes b = ab$, $a \otimes N = N \otimes a = N$ and $N \otimes N = \bigoplus_{a\in A} a$. Its associators are fixed by a symmetric non-degenerate bicharacter $\chi : A \times A \to \mathrm{U}(1)$ and the sign of $\pm 1/\sqrt{|A|}$ (Bhardwaj and Tachikawa, (5.21) and after). A *bicharacter* is multiplicative in each variable:

$$\begin{aligned} \chi(ab, c) &= \chi(a, c)\chi(b, c), \\ \chi(a, bc) &= \chi(a, b)\chi(a, c) \\ &(a, b, c \in A). \end{aligned}$$

It is symmetric if $\chi(a, b) = \chi(b, a)$, and non-degenerate if $\chi(a, b) = 1$ for every $b \in A$ forces $a = 1$. For $A = \mathbb{Z}_2 = \{1, \eta\}$, $\chi(\eta, \eta) = -1$.

Theorem 13.7 (The Kramers–Wannier line · *cited*)**.** *In the Ising conformal field theory, the continuum limit at $g = 1$, there is a topological line N with*

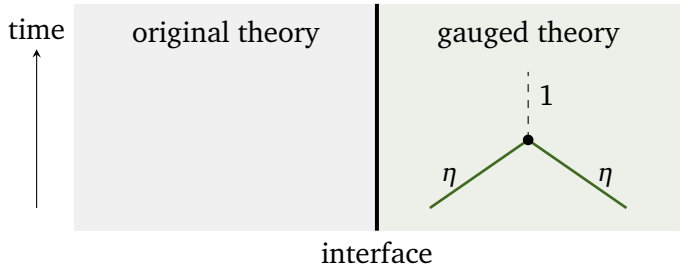
$$N \otimes N = 1 \oplus \eta,$$
$$N \otimes \eta = \eta \otimes N = N.$$

The lines $1, \eta, N$ form the Tambara–Yamagami category $\mathrm{TY}(\mathbb{Z}_2, \chi, +1/\sqrt{2})$, the Ising fusion category with $\kappa = +1$ (Fröhlich, Fuchs, Runkel and Schweigert; Shao §3.1, (3.13); Bhardwaj and Tachikawa §5.6).

The parameter $+1/\sqrt{2}$ is the vacuum entry of the F-matrix in Theorem 6.7. Multiplying this entry by $d_\sigma = \sqrt{2}$ gives the Frobenius–Schur indicator $\kappa_\sigma = +1$. The negative parameter gives $\mathrm{SU}(2)_2$'s fusion category (Lecture 9). The line N is *non-invertible*: no line N' has $N \otimes N' = 1$.

Gauging $\mathbb{Z}_2$ replaces its symmetry category by the dual $\mathrm{Rep}(\mathbb{Z}_2)$ of Example 12.3. At criticality the gauged theory is the same theory. Gauging η in half of space and not in the other half leaves an interface between the two, and at $g = 1$ the two halves are the same theory, so the interface is a topological line in one theory: this is N (Shao §6.3.1, (6.11)).

The following spacetime drawing shows what "half of space" means. Time runs upward, and the vertical seam separates the original theory from its gauged version. The right-hand network is one illustrative term in the sum of symmetry-line networks. At the marked junction two η lines merge to the unit line, using the multiplication of $1 \oplus \eta$.



Gauging sums over allowed labels and junctions, with the consistency data of the algebra; this single drawing does not specify all weights of that sum. The interface is a *defect*: it separates two theories. At the self-dual critical point the two theories are identified and this interface gives the line N in one theory, by the cited continuum result. No claim about all periodic lattice sectors follows from this drawing.

On the lattice the operator D realising the map does not obey this fusion rule. With the translation $\mathcal{T}$, $\mathcal{T} O_j \mathcal{T}^{-1} = O_{j+1}$,

$$D^2 = (1 + \eta)\, \mathcal{T}^{-1},$$
$$\eta D = D\eta = D, \qquad D X_j = Z_{j-1} Z_j D$$

(Seiberg, Seifnashri and Shao, (1.4)–(1.5)). Together with $D Z_{j-1} Z_j = X_{j-1} D$, these relations fix D up to a scalar.[1]

[1] Comparing SSS (2.36) with Shao (3.46)–(3.48) gives

$$D_{\mathrm{Shao}} = D^\dagger / \sqrt{2}.$$

Thus Shao's (3.51) reads

$$D^2_{\mathrm{Shao}} = \frac{1}{2}(1 + \eta)\mathcal{T}.$$

The factor $1+\eta$ has a separate origin from the site shift. Decompose the periodic-chain Hilbert space by the eigenvalues of η:

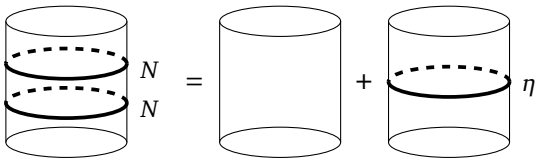
$$\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-, \qquad P_+ = \frac{1}{2}(1+\eta), \quad P_- = \frac{1}{2}(1-\eta).$$

The relations $D\eta = \eta D = D$ say that D vanishes on $\mathcal{H}_-$ and maps $\mathcal{H}_+$ to itself. In the chosen SSS normalization, its square on $\mathcal{H}_+$ is twice the backward translation. Equivalently, on the whole Hilbert space,

$$D^2 = 2P_+\mathcal{T}^{-1}, \qquad (D/\sqrt{2})^2 = P_+\mathcal{T}^{-1}.$$

Thus $1+\eta$ is twice a projector, rather than the projector itself. The intertwining assignments determine D only up to a scalar; the printed square chooses that scalar up to sign. The coefficient 2 would change under a different rescaling of D. The odd-sector kernel and the shift direction would remain. Here $1+\eta$ is addition of operators on one Hilbert space. In the continuum fusion rule $1 \oplus \eta$ is a direct sum of line objects, which becomes addition when the lines act as operators.

The square of the lattice line contains a translation, which is not an internal symmetry: its defects add or remove sites, so the lattice symmetry mixes with space and is not a fusion category (SSS §1.2). Only the continuum limit, where $\mathcal{T} \to 1$ on the relevant states, gives $N \otimes N = 1 \oplus \eta$. Drawn on the cylinder of space and time, a line wraps the spatial circle, and in the continuum two parallel N lines are the sum of no line and an η line:



(13.1)

On the periodic chain the right side acquires $\mathcal{T}^{-1}$.

13.4 Full gauging, Rep(G), anomaly (statement)

Definition 13.8 (Full gauging, anomaly)**.** *Full gauging* uses the symmetry lines of the whole group in the network construction above. A fusion category is *anomalous* if it has no module category with one simple object (Thorngren and Wang, p. 13). By Theorem 11.6 with $H = G$, Vec_G^ω is anomalous exactly when $[\omega] \neq 0$.

Theorem 13.9 (Full gauging · *sketched*)**.** *Gauging all of a* Vec_G^ω *symmetry uses a rank-one module category, a* ψ *solving the module pentagon of Theorem 11.6 with* $H = G$*, which exists if and only if* $[\omega] = 0$*; the choices of* ψ *then differ by classes in* $H^2(G, \mathbb{C}^\times)$*. For* $\omega = \psi = 1$ *the gauged theory has the dual symmetry* $(\mathrm{Vec}_G)^*_{\mathrm{Vect_{fd}}} \simeq \mathrm{Rep}(G)$ *(Bhardwaj and Tachikawa §§4.2–4.6).*

Sketch. Gauging inserts a fine network of symmetry lines, glued by an algebra in the symmetry category; for all of G this is $\bigoplus_g \delta_g$ with a product twisted by ψ, whose module category is the rank-one module (Lecture 11), and which is an algebra only when $[\omega] = 0$. The lines that remain topological in the gauged theory are the module endofunctors of that module category, its dual (Theorem 12.2). ∎

For $\mathbb{Z}_2$ this is Kramers–Wannier again: gauging η gives a theory with symmetry $\mathrm{Rep}(\mathbb{Z}_2) \simeq \mathrm{Vec}_{\mathbb{Z}_2}$. For a non-abelian group the dual is not a group at all: gauging S_3 gives the symmetry

$\mathrm{Rep}(S_3)$, in which std $\otimes$ std $\cong$ **1** $\oplus$ sgn $\oplus$ std (Lecture 3), so the line std has no inverse. Gauging an ordinary symmetry can produce a non-invertible one. The semion's $\mathrm{Vec}^{\omega}_{\mathbb{Z}_2}$ cannot be gauged: its only module category is the regular one. Gauging a subgroup uses the other module categories, beyond this course.

Summary.

- *Boundaries:* the toric code has the smooth (m condenses, excitations $\{1, e\}$) and the rough (e condenses, excitations $\{1, m\}$) lattice boundaries (Theorem 13.1); by the physical principle they are its only gapped ones.
- *Lagrangian algebras:* connected étale L with $d(L)^2 = \dim \mathcal{Z}$; in the toric code exactly $1 \oplus e$ and $1 \oplus m$ (Theorem 13.3); they match the module categories, and the excitations are the duals with the tensor product reversed (Theorems 13.4 and 13.5).
- *Non-invertible symmetry:* the critical Ising field theory has the line N with $N \otimes N = 1 \oplus \eta$, the category $\mathrm{TY}(\mathbb{Z}_2, \chi, +1/\sqrt{2})$ (Theorem 13.7); on the periodic chain $D^2 = (1 + \eta)\mathcal{T}^{-1}$ instead.
- *Gauging:* full gauging of $\mathrm{Vec}^{\omega}_{G}$ needs $[\omega] = 0$ and, for $\omega = \psi = 1$, yields $\mathrm{Rep}(G)$ (Theorem 13.9).
- *Next:* Lecture 14 assembles the course into topological quantum field theory.

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Block VI

Synthesis

Lecture 14

Topological quantum field theory: a synthesis

Objectives. *The course has built duals (Lecture 4), F-symbols (Lecture 6), modular data (Lecture 10), algebras (Lecture 11) and centres (Lecture 12). A topological quantum field theory assigns vector spaces to spaces and linear maps to spacetimes between them. In one dimension it is determined by a dual pair. In two dimensions it is determined by a commutative Frobenius algebra. Finite groups give examples called Dijkgraaf–Witten theories. In three dimensions a spherical fusion category gives a Turaev–Viro state sum. For a unitary category its Hamiltonian form is a string net. A modular category gives the Reshetikhin–Turaev theory, related to the state sum through the centre. These constructions cover selected structures of the course. Turaev–Viro theories are non-chiral;* $\mathrm{Rep}(\mathrm{SU}(2))$ *and generic* U_q *are not finite inputs.*

By the end of this lecture you will be able to:

- Define a TQFT as a symmetric monoidal functor, and prove that its state spaces are finite-dimensional, with $Z(\Sigma \times S^1) = \dim Z(\Sigma)$.
- Define Frobenius algebras in pictures, state the classification of 2D TQFTs, and compute partition functions with the handle element.
- Compute 2D Dijkgraaf–Witten partition functions by Mednykh's formula.
- State the Turaev–Viro and Reshetikhin–Turaev constructions and their relation through the centre, with the framing anomaly, and count torus states.

14.1 Atiyah's axioms; 1D

The *bordism category* Bord_n has as objects closed oriented $(n-1)$-manifolds, and as morphisms $\Sigma \to \Sigma'$ the compact oriented n-manifolds M with $\partial M = \bar{\Sigma} \sqcup \Sigma'$, up to diffeomorphism fixing the boundary. Here $\bar{\Sigma}$ is Σ with its orientation reversed. Composition glues along the common boundary. Disjoint union $\sqcup$, with the swap of components, makes this category symmetric monoidal (Kock 1.2–1.3; Carqueville–Runkel §2). The category Cob_2 of Lecture 1 is Bord_2. Time runs upward, as everywhere in the course.

Definition 14.1 (TQFT). An n-dimensional *topological quantum field theory* is a symmetric monoidal functor $Z\colon \mathrm{Bord}_n \to \mathrm{Vect}$ (Kock 1.3.32; Carqueville–Runkel Def. 2.1).

The target is *symmetric*, not braided: exchanging two spaces is the flip of vector spaces. A closed n-manifold is a bordism $\emptyset \to \emptyset$, so its image is a number, the *partition function*.

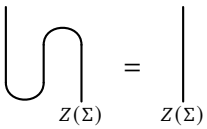
The dimension of the bordism decides what kind of value Z assigns. A surface is a spacetime in 2D and a space in 3D.

Theory	geometric input	linear-algebra output
nD	closed $(n-1)$-manifold Σ	vector space $Z(\Sigma)$
nD	bordism $M\colon \Sigma \to \Sigma'$	map $Z(M)\colon Z(\Sigma) \to Z(\Sigma')$
nD	closed n-manifold M	scalar $Z(M)\colon \mathbb{C} \to \mathbb{C}$
2D	circle S^1; closed torus T^2	space $Z(S^1)$; number $Z(T^2)$
3D	closed torus T^2; closed 3-manifold M	space $Z(T^2)$; number $Z(M)$

Thus the notation $Z(T^2)$ must always be read with the theory's dimension. Taking its dimension makes sense in the 3D row; in the 2D row it is already a scalar.

Theorem 14.2 (Atiyah's axioms, consequences · *sketched; (i)–(ii) proved*)**.** *(i) Each $Z(\Sigma)$ is finite-dimensional, with $Z(\bar{\Sigma}) \cong Z(\Sigma)^*$. (ii) $Z(\Sigma \times S^1) = \dim Z(\Sigma)$. (iii) A one-dimensional TQFT is determined, up to isomorphism, by the finite-dimensional space $V = Z(\mathrm{pt}_+)$, and every such V occurs (Carqueville–Runkel Thm. 3.2).*

Proof. (i) The cylinder $\Sigma \times [0,1]$, bent, is a bordism $\bar{\Sigma} \sqcup \Sigma \to \emptyset$ and one $\emptyset \to \Sigma \sqcup \bar{\Sigma}$; straightening the bend is a diffeomorphism, so their images ev and coev satisfy the snake equations of Definition 4.1. In one dimension these are pictures of the course's cups and caps:



(14.1)

Write $\mathrm{coev}(1) = \sum_{i=1}^{k} a_i \otimes b_i$, a finite sum, with $a_i \in Z(\Sigma)$ and $b_i \in Z(\bar{\Sigma})$. The snake equation says $v = \sum_i a_i \,\mathrm{ev}(b_i \otimes v)$ for every $v \in Z(\Sigma)$, so $a_1, \dots, a_k$ span $Z(\Sigma)$; and ev identifies $Z(\bar{\Sigma})$ with $Z(\Sigma)^*$. (ii) $\Sigma \times S^1$ is the cylinder with its ends glued, and the gluing passes one copy of Σ past the other: it is $\mathrm{ev} \circ P \circ \mathrm{coev}$, with P the flip of Vect, so $\sum_i \mathrm{ev}(b_i \otimes a_i)$, the trace of the identity, which is $\dim Z(\Sigma)$:



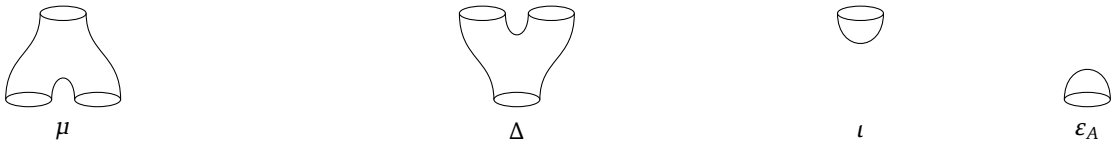
$$= \dim Z(\Sigma). \tag{14.2}$$

(iii) Every 1-manifold with boundary is a composite of the elbows and the identities, so Z is fixed by V, ev and coev; the relations between them are the snake equations, so any dual pair defines a theory. ∎

So a one-dimensional theory is Lecture 4's duality, and the circle is the loop $\dim V$ of Equation (4.7).

14.2 Frobenius algebras; 2D TQFT is commutative Frobenius

In two dimensions the objects are disjoint unions of circles, and every surface with boundary is glued from a few pieces:

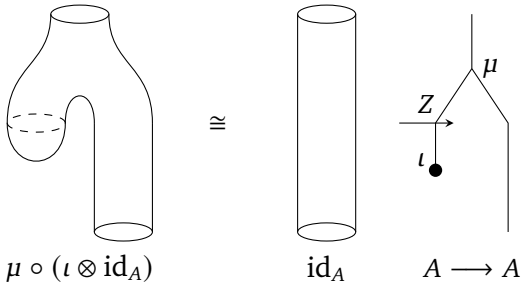


the pair of pants, read upward, merges two circles into one; turned over it splits one into two; the disc with its circle on top creates a circle from nothing; the disc with its circle below annihilates one. (Kock calls the first disc a "cap"; here the words cup and cap stay

with Lecture 4.) With $A = Z(S^1)$ they are maps $\mu\colon A\otimes A\to A$, $\Delta\colon A\to A\otimes A$, $\iota\colon \mathbb{C}\to A$ and $\varepsilon_A\colon A\to\mathbb{C}$.

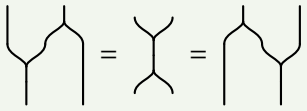
A glued surface, its graph and its map

Filling one incoming circle of the merging pants with a disc gives a cylinder. The dashed ellipse marks the glued circle, which is internal after gluing. The other incoming circle and the outgoing circle remain. The centre-line graph records the same composition after applying Z.



The surface equivalence is the unit relation in the cited presentation of Bord_2. Its image is the algebra's unit law. The graph is bookkeeping for the map; its edges stand for vector spaces, while the surface's boundary components are circles.

Definition 14.3 (Frobenius algebra)**.** A *Frobenius algebra* is $(A,\mu,\iota,\Delta,\varepsilon_A)$, an algebra (Definition 11.1) and a coalgebra on the same space (Δ coassociative with counit ε_A: the algebra axioms with the arrows reversed), satisfying the *Frobenius relation*



(14.3)

that is, $(\mathrm{id}\otimes\mu)(\Delta\otimes\mathrm{id}) = \Delta\circ\mu = (\mu\otimes\mathrm{id})(\mathrm{id}\otimes\Delta)$ (Kock Lemma 2.3.19). It is *commutative* if $\mu\circ P=\mu$, and *special* if $\mu\circ\Delta=\mathrm{id}$ (Heunen and Vicary Def. 5.5).

Equivalently, $\varepsilon_A\circ\mu$ is a nondegenerate pairing with $\varepsilon_A(xy\cdot z)=\varepsilon_A(x\cdot yz)$, and then Δ is determined (Kock 2.2.5, Props. 2.3.22 and 2.3.24). For instance $\mathbb{C}[G]$ with $\varepsilon_A(g)=\delta_{g,1}$, or the algebra of $n\times n$ matrices with the trace.

Theorem 14.4 (2D TQFTs · *cited; the bijection proved from the cited presentation*)**.** *$Z\mapsto Z(S^1)$ is a bijection, up to isomorphism, between 2D TQFTs and commutative Frobenius algebras (Abrams; Kock Thm. 3.3.2; Carqueville–Runkel Thm. 3.5, which make it an equivalence of groupoids).*

Proof. Bord_2 is generated, as a symmetric monoidal category, by the four pieces above and the cylinder, subject to relations which say exactly that they form a commutative Frobenius object: associativity and unit, coassociativity and counit, the Frobenius relation, and commutativity, which uses the swap of the two legs of the pants (Kock Prop. 1.4.13 and 1.4.24–1.4.40; Carqueville–Runkel Thm. 3.6) · *cited*. A symmetric monoidal functor therefore sends S^1 to a commutative Frobenius algebra; conversely a commutative Frobenius algebra gives values on the generators that satisfy the relations, hence a functor. ∎

Commutativity comes from the swap of the legs of the pants, which a symmetric Z sends to the flip of Vect; a Z that is only monoidal can give graded-commutative algebras (Kock

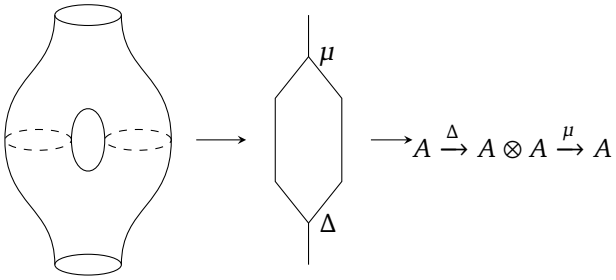
3.3.3). Every closed surface is a composite, and the genus-g surface is

$$Z(\Sigma_g) = \varepsilon_A\big(w^g\big), \qquad w = \mu \circ \Delta \circ \iota = \tag{14.4}$$



with the *handle element* w: each handle is a split followed by a merge.

The split followed by the merge has a different surface from a cylinder: it has a handle and two boundary circles. The dashed ellipses show the two circles at which the pants were glued. Its centre-line graph has a cycle.



Close its incoming circle with ι and its outgoing circle with ε_A to obtain the torus. Repeating the handle g times and closing gives the typed formula

$$Z(\Sigma_g) = \varepsilon_A \circ (\mu \circ \Delta)^g \circ \iota \colon \mathbb{C} \longrightarrow \mathbb{C}.$$

The Frobenius relation makes $\mu \circ \Delta$ multiplication by w. This explains the power w^g in (14.4). A split–merge is not generally the identity; that extra condition would be specialness.

Example 14.5 (Semisimple theories)**.** Let $A = \mathbb{C}^n$, with orthogonal idempotents $e_1, \dots, e_n$ and $\varepsilon_A(e_i) = \lambda_i \neq 0$. The pairing $\varepsilon_A(e_i e_j) = \delta_{ij}\lambda_i$ is nondegenerate, so this is a commutative Frobenius algebra, with $\Delta(e_i) = \lambda_i^{-1} e_i \otimes e_i$ and $w = \sum_i \lambda_i^{-1} e_i$. Hence

$$Z(\Sigma_g) = \sum_i \lambda_i^{1-g},$$

so $Z(S^2) = \sum_i \lambda_i$ and $Z(T^2) = n = \dim A$, as Theorem 14.2 requires. Each idempotent alone gives a theory with one-dimensional state spaces and $Z(\Sigma_g) = \lambda_i^{1-g}$, and on connected surfaces Z is their sum. Every commutative semisimple algebra over $\mathbb{C}$ is some $\mathbb{C}^n$, so this is the whole story for semisimple A, the case of the next section.

For a numerical example take $A = \mathbb{C}^2$ with $\varepsilon_A(e_1) = 1$ and $\varepsilon_A(e_2) = 2$. The unit is $e_1 + e_2$ and the handle element is $w = e_1 + \frac{1}{2}e_2$. Since the idempotents are orthogonal, the three closed surfaces give

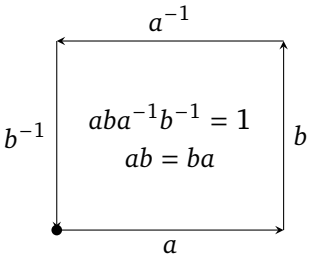
$$\begin{aligned} Z(\Sigma_0) &= \varepsilon_A(e_1 + e_2) = 1 + 2 = 3, \\ Z(\Sigma_1) &= \varepsilon_A(e_1 + \tfrac{1}{2}e_2) = 1 + 1 = 2, \\ Z(\Sigma_2) &= 1 + \frac{1}{2} = \frac{3}{2}. \end{aligned}$$

The sphere tests the counit on the unit. The torus counts the two basis idempotents. The second handle changes the value again; a closed-spacetime value need not be an integer dimension.

14.3 Dijkgraaf–Witten

A group label on a closed path records transport around that path. Following two paths multiplies their labels; reversing a path takes the inverse. Deforming a path without crossing a hole does not change its label in a flat assignment. The group $\pi_1\Sigma_g$ records closed paths from a chosen base point, with deformations identified and multiplication given by following paths. A homomorphism to G is therefore a consistent choice of such labels.

For the torus, cut along two generating paths to get a square. The boundary traversal shown below must have trivial transport when the square is filled.



The labels on opposite sides refer to the same identified path, traversed in opposite directions around the square. Thus a torus assignment is a commuting pair $(a, b) \in G^2$. More generally a genus-g surface gives tuples satisfying

$$\prod_{i=1}^{g} a_i b_i a_i^{-1} b_i^{-1} = 1.$$

For the sphere there are no such generators; the empty product is 1, so there is one assignment. This is the concrete meaning of the homomorphism count below. The surface presentation is the topological input, as in the cited Mednykh formula; no further topology is needed to enumerate these tuples.

A gauge change at the base point means conjugating every label by the same $k \in G$. The partition function divides the number of assignments by $|G|$. It is generally *not* the number of conjugacy orbits: an orbit with stabilizer $\mathrm{Stab}(a, b)$ contributes

$$\frac{|\mathrm{orbit}(a,b)|}{|G|} = \frac{1}{|\mathrm{Stab}(a,b)|}.$$

This equality is the orbit–stabilizer formula. For S_3 there are 6 elements commuting with the identity, 2 with each transposition and 3 with each 3-cycle. Consequently the torus has $6 + 3 \cdot 2 + 2 \cdot 3 = 18$ assignments and

$$Z(T^2) = \frac{18}{6} = 3.$$

The sphere instead has value $1/|G|$ in this normalization. The character formula gives a practical way to perform the same weighted count at every genus.

Theorem 14.6 (2D Dijkgraaf–Witten · *cited; the character sum sketched*)**.** *For a finite group G, the class functions on G with the convolution $(f * h)(x) = \sum_{yz=x} f(y)h(z)$ and*

$\varepsilon_A(f) = f(1)/|G|$ form a commutative Frobenius algebra, and its TQFT has

$$Z(\Sigma_g) = \frac{\#\operatorname{Hom}(\pi_1\Sigma_g, G)}{|G|} = \sum_{\chi} \left(\frac{|G|}{\dim \chi}\right)^{2g-2},$$

the sum over the irreducible characters (Mednykh; Snyder eq. (1); Freed Ex. 4.17).

Sketch. Character orthogonality (*Symmetries*, Lecture 5) gives the central idempotents as class functions:

$$e_\chi(x) = \frac{\dim \chi}{|G|}\,\chi(x^{-1}) \qquad (x \in G).$$

They satisfy $e_\chi * e_{\chi'} = \delta_{\chi\chi'} e_\chi$. Since $\chi(1) = \dim\chi$, the counit is

$$\varepsilon_A(e_\chi) = \frac{e_\chi(1)}{|G|} = \left(\frac{\dim\chi}{|G|}\right)^2.$$

In this basis $\Delta(e_\chi) = \varepsilon_A(e_\chi)^{-1} e_\chi \otimes e_\chi$, so $w = \sum_\chi \varepsilon_A(e_\chi)^{-1} e_\chi$ and $Z(\Sigma_g) = \sum_\chi \varepsilon_A(e_\chi)^{1-g}$. The first expression is Mednykh's theorem (cited; Sheet 14 proves it): it counts flat G-connections, weighted by $1/|G|$ for the gauge symmetry. ∎

For $G = S_3$, with $\dim\chi = 1, 1, 2$:

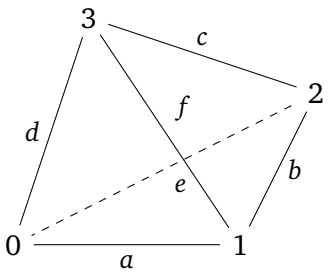
$$Z(\Sigma_2) = 6^2 + 6^2 + 3^2 = 81,$$

and $\#\operatorname{Hom}(\pi_1\Sigma_2, S_3) = 486 = 81 \cdot 6$; at genus 1 the value is 3, the number of conjugacy classes (the script counts both). Scaling ε_A by a nonzero constant multiplies $Z(\Sigma_g)$ by that constant to the power $1 - g$. Thus the counit extracting $f(1)$ instead of $f(1)/|G|$ gives a different answer.

14.4 Turaev–Viro and string nets

In three dimensions the input is a spherical fusion category C, and the F-symbols of Lecture 6 become the weights of a state sum.

The six edge labels of one tetrahedron are the six labels of a recoupling coefficient. Choose ordered vertices $0, 1, 2, 3$ and direct each edge from the smaller vertex to the larger one. The following assignment displays the correspondence with $[F_d^{abc}]_{ef}$.



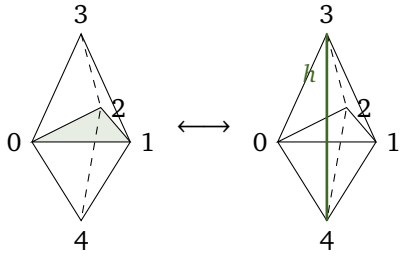
$a \otimes b \to e$
$e \otimes c \to d$
$b \otimes c \to f$
$a \otimes f \to d$

The faces 012 and 023 give the fusion tree with intermediate label e. The faces 123 and 013 give the tree with intermediate label f. Their comparison is precisely the associator with external labels a, b, c, d from Lecture 6. Reversing an edge replaces its label by the dual.

This correspondence identifies the *data* in a tetrahedral weight. It does not identify every oriented tetrahedral weight with the bare matrix entry $[F_d^{abc}]_{ef}$ in our vertex normalization. The state sum uses normalized tetrahedral evaluations, with dimension factors and duality conventions attached to the chosen face bases. The cited construction below fixes those factors consistently. The bipyramid picture then shows why the pentagon, with its intermediate label sum, is the relation needed for the 2–3 move.

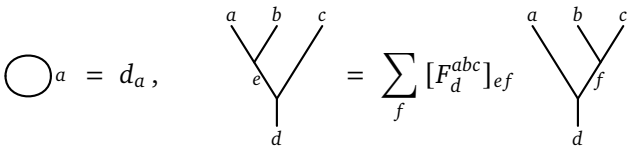
Theorem 14.7 (Turaev and Viro · *sketched*)**.** *Let C be a spherical fusion category with $\dim C \neq 0$, multiplicity-free as in Lecture 6. Colour the edges of a triangulation of a closed 3-manifold M by simple objects, weight each tetrahedron by a normalised F-symbol and each edge by its dimension, and sum over colourings with a factor $(\dim C)^{-1}$ for each vertex. (In general one also sums over bases of the fusion spaces on the faces, Turaev–Virelizier pp. 273–275.) The result $Z_{\mathrm{TV}}(M)$ does not depend on the triangulation, and extends to a 3D TQFT (Turaev–Virelizier Thms. 13.1 and 13.5; Simon (23.1)).*

Sketch. Any two triangulations are related by the Pachner moves. The 2–3 move fills the same triangular bipyramid in two ways:



The two tetrahedra share face 012. The three tetrahedra share the new edge 34. Boundary edge labels are fixed. Summing the label h of the new edge gives the intermediate-channel sum in the pentagon (Theorem 6.3), so the two sets of weights agree. The 1–4 move is absorbed by the vertex factor, by the orthogonality of the F-symbols (Simon p. 319). No braiding is used. ∎

For unitary C its Hamiltonian form is the *string net* of Levin and Wen. Put a label of C on each edge of a trivalent lattice in a surface; the ground states are the superpositions of labelled graphs invariant under local moves, the graphical calculus of Lectures 5 and 6: a loop is d_a, a bubble is removed, and the F-move of Equation (6.1) relates the two ways of splitting d into a, b, c,



(14.5)

(Levin and Wen, (4)–(7)). The Hamiltonian $H = -\sum_v Q_v - \sum_p B_p$ has commuting projectors as terms: Q_v asks that the three labels at a vertex fuse to 1, and B_p inserts the loop $\sum_s (d_s/\dim C)\, s$ inside a plaquette and fuses it into the edges with F-moves. It projects onto the subspace in which the plaquette can be filled without a puncture (Kirillov Thm. 3.9). Its coefficients are those of Lecture 10's killing loop, but the action is different: a loop is added to the lattice and resolved by F-moves. No braiding of the input C is needed. The strand-encircling projector in Lecture 10 instead uses a braiding and non-degeneracy. The ground space on a surface Σ is $Z_{\mathrm{TV}}(\Sigma)$ (Kirillov Thm. 5.1). For $C = \mathrm{Vec}_{\mathbb{Z}_2}$ it is the toric code, with four ground states on the torus (*Quantum Computing* Thm. 14.2).

14.5 RT and TV = RT ∘ $\mathcal{Z}$; the outlook

Theorem 14.8 (Reshetikhin and Turaev; Turaev and Virelizier · *cited*)**.** *(i) A modular category C with $p_+ = p_-$ gives a 3D TQFT RT_C; if $p_+ \neq p_-$ the gluing holds only up to a phase, the* framing anomaly *(Bakalov–Kirillov Thm. 4.4.3 and Rem. 4.4.5). (ii) For a spherical fusion category with $\dim C \neq 0$, $Z_{\mathrm{TV},C} \cong \mathrm{RT}_{\mathcal{Z}(C)}$ (Turaev–Virelizier Thm. 17.1, Cor. 17.5). (iii) $\dim \mathrm{RT}_C(T^2)$ is the number of simple objects of C (Bakalov–Kirillov Ex. 4.5.3).*

For a unitary C, $p_- = \overline{p_+}$, so $p_+/p_- = e^{2\pi ic/4}$ by Theorem 10.9: the anomaly vanishes exactly when $c \equiv 0 \pmod 4$, if the square root of $\dim C$ that normalises RT may be taken negative, as Bakalov and Kirillov allow. With the positive root, as unitarity asks, the gluing phase $p_+/\sqrt{\dim C} = e^{2\pi ic/8}$ is 1 exactly when $c \equiv 0 \pmod 8$. A centre has $c = 0$ (Lecture 12), so $\mathrm{RT}_{\mathcal{Z}(C)}$ is anomaly-free, as (ii) needs. Fibonacci has $c = \frac{14}{5}$: $\mathrm{RT}_{\mathrm{Fib}}$ is anomalous, and Fibonacci is not a centre, so no Turaev–Viro theory produces it alone; the closest is $Z_{\mathrm{TV,Fib}} = \mathrm{RT}_{\mathrm{Fib}\boxtimes\mathrm{Fib}^{\mathrm{rev}}}$, the doubled Fibonacci of Levin and Wen (Turaev–Virelizier Cor. 17.5 with Theorem 12.10). By (iii), the torus carries rank $\mathcal{Z}(C)$ states: 4 for $\mathrm{Vec}_{\mathbb{Z}_2}$, the toric code's degeneracy of *Quantum Computing*, and 8 for Vec_{S_3}, where 2D Dijkgraaf–Witten gave $Z(T^2) = 3$ (the script checks both).

From processes to computable invariants

In Lecture 1 the category Cob_2 described circles and surfaces as systems and processes, but no numerical value had been assigned to a surface. We can now choose a commutative Frobenius algebra, turn each disc and pair of pants into a typed linear map, and compute the value of a glued surface. The calculation for $A = \mathbb{C}^2$ above gives such a functor explicitly on its generators and evaluates its sphere, torus and genus-two surface. The cited presentation ensures that a different decomposition gives the same map.

The three-dimensional constructions use more of the course. Fusion trees describe state spaces, F-symbols change their bases, and the pentagon relates the local tetrahedral moves. The centre identifies the bulk anyons of a string net; the module-category calculation distinguishes boundaries that absorb different charges. For S_3, the value 3 of a closed torus in the 2D theory and the dimension 8 of the torus state space in the 3D theory answer different typed questions. The finite calculations are explicit; the general TV/RT and bulk–boundary constructions retain the cited or sketched status stated above. These are concrete uses of the objects, morphisms and functors with which the course began.

Optional outlook

For further study, a *fully extended* TQFT assigns data in every lower dimension, down to points. Framed theories also retain a continuously chosen frame of tangent vectors. The cobordism hypothesis classifies such theories by a higher-dimensional duality condition called *full dualizability* (Lurie, proof outline, Thm. 1.2.16). Fusion categories over $\mathbb{C}$ satisfy this condition and give framed 3D theories with a fusion category assigned to a point (Douglas, Schommer-Pries and Snyder, Thm. 1). No result here uses this classification.

Summary.

- *TQFTs:* symmetric monoidal functors $\mathrm{Bord}_n \to \mathrm{Vect}$; state spaces are finite-dimensional, $Z(\Sigma \times S^1) = \dim Z(\Sigma)$, and 1D theories are dual pairs (Definition 14.1 and Theorem 14.2).

- *2D:* commutative Frobenius algebras (Theorem 14.4); $Z(\Sigma_g) = \varepsilon_A(w^g)$ with the handle element.
- *Dijkgraaf–Witten:* $Z(\Sigma_g) = \sum_\chi (|G|/\dim\chi)^{2g-2}$; 81 for S_3 at genus 2 (Theorem 14.6).
- *3D:* Turaev–Viro from spherical fusion C, as string nets for unitary C; RT from modular C, anomalous unless $c \equiv 0 \pmod 4$, or modulo 8 with the positive root of $\dim C$; $Z_{\mathrm{TV},C} = \mathrm{RT}_{\mathcal{Z}(C)}$, and the torus counts rank $\mathcal{Z}(C)$ (Theorems 14.7 and 14.8).

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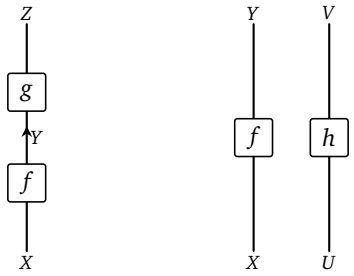
Kock, *Frobenius Algebras and 2D Topological Quantum Field Theories*: TQFTs, 1.2.23–1.3.32, pp. 30–54; the presentation of 2Cob, Prop. 1.4.13 and 1.4.24–1.4.40, pp. 62–77; Frobenius algebras, 2.2.5, p. 95, Lemma 2.3.19 and Props. 2.3.22 and 2.3.24, pp. 117–119; the classification, Thm. 3.3.2 and 3.3.3, pp. 173–175. Carqueville and Runkel, *Introductory lectures on topological quantum field theory*: Def. 2.1, p. 8; Prop. 2.6, p. 13; Thms. 3.2, 3.5 and 3.6, pp. 21–28. Heunen and Vicary, *Categories for Quantum Theory*, Def. 5.5, p. 150. Snyder, math/0703073, eq. (1), p. 1. Freed, *Lectures on Field Theory and Topology*, Ex. 4.17, p. 59. Turaev and Virelizier, *Monoidal Categories and Topological Field Theory*: multiplicities, pp. 273–275; Thms. 13.1 and 13.5, pp. 275–286; Thm. 17.1, p. 432, and Cor. 17.5, p. 441. Bakalov and Kirillov: Thm. 4.4.3, p. 83, Rem. 4.4.5, p. 87, and Ex. 4.5.3, p. 89. Simon, *Topological Quantum*: state sums, (23.1), p. 317, and p. 319. Levin and Wen, cond-mat/0404617, (4)–(7), p. 5. Kirillov, arXiv:1106.6033, Thm. 3.9, p. 9, and Thm. 5.1, p. 11. Lurie, arXiv:0905.0465, Thm. 1.2.16, p. 14. Douglas, Schommer-Pries and Snyder, arXiv:1312.7188, Thm. 1, p. 4.

Diagram reference

A diagram denotes a morphism with specified input and output. In a process diagram, time runs upward. Read inputs from left to right along the bottom, and outputs from left to right along the top. An orientation arrow on a strand distinguishes an object from its dual. It does not reverse the direction in which the composite is read. A spatial lattice or a surface drawing is identified as such in its caption.

Composition and tensor product

A box $f\colon X \to Y$ followed by $g\colon Y \to Z$ denotes $g \circ f\colon X \to Z$. Two boxes beside one another denote a tensor product. The pictures below have types $X \to Z$ and $X \otimes U \to Y \otimes V$, respectively.



The composite has f below g. In the tensor product the first factor is the left strand. Boxes on independent strands can change their relative height by interchange (Theorem 3.4). This does not exchange the strands.

The unit and scalars

The tensor unit is drawn by the absence of a strand. A scalar is a morphism $1 \to 1$, so its box has no incident strands. Scalars commute with one another (Theorem 3.6). In a general monoidal category a scalar can slide on the same side of a strand, but cannot be moved to the other side merely by a planar deformation. In the linear fusion categories used for the anyon calculations, a scalar is a complex number and floats on either side. These statements have different hypotheses.

Bends and duality

A cup has type $1 \to X \otimes X^*$ and a cap has type $X^* \otimes X \to 1$. Their snake composites are identities (Definition 4.1). Before straightening a bend, check the order of its labels and the chosen cap. The four-bend atlas in Lecture 4 (Figure 4.1) also includes the pivotal closing maps, which have the opposite order. Turning a box through a half-turn with the stated dual bends gives $f^*\colon Y^* \to X^*$ (Theorem 4.2). In chosen dual bases this is the transpose matrix. In a dagger category, reflecting the picture gives $f^\dagger\colon Y \to X$. In finite-dimensional

Hilbert spaces its matrix in orthonormal bases is the conjugate transpose. These are different operations.

Pivotal and spherical closures

A pivotal structure supplies the comparison with the double dual needed to close a strand. The resulting trace is cyclic. Its value on the identity is the quantum dimension (Definition 4.6 and Theorem 4.7). Name the pivotal structure when assigning a number to a loop. A spherical structure makes the two trace closures agree. Neither assertion licenses replacing an arbitrary cap by a dagger cap. A flag on a self-dual bend records the self-duality identification; its reversal may produce the Frobenius–Schur sign (Definition 4.9).

Vertices and F-moves

A splitting vertex is a map $c \to a \otimes b$ and its dagger is the fusion vertex. The bubble and completeness relations have the coefficients printed in Equations (5.1) and (5.3). They depend on the chosen vertex normalization. A fusion tree also specifies a basis vector in a fusion space with fixed external and total charges.

An F-move expands one tree basis in another (Definition 6.1). The two branching graphs are different. An F-move is not a planar deformation and does not measure the intermediate charge. Use the inverse basis change when returning to the original basis. The three-particle calculation in Lecture 7 distinguishes these basis changes from the physical exchange between them.

Crossings and ribbons

The crossing $c_{X,Y} : X \otimes Y \to Y \otimes X$ puts the strand from bottom left to top right over the other. A crossing followed by its inverse cancels by invertibility, the Reidemeister II move. A double crossing of the same handedness does not generally cancel. Naturality lets a box pass through a crossing, and the hexagons imply Reidemeister III (Definition 7.2 and Theorem 7.4). A strand passing over a pair can be expanded as a crossing over each strand by the hexagon.

A ribbon remembers its framing. A curl equals a twist on a straight strand, not an unchanged strand (Equation (9.2)). The twist is a specified endomorphism. Removing it would discard information. The marked-band picture in Lecture 9 shows the framing that the thin-strand notation records.

Walls, surfaces and spatial graphs

A module-category picture has a region for a category and a wall for one of its objects. It is not a fusion vertex between three bulk particles. Its joining rules are the module action and module associator (Definition 11.3). Keep the bulk, the module category and the category of boundary excitations separate, as in the role diagram of Lecture 12 and the boundary table of Lecture 13.

In a bordism picture the input and output are actual boundary spaces. A circle can represent a space assigned a vector space, whereas a closed surface can represent a spacetime assigned a number. The surface-to-map dictionary of Lecture 14 states the dimension of the theory. In a string-net picture the graph lies in a spatial surface; the F relation changes the local graph description. The state-sum figure instead depicts a triangulation of spacetime. Its tetrahedron labels and normalization are additional data, specified by the construction.

Bibliography

The per-lecture *References* sections cite the works below by author and short title, with printed pages. Several use conventions different from this course's (diagrams read downward, unit-norm vertices, other names for the global dimension); the lectures translate at the point of use.

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